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(54) **USE OF VAPOR DEPOSITION COATED FLOW PATHS FOR IMPROVED CHROMATOGRAPHY OF METAL INTERACTING ANALYTES**

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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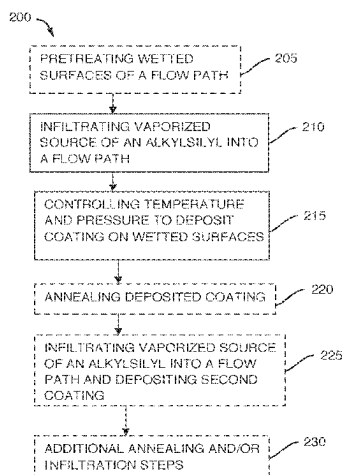
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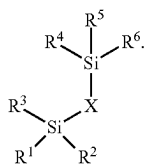
(57) **ABSTRACT**

A device for separating analytes is disclosed. The device has a sample injector, sample injection needle, sample reservoir container in communication with the sample injector, chromatography column downstream of the sample injector, and fluid conduits connecting the sample injector and the column. The interior surfaces of the fluid conduits, sample injector, sample reservoir container, and column form a flow path having wetted surfaces. A portion of the wetted surfaces of the flow path are coated with an alkylsilyl coating that is

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inert to at least one of the analytes. The alkylsilyl coating has the Formula I:



R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 are each independently selected from (C_1-C_6) alkoxy, $-NH(C_1-C_6)$ alkyl, $-N((C_1-C_6)alkyl)_2$, OH, OR^d , and halo. R^d represents a point of attachment to the interior surfaces of the fluidic system. At least one of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is OR^d . X is (C_1-C_{20}) alkyl, $-O[(CH_2)_2O]_{1-20}-$, $-(C_1-C_{10})[NH(CO)NH(C_1-C_{10})]_{1-20}-$, or $-(C_1-C_{10})[alkylphenyl(C_1-C_{10})alkyl]_{1-20}-$.

17 Claims, 34 Drawing Sheets

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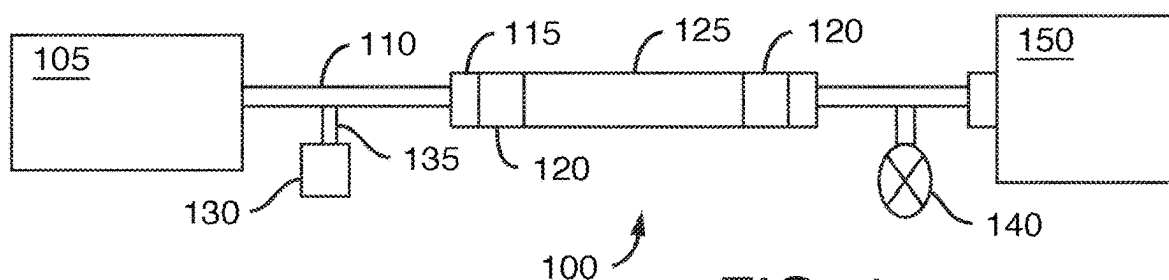


FIG. 1

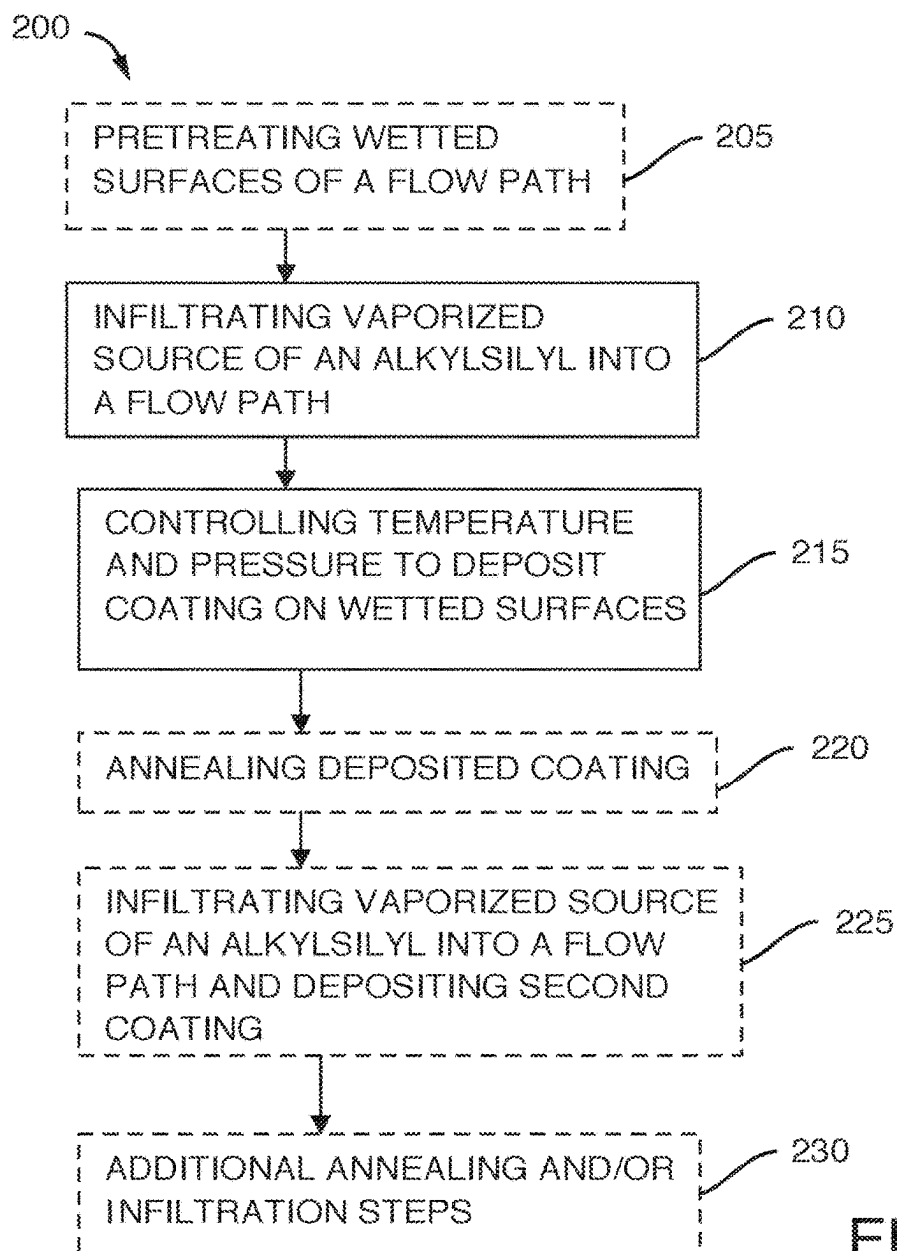


FIG. 2

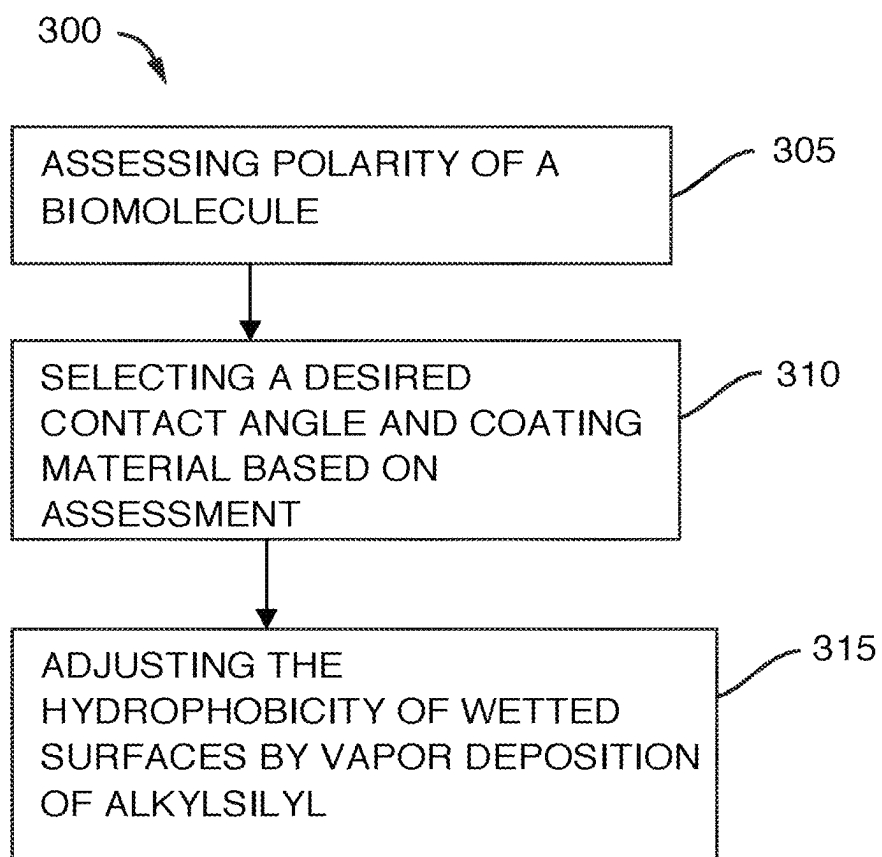


FIG. 3

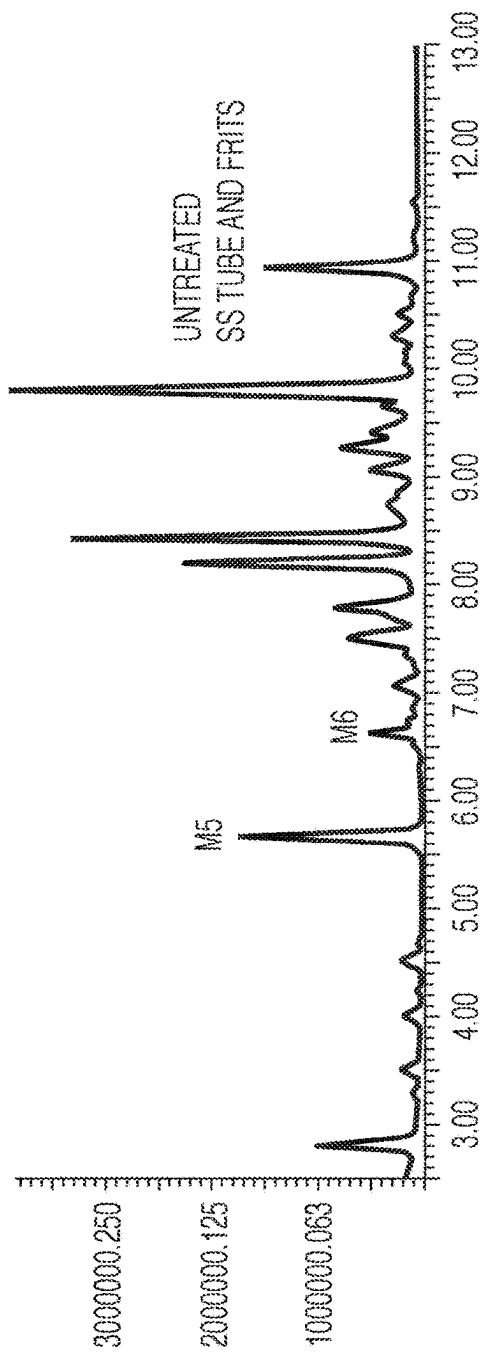


FIG. 4A

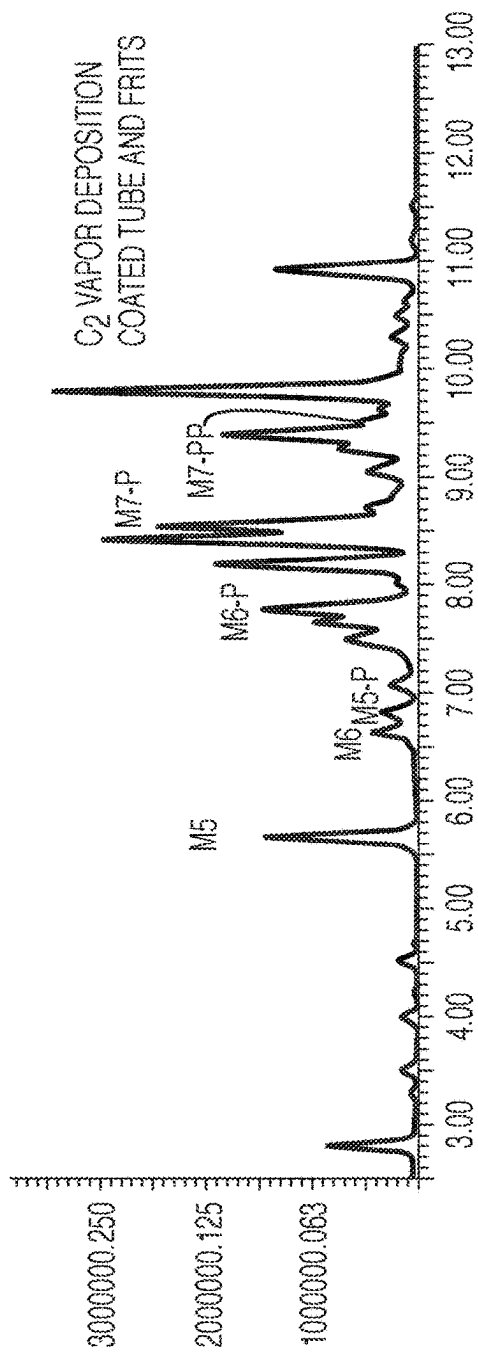


FIG. 4B

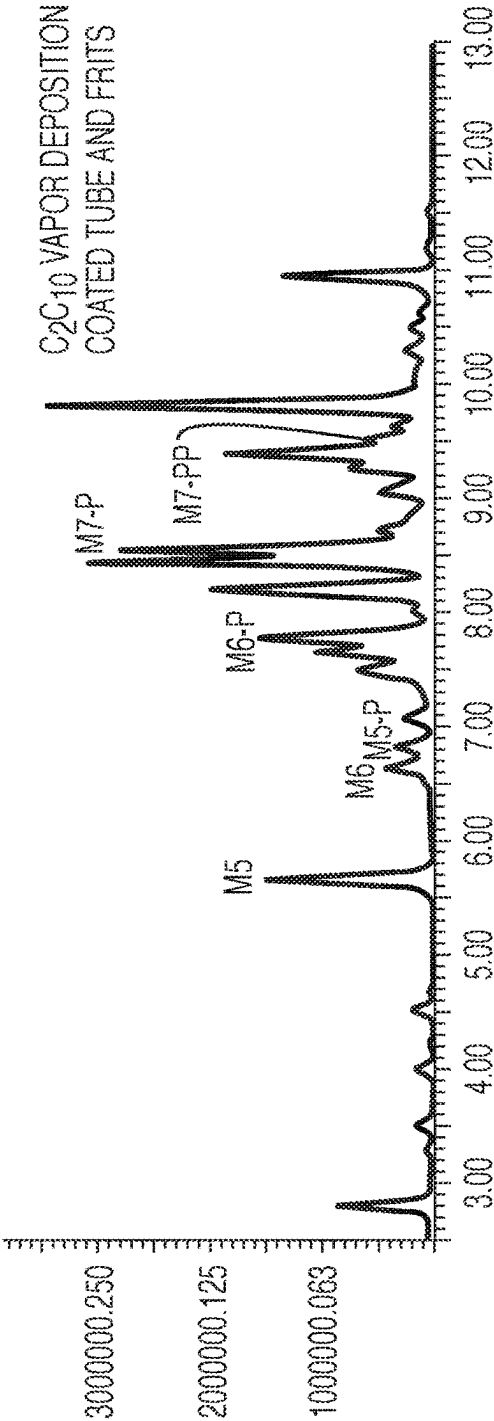


FIG. 4C

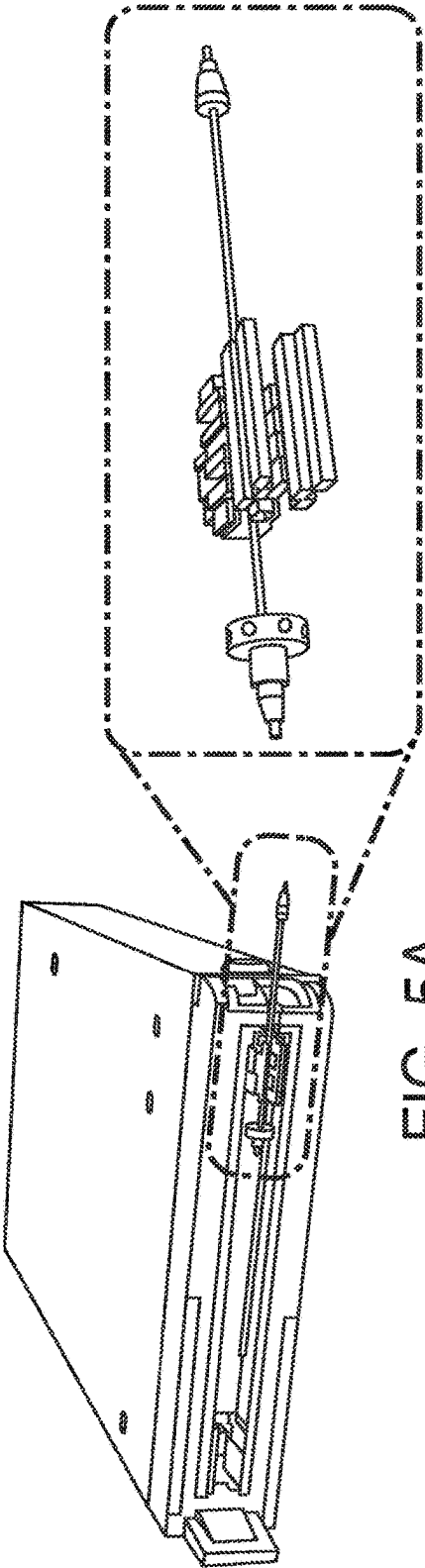


FIG. 5A

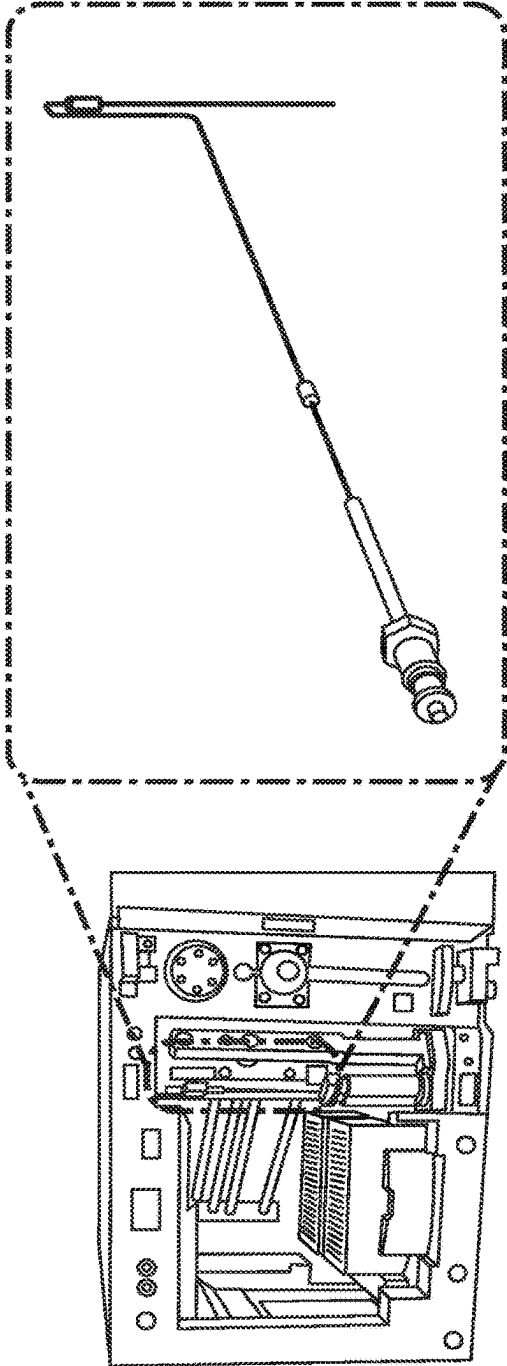


FIG. 5B

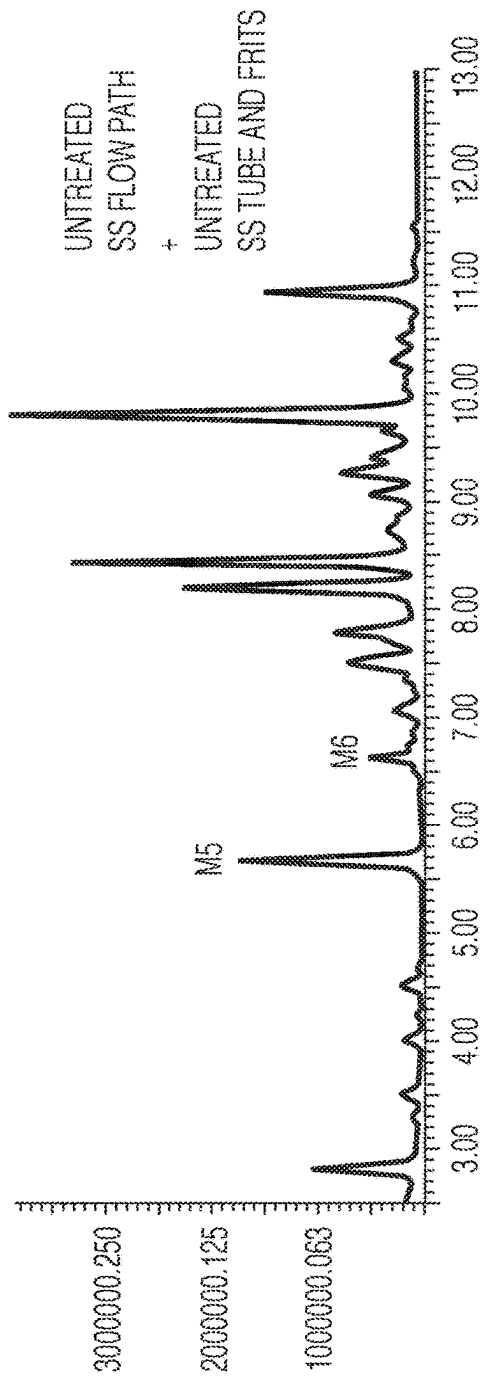


FIG. 6A

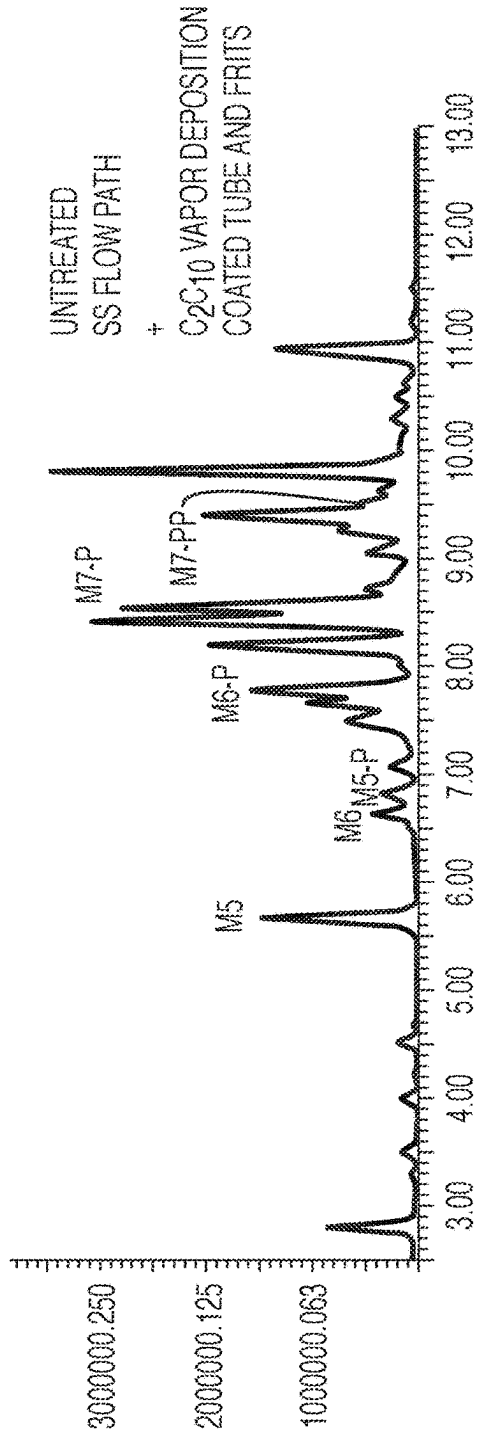


FIG. 6B

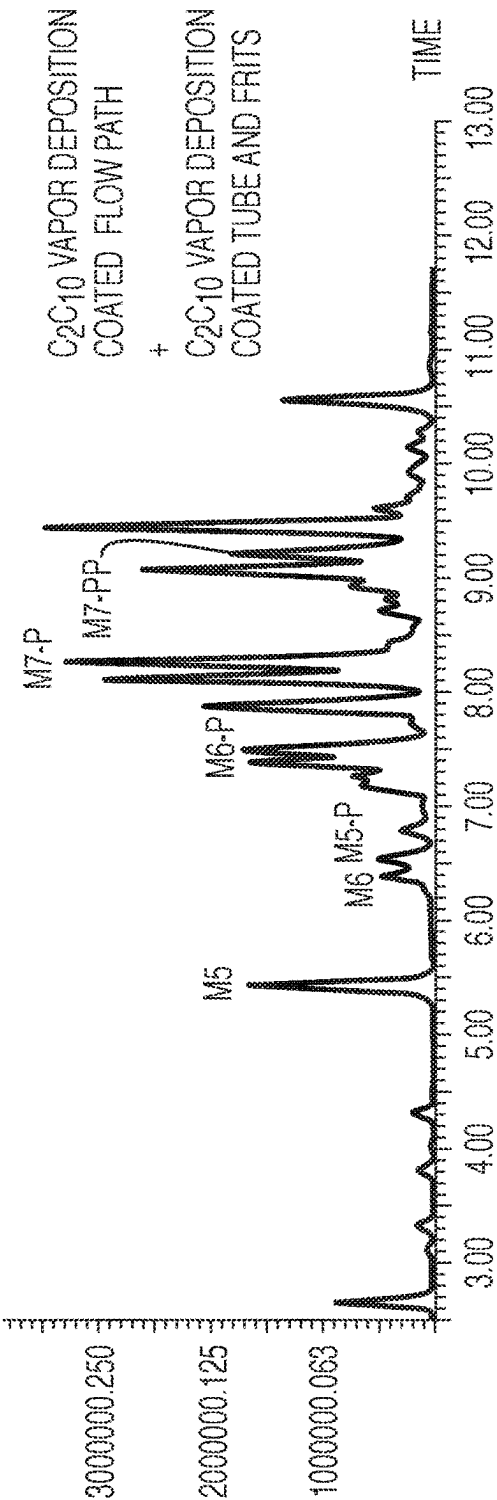


FIG. 6C

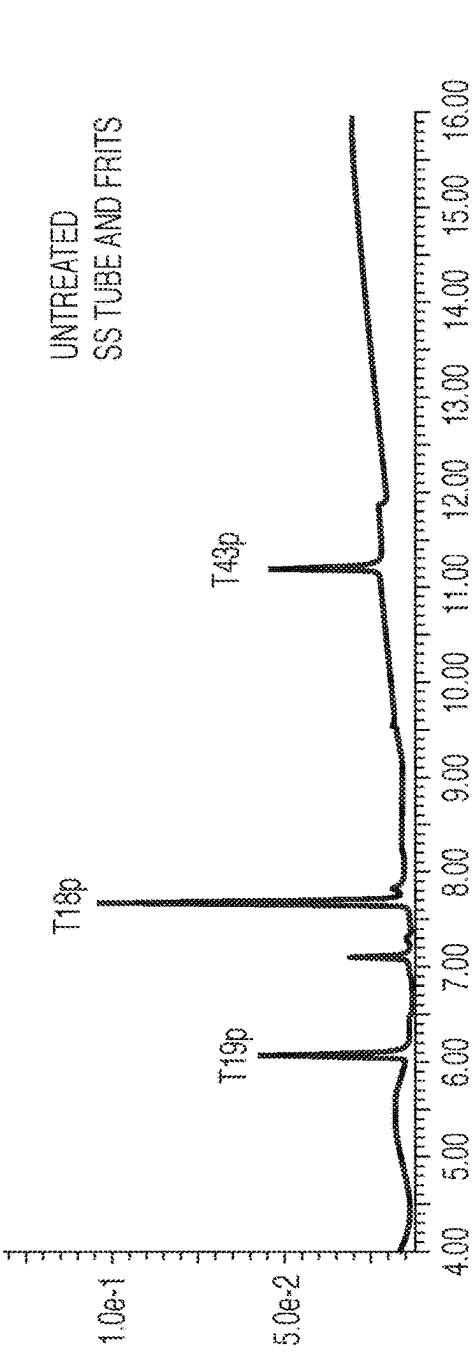


FIG. 7A

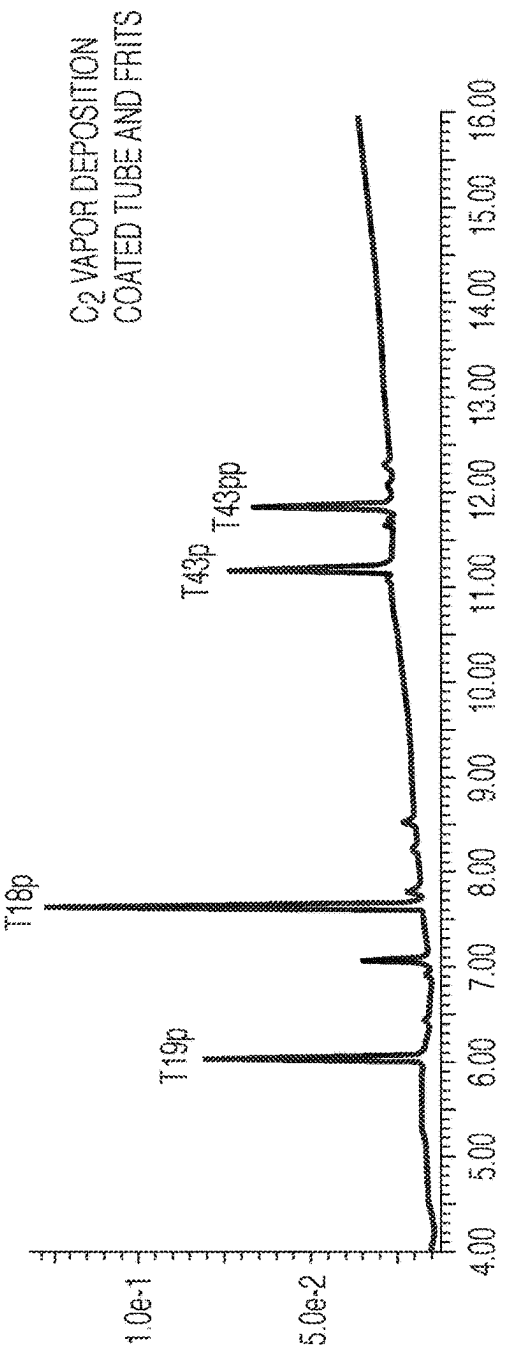


FIG. 7B

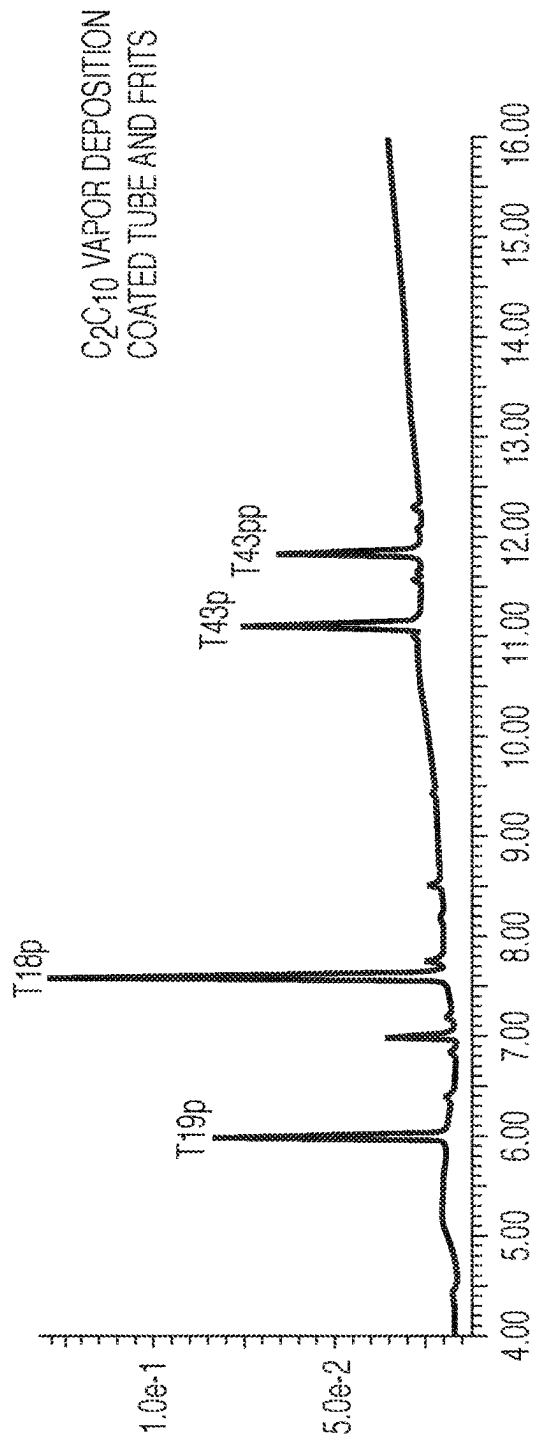


FIG. 7C

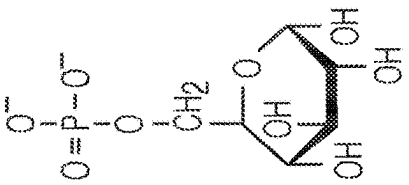
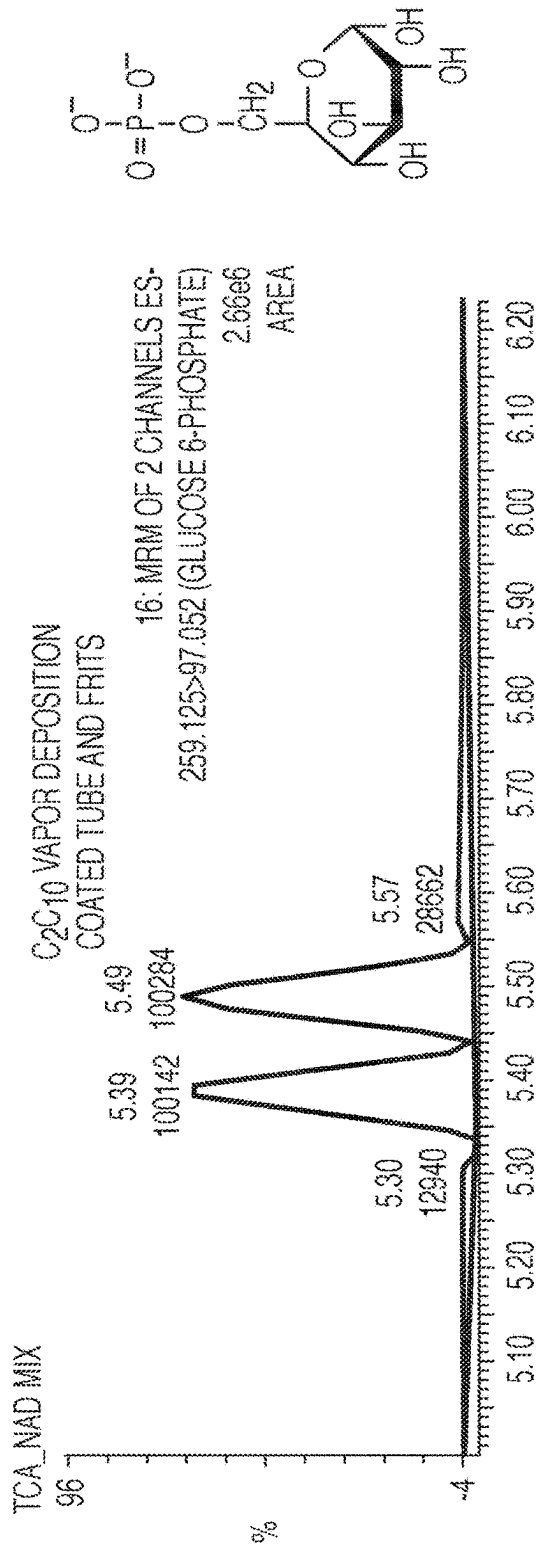


FIG. 8A

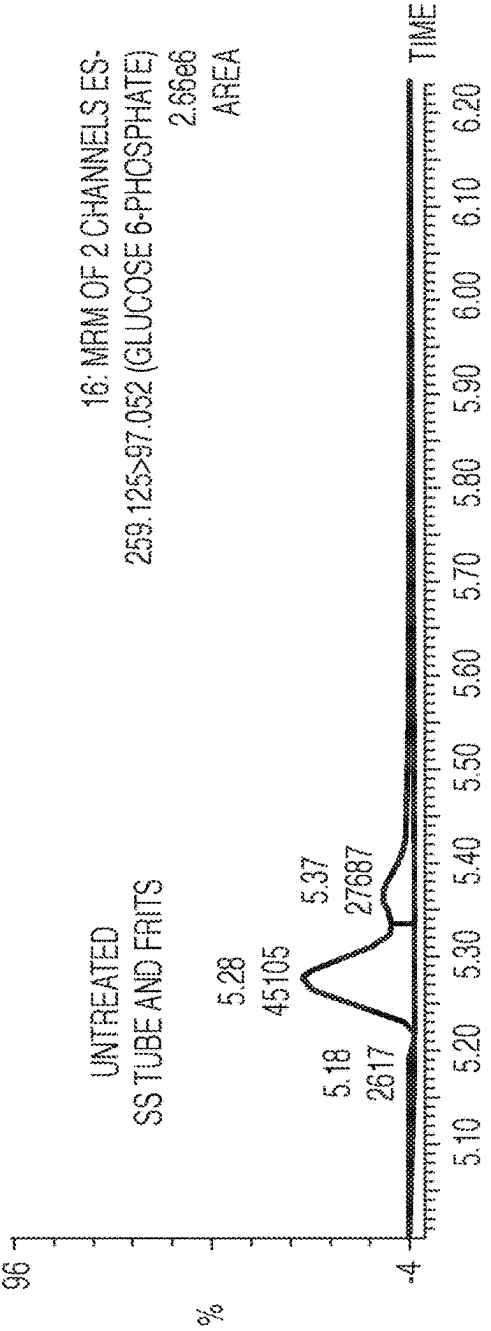


FIG. 8B

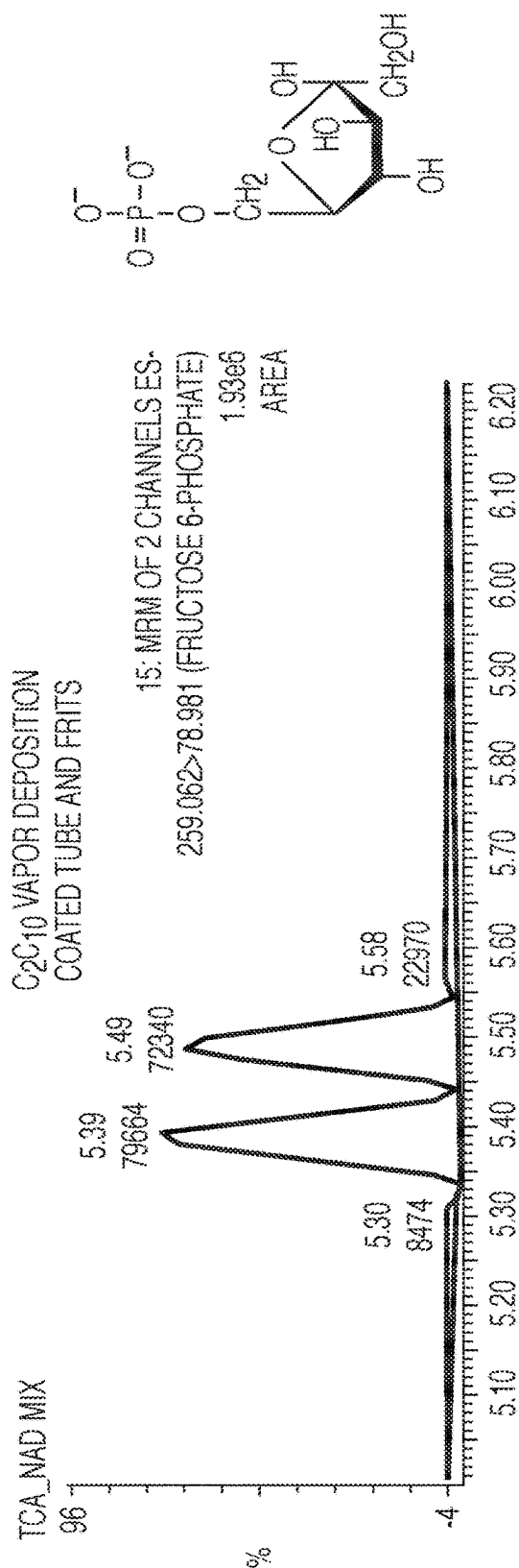


FIG. 9A

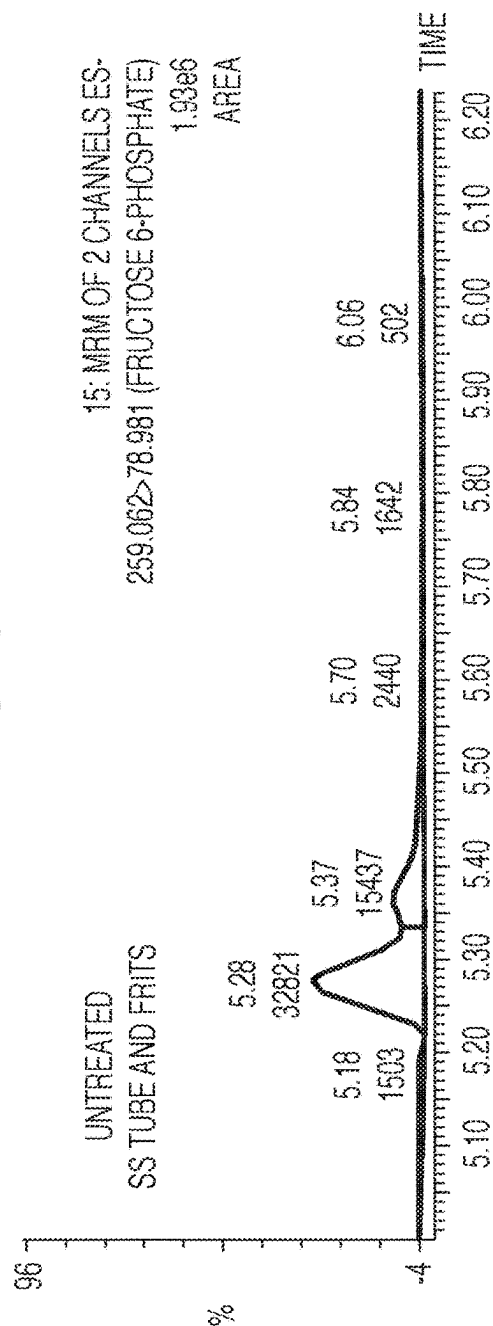


FIG. 9B

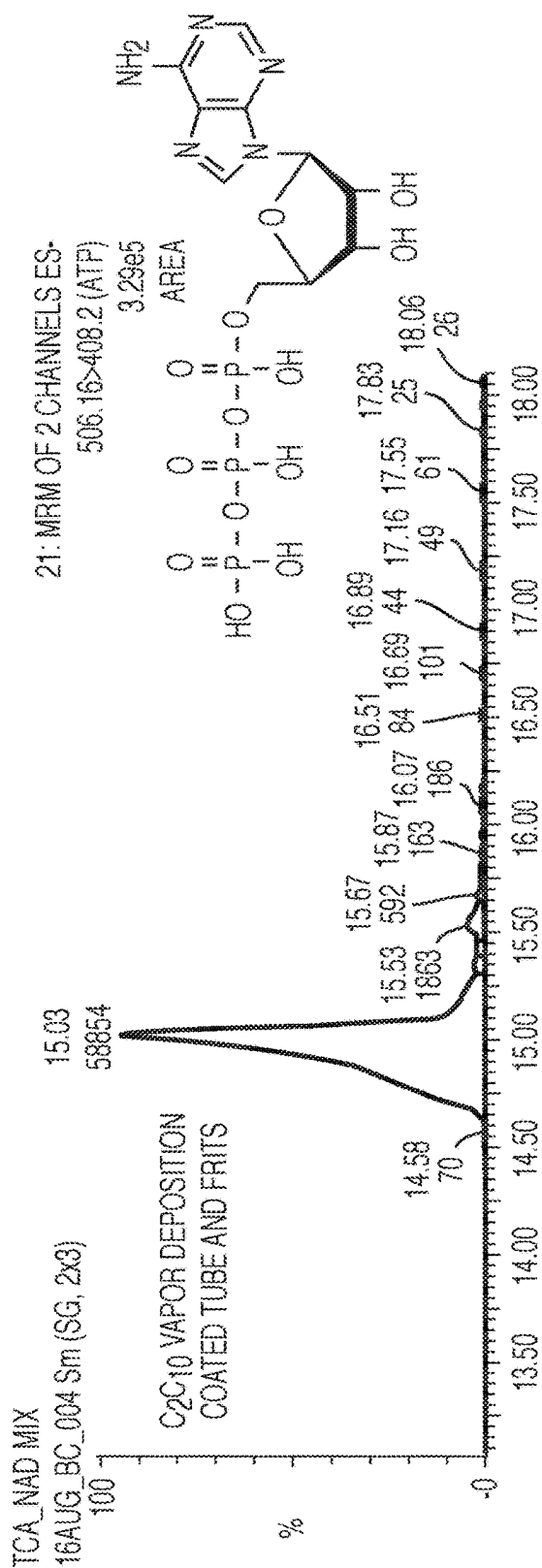


FIG. 10A

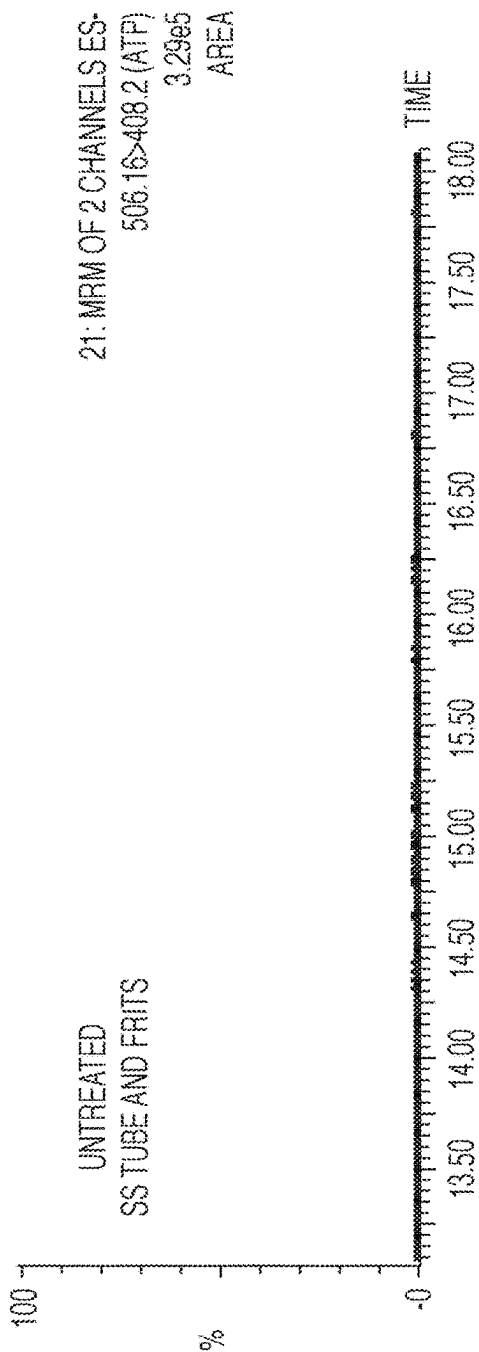


FIG. 10B

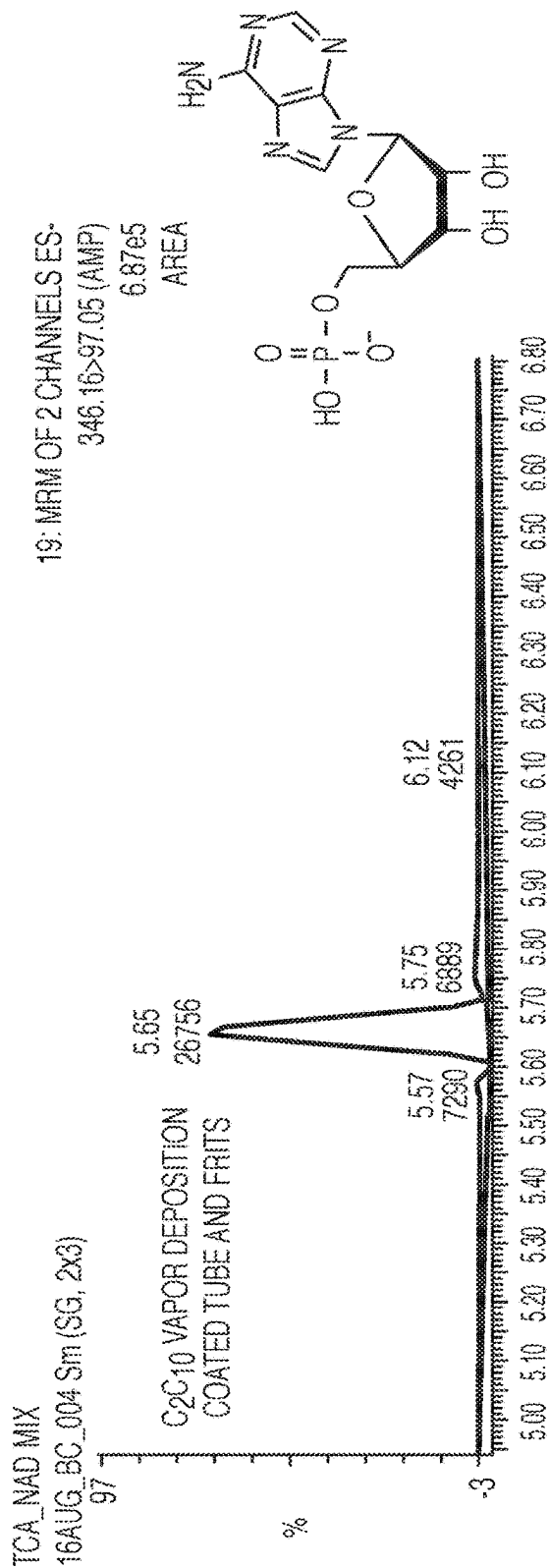


FIG. 11A

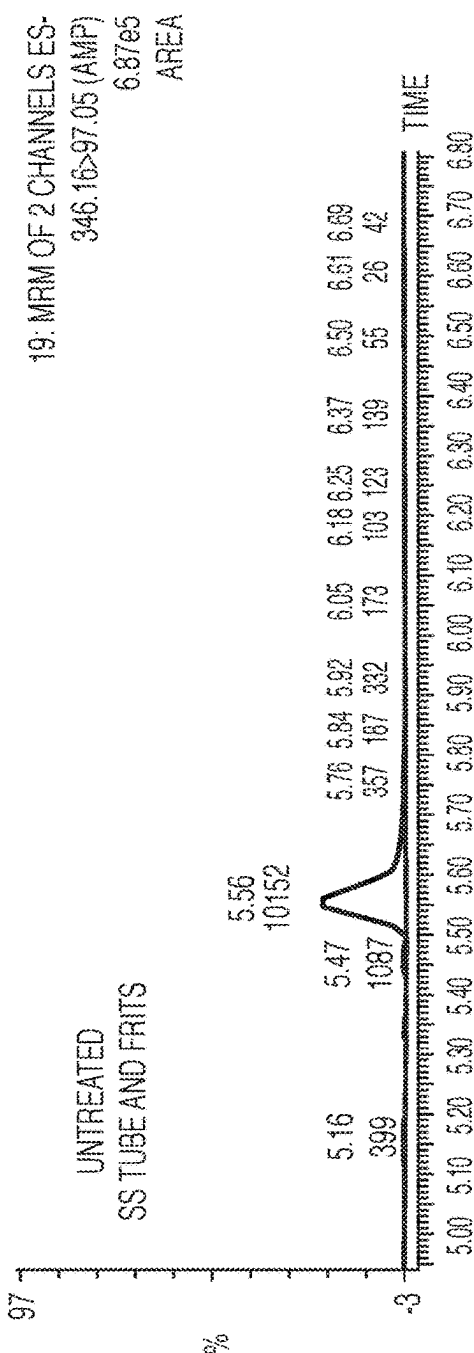


FIG. 11B

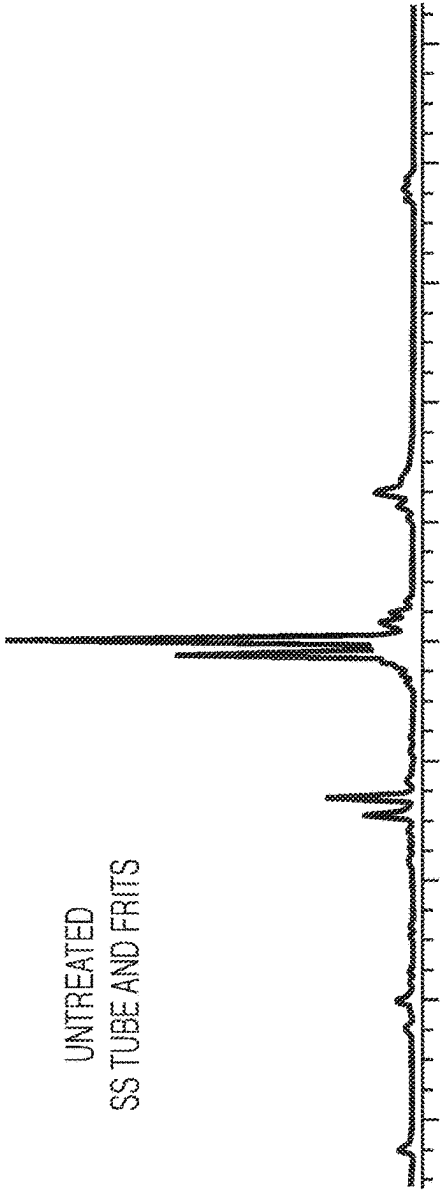


FIG. 12A

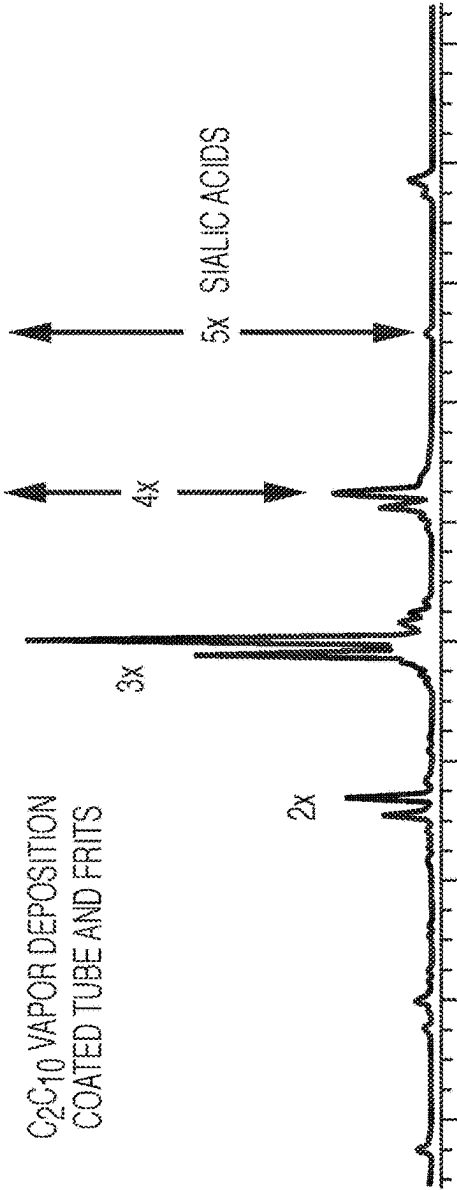
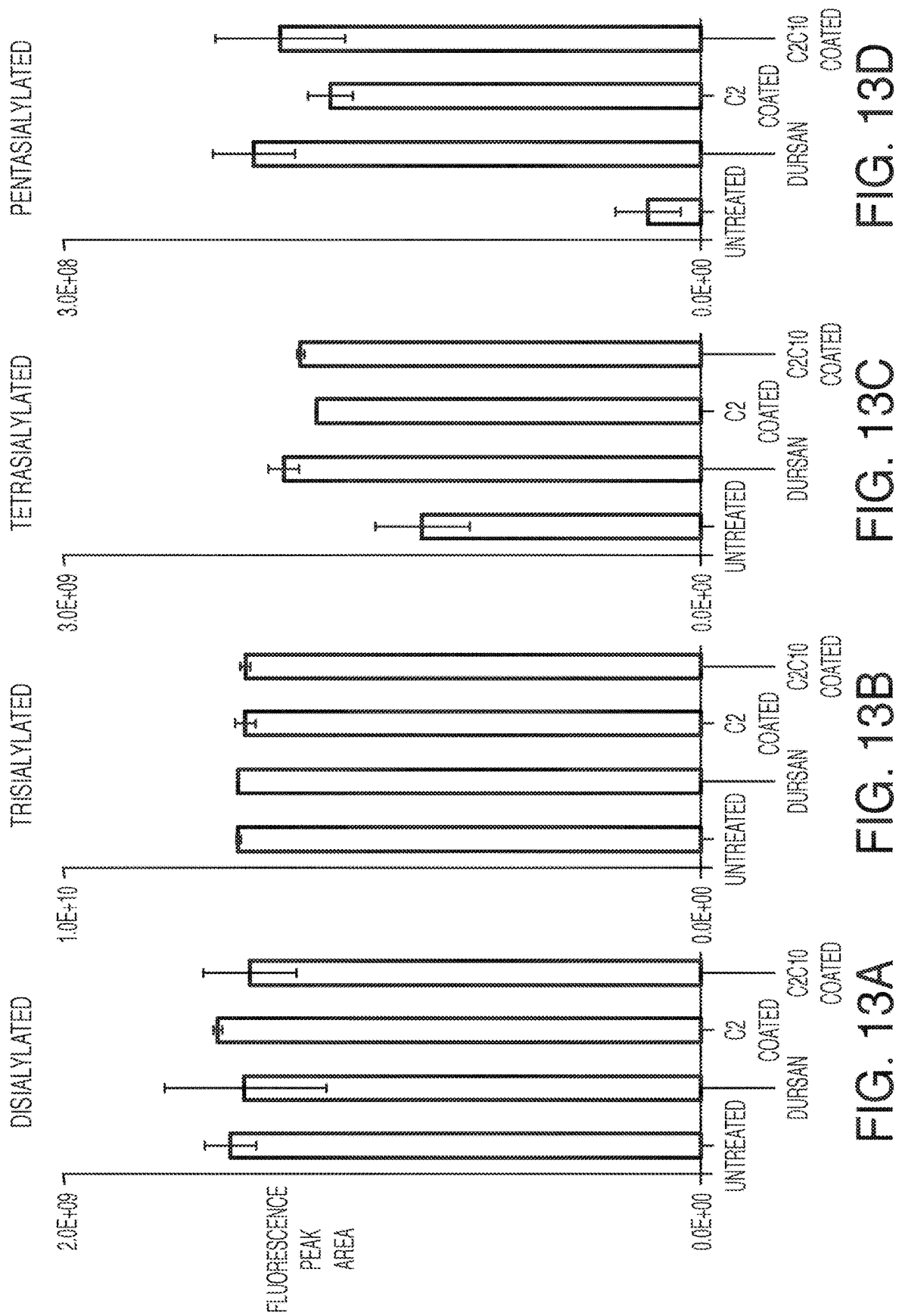


FIG. 12B



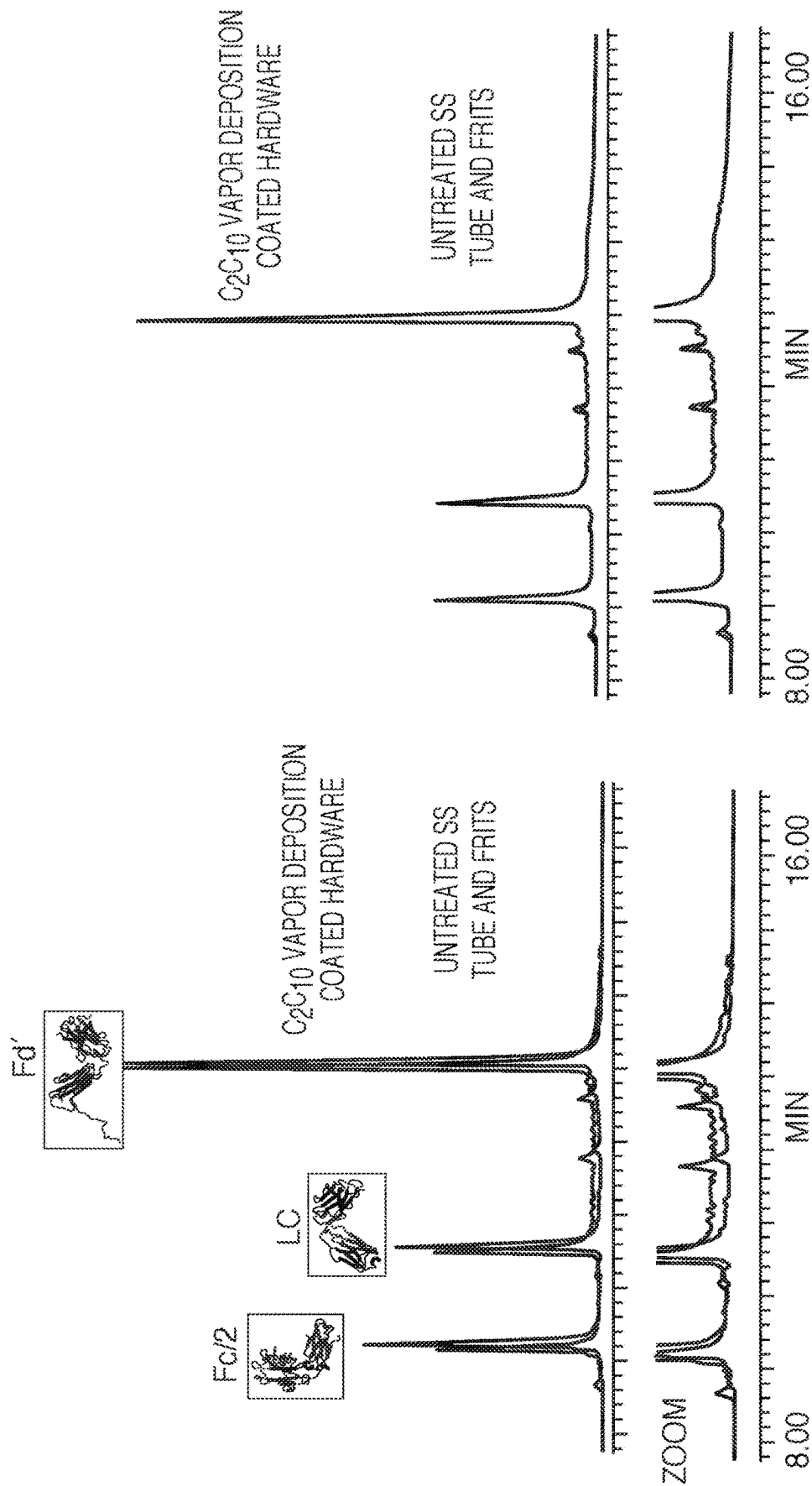


FIG. 14B

FIG. 14A

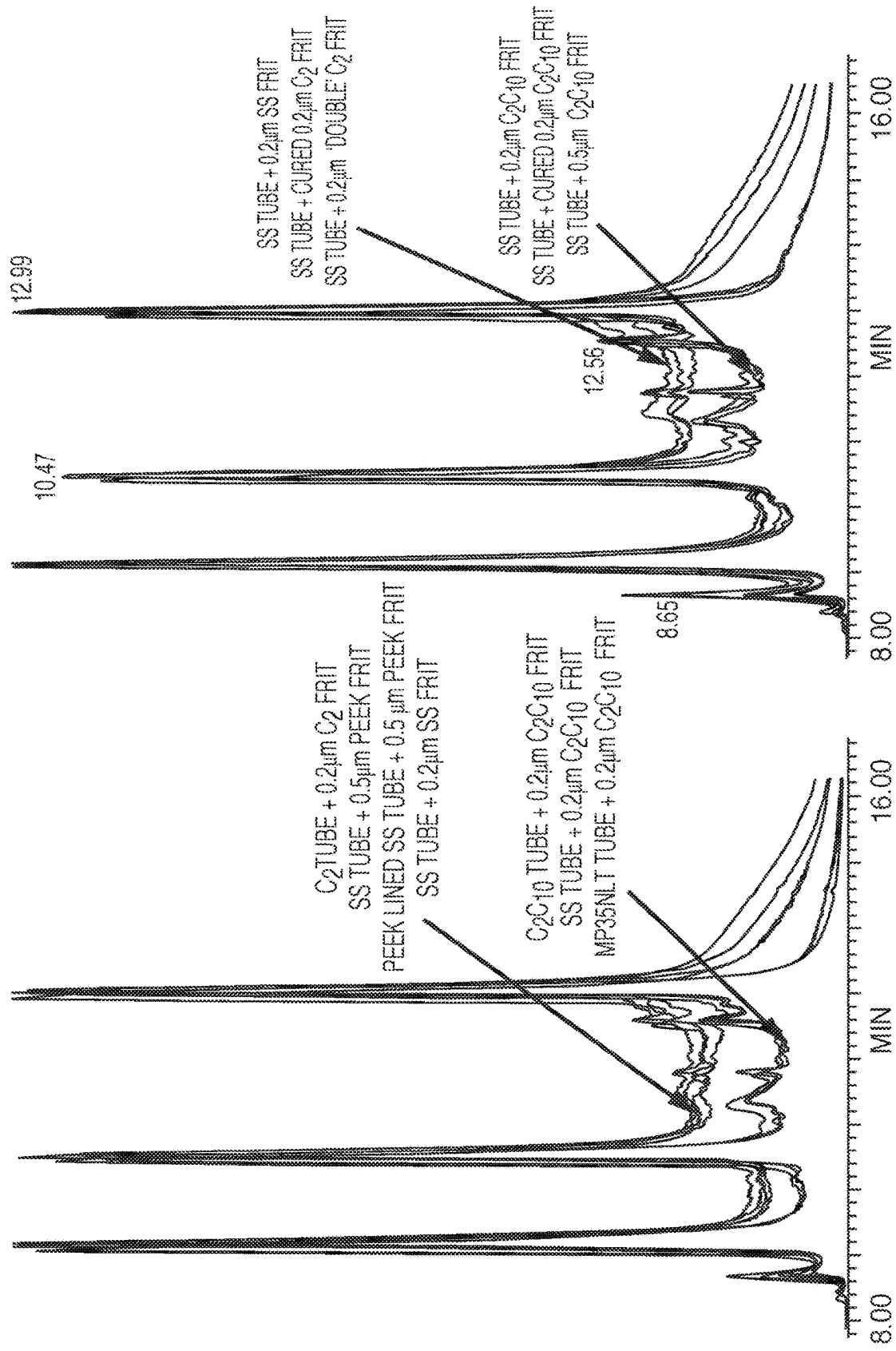


FIG. 15B

FIG. 15A

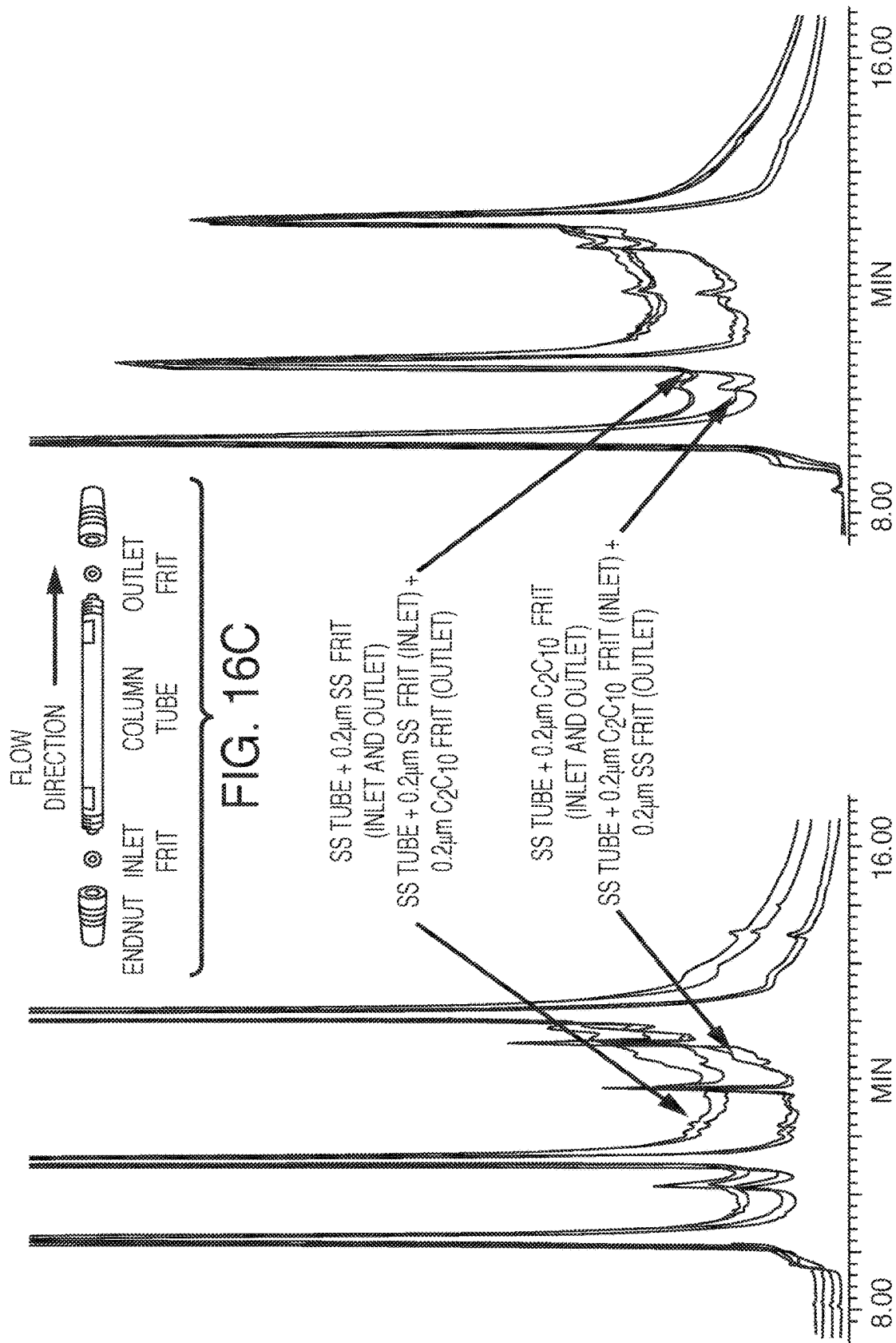
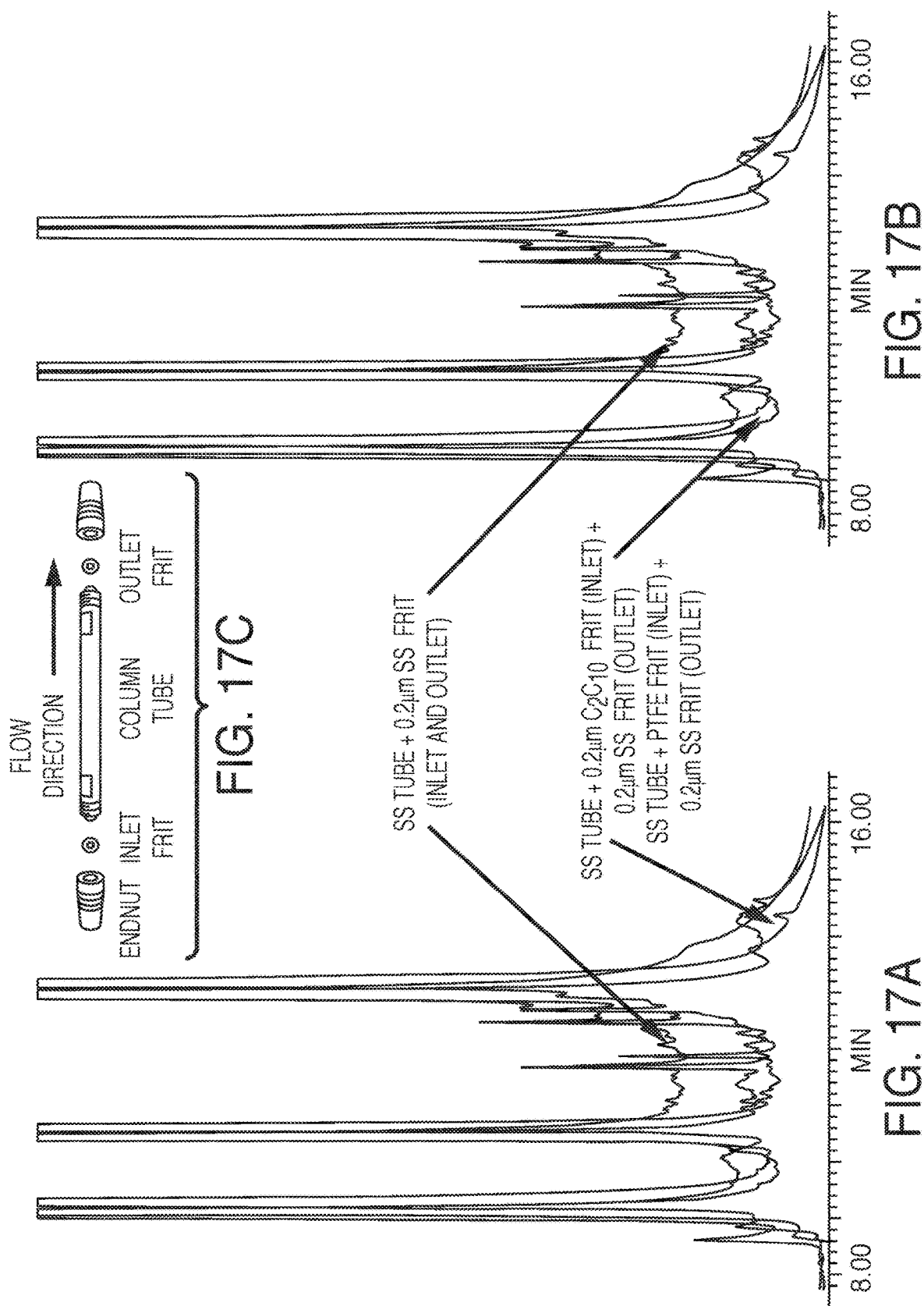


FIG. 16C

FIG. 16B

FIG. 16A



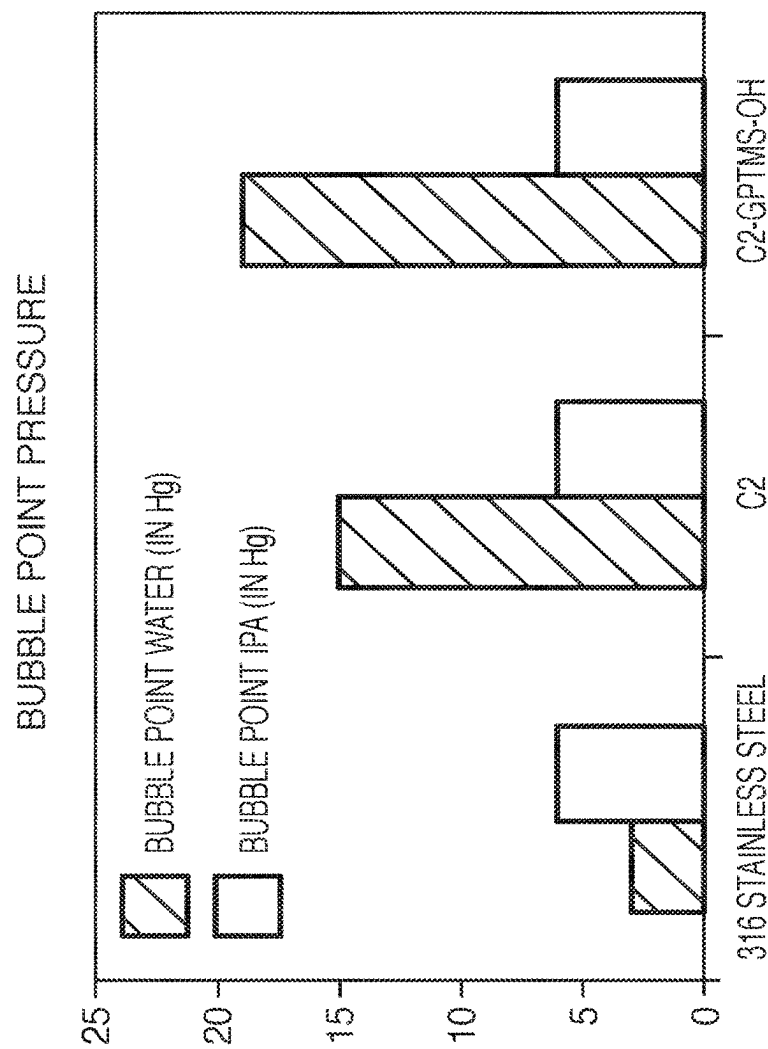


FIG. 18

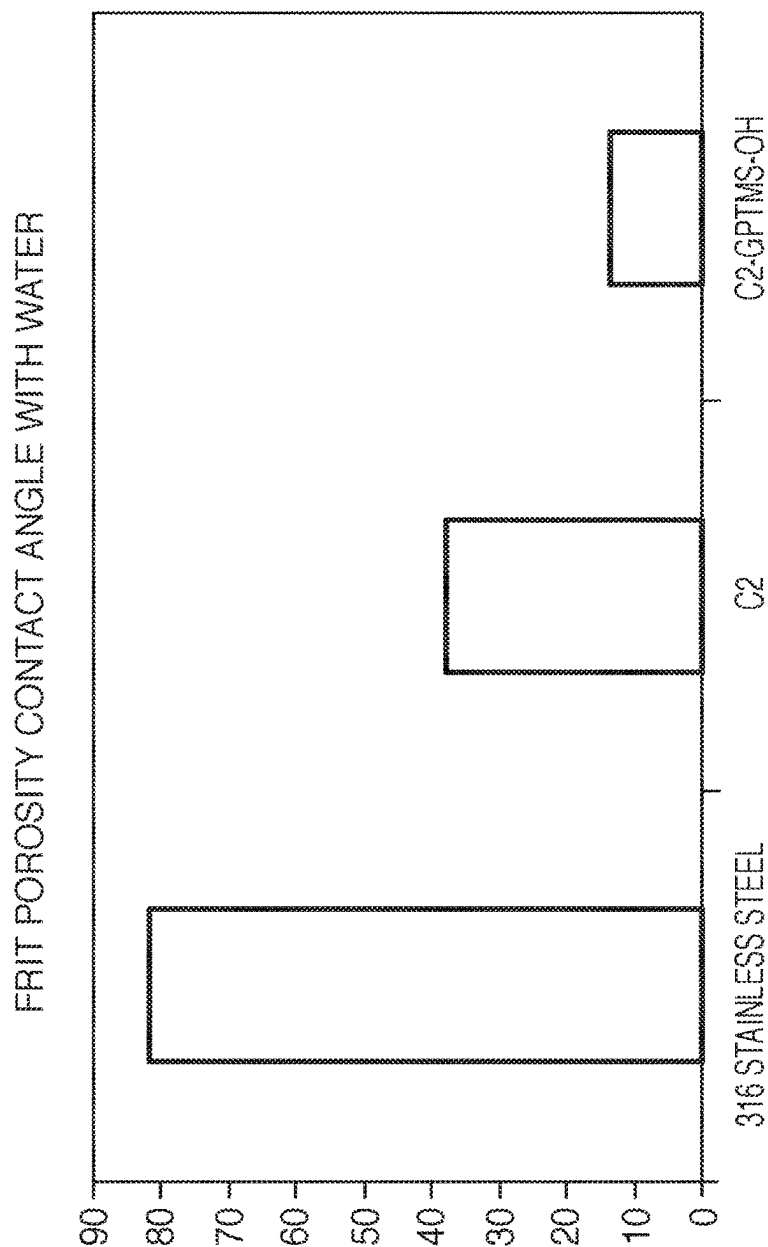


FIG. 19

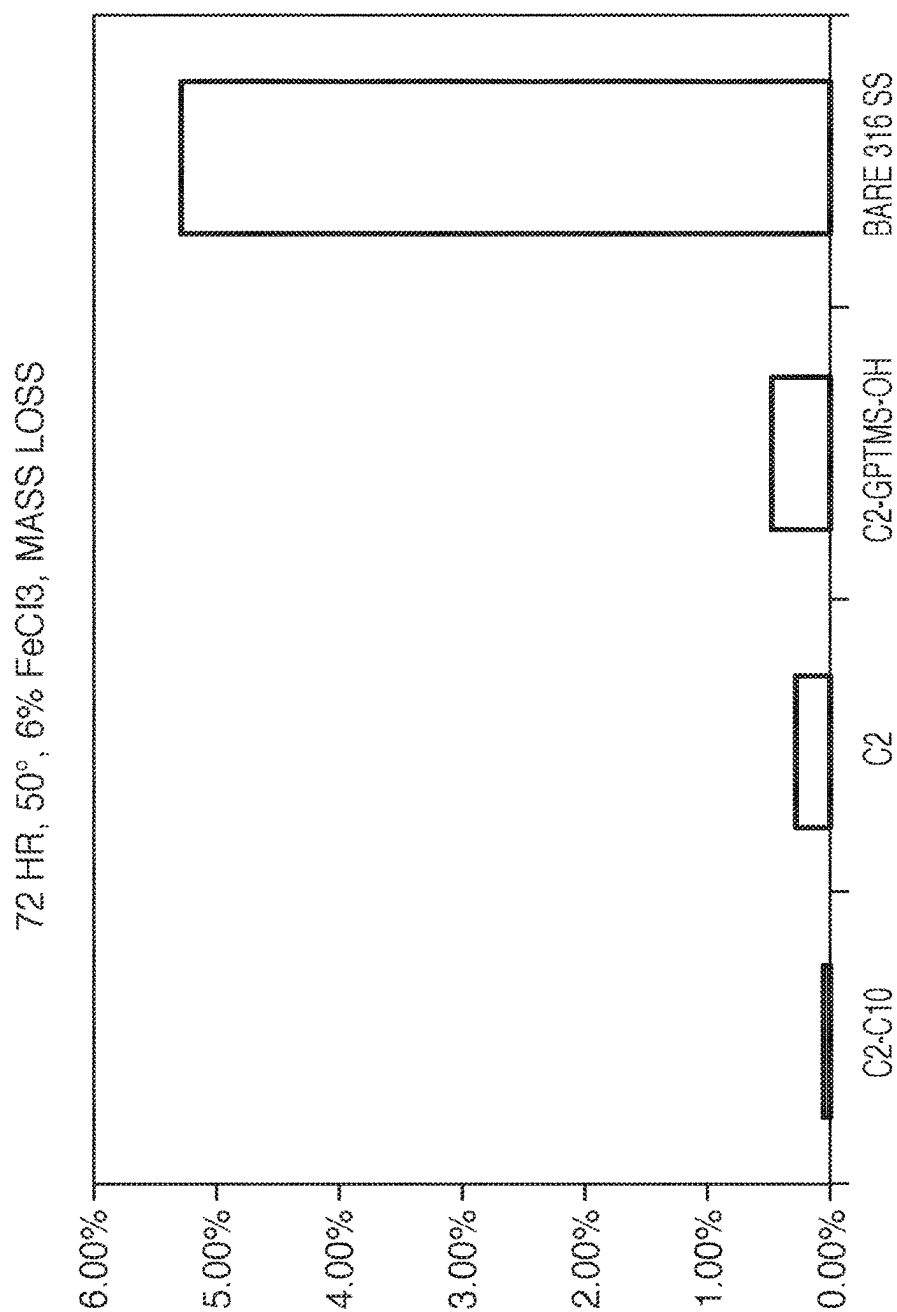


FIG. 20

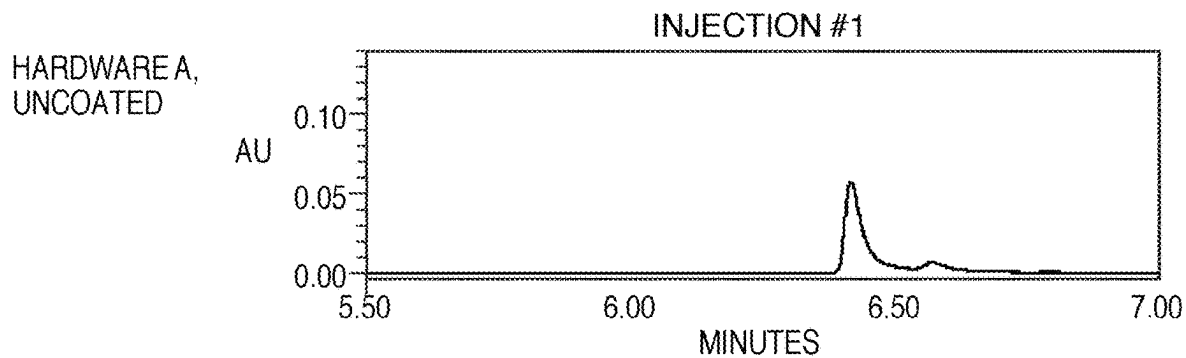


FIG. 21A

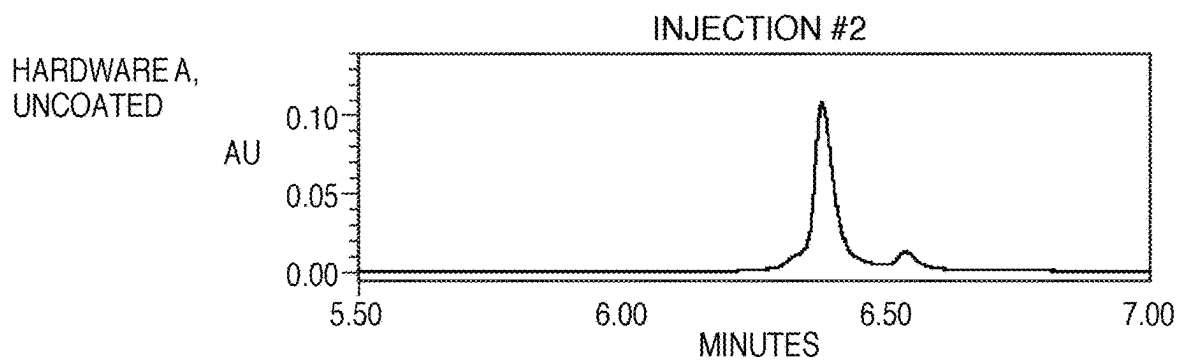


FIG. 21B

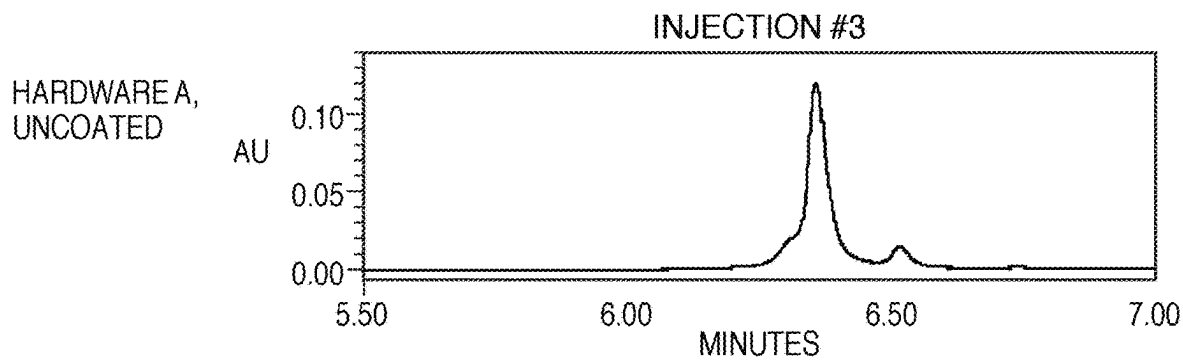


FIG. 21C

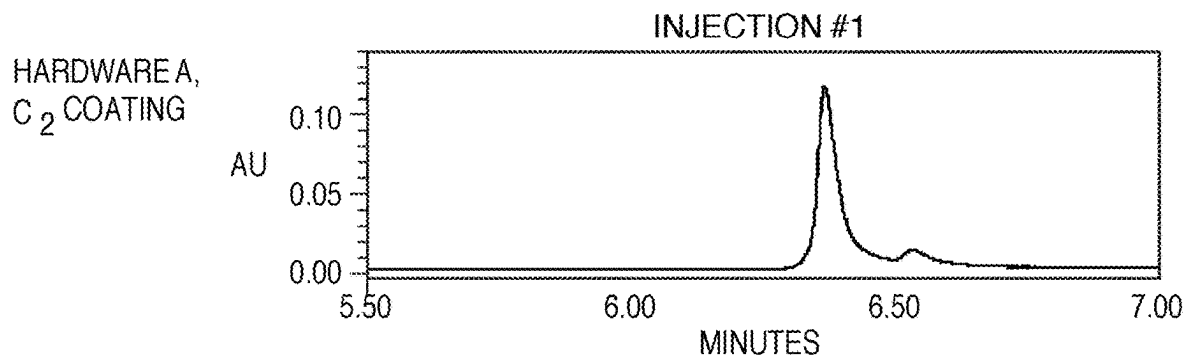


FIG. 21D

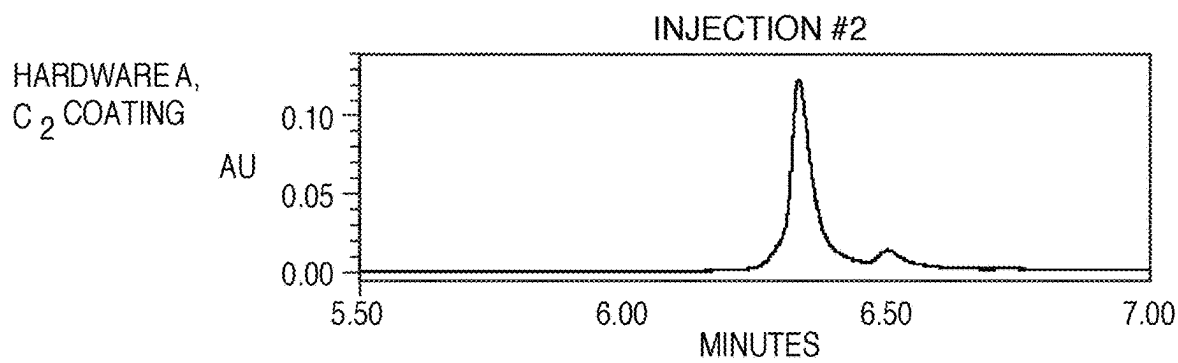


FIG. 21E

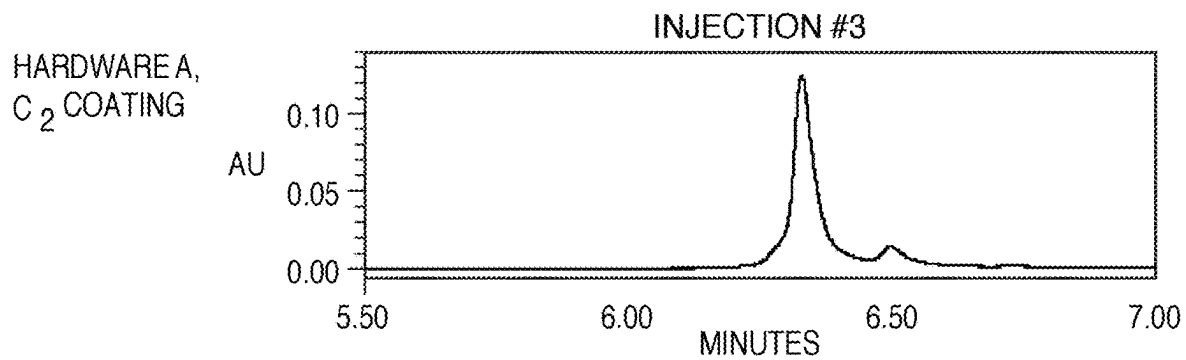


FIG. 21F

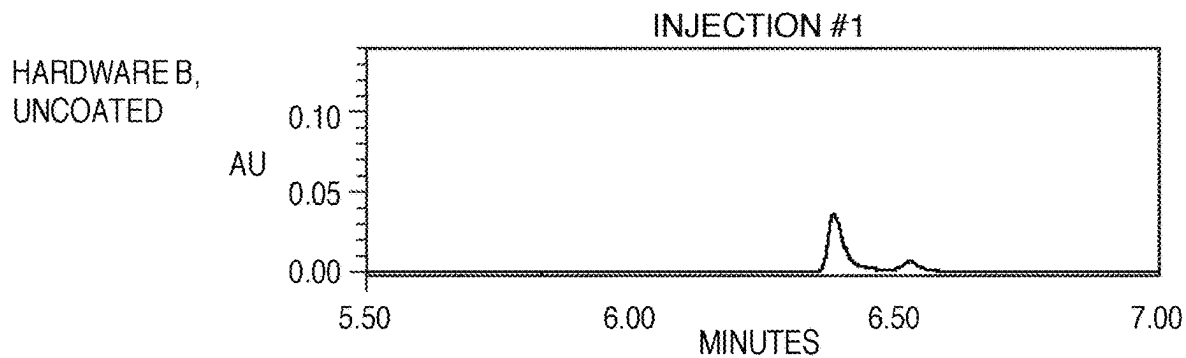


FIG. 21G

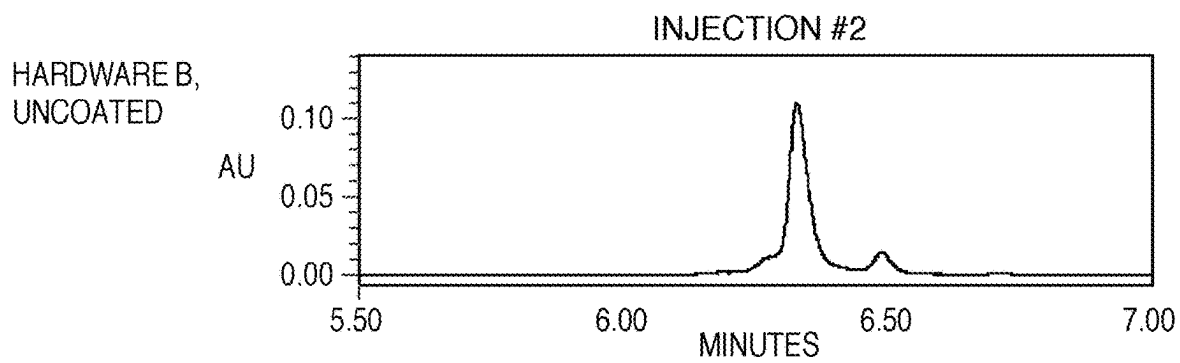


FIG. 21H

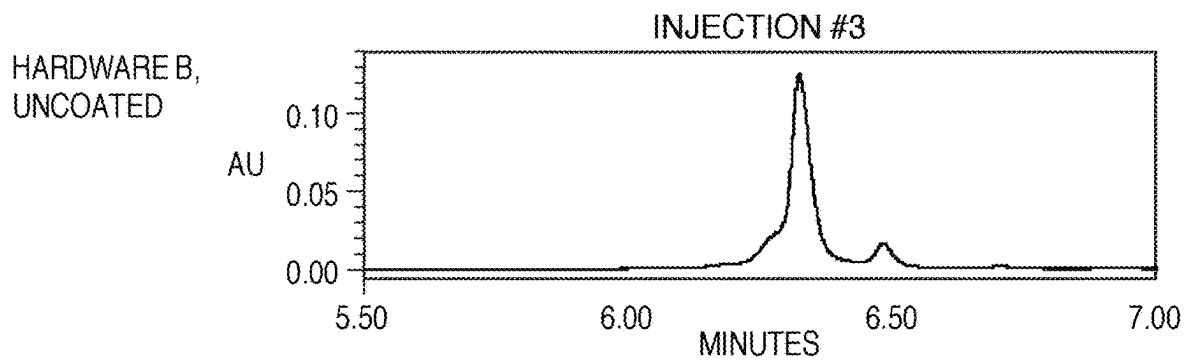


FIG. 21I

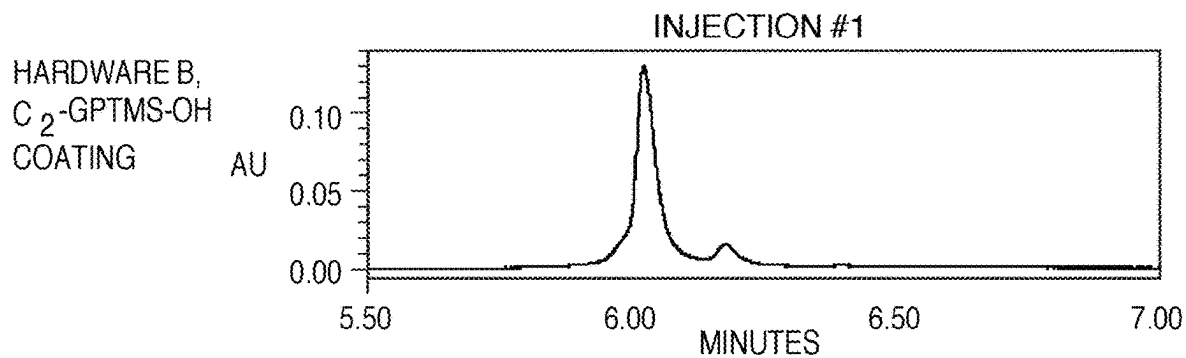


FIG. 21J

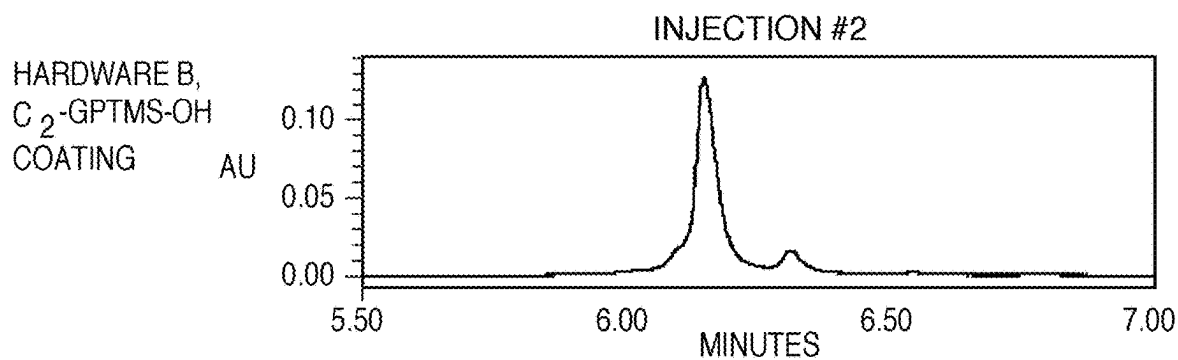


FIG. 21K

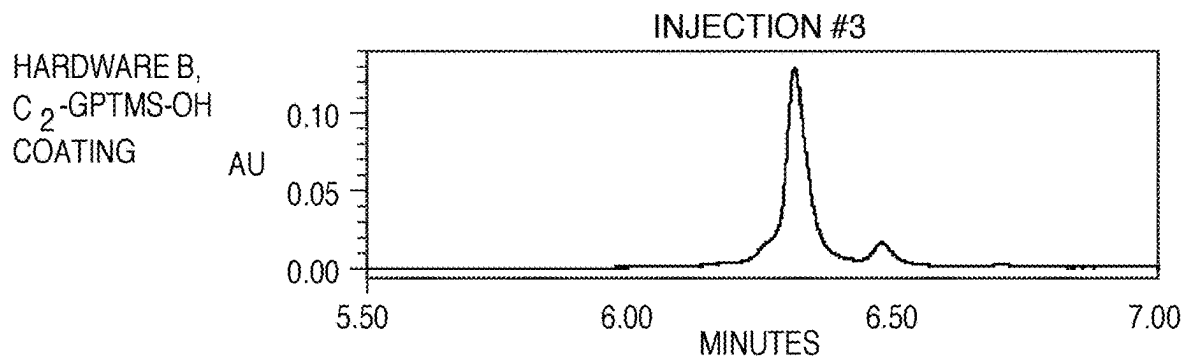
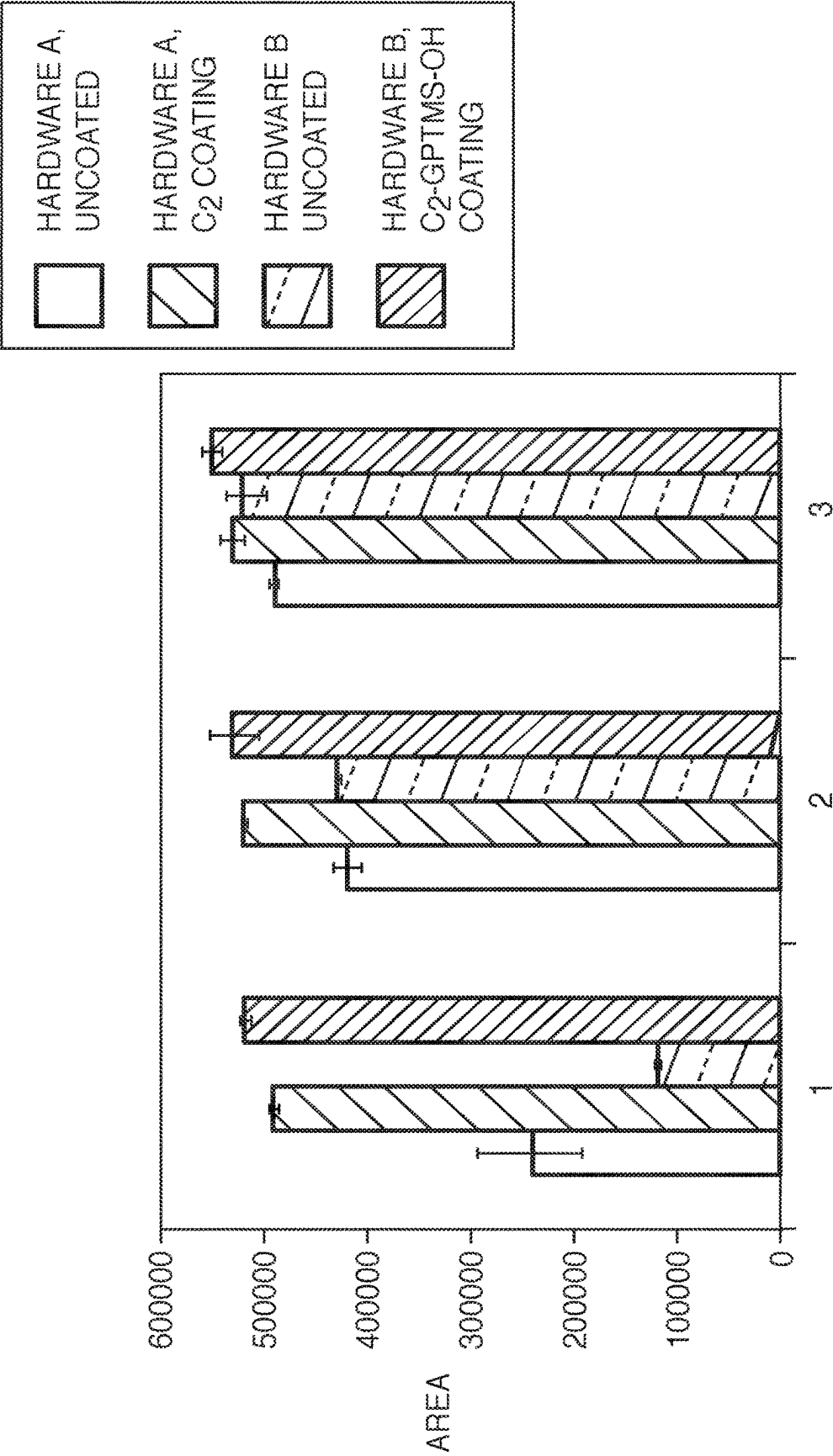
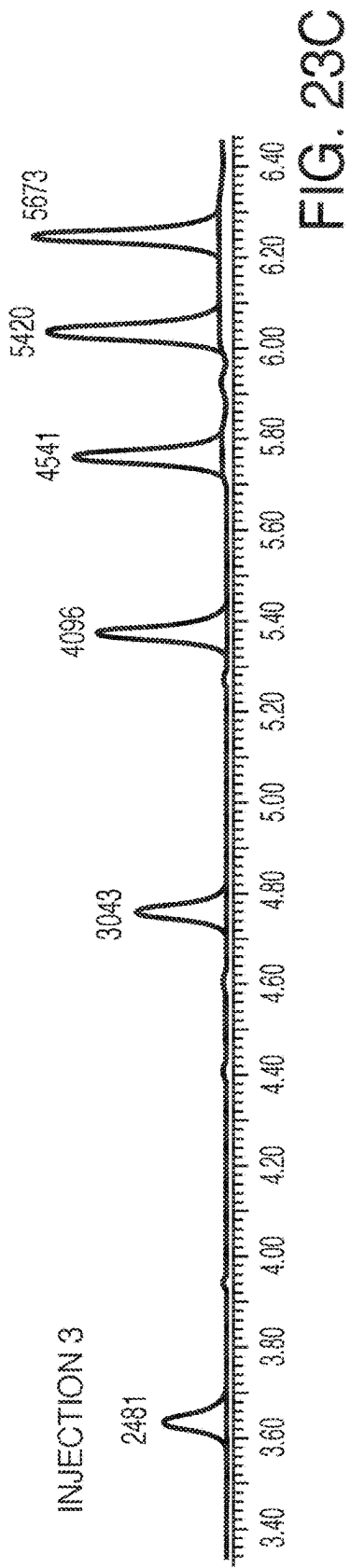
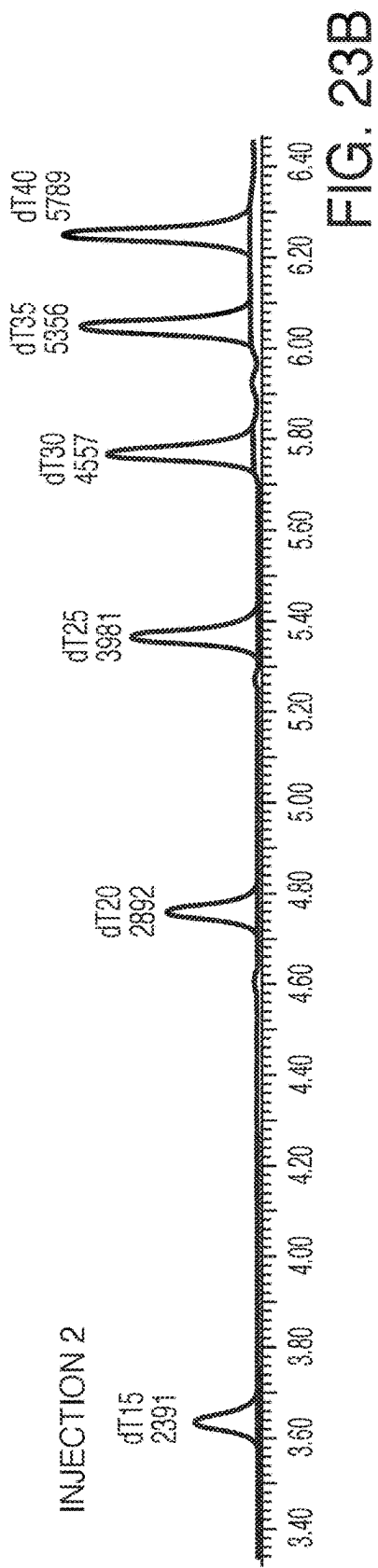
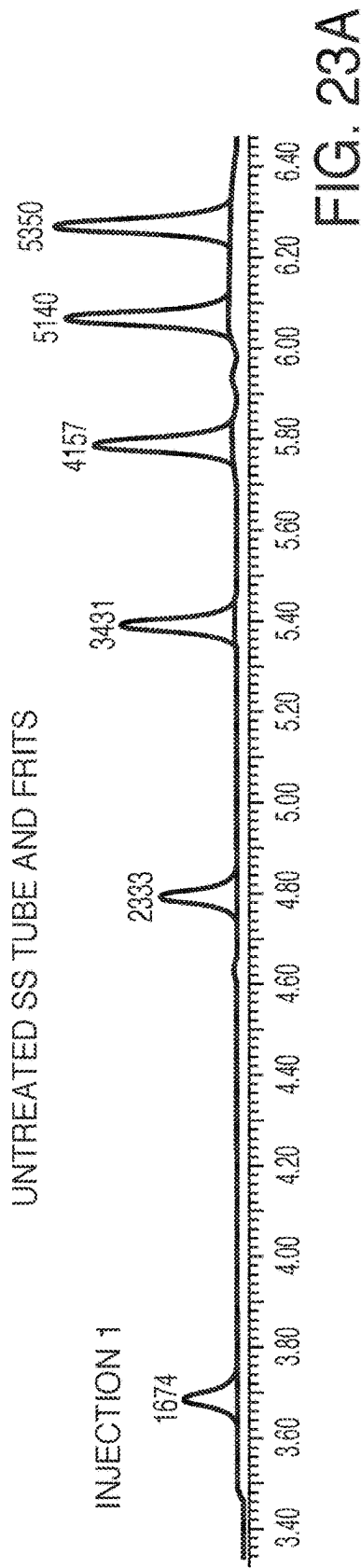


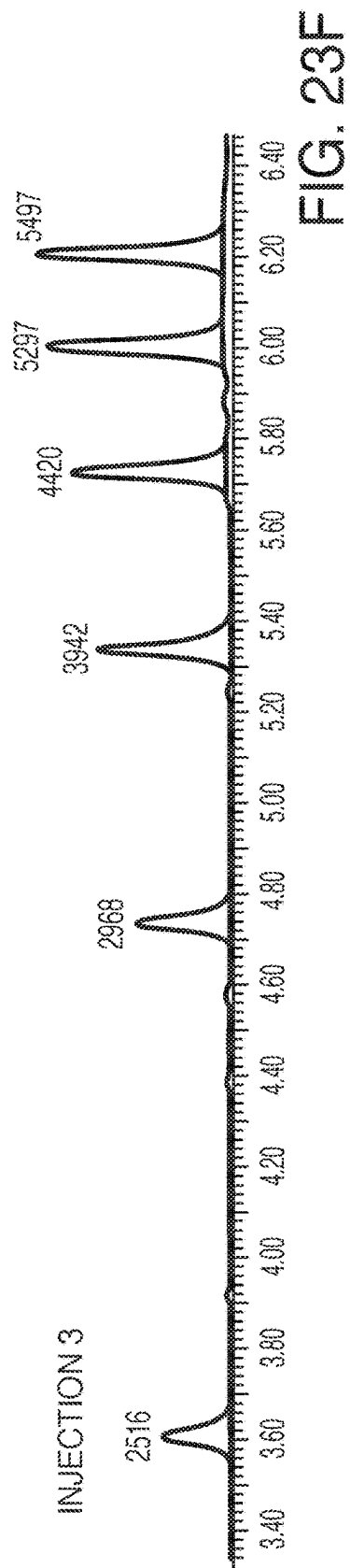
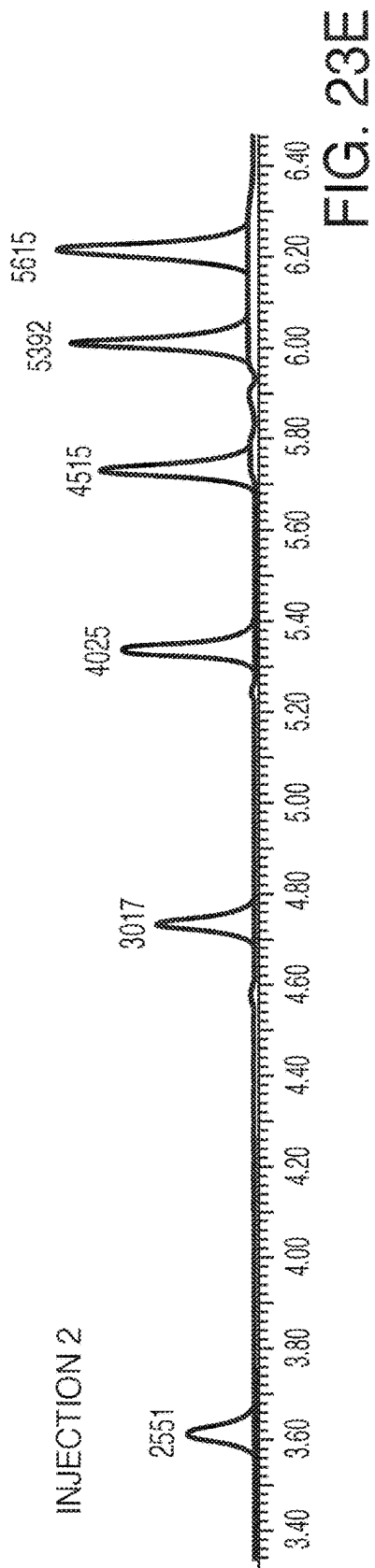
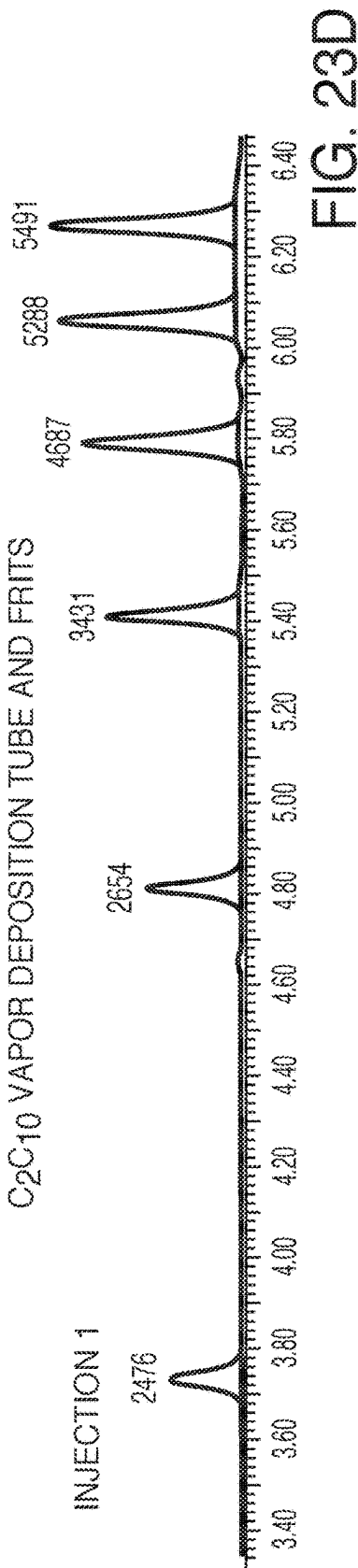
FIG. 21L



INJECTION #
FIG. 22



C₂C₁₀ VAPOR DEPOSITION TUBE AND FRITS



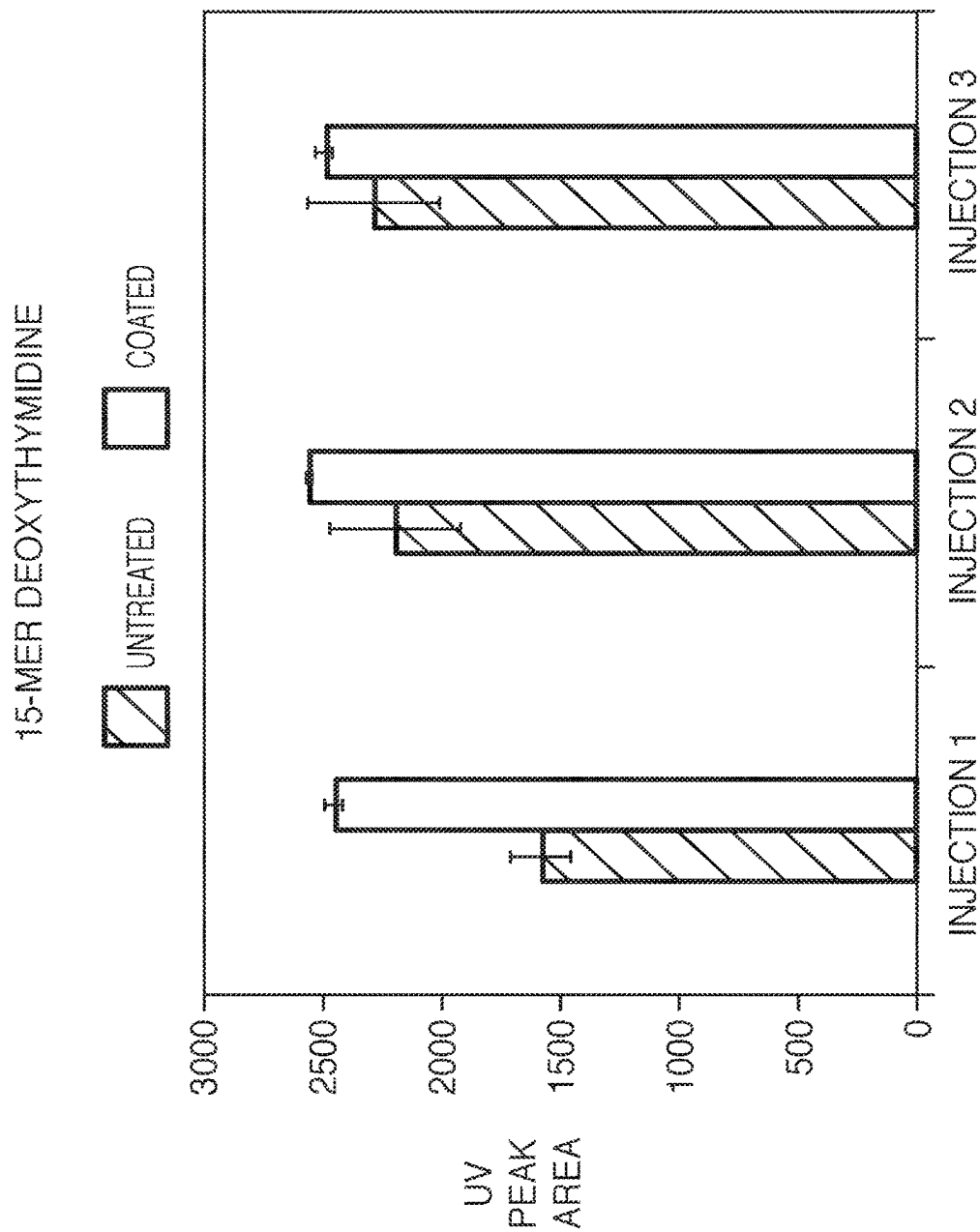
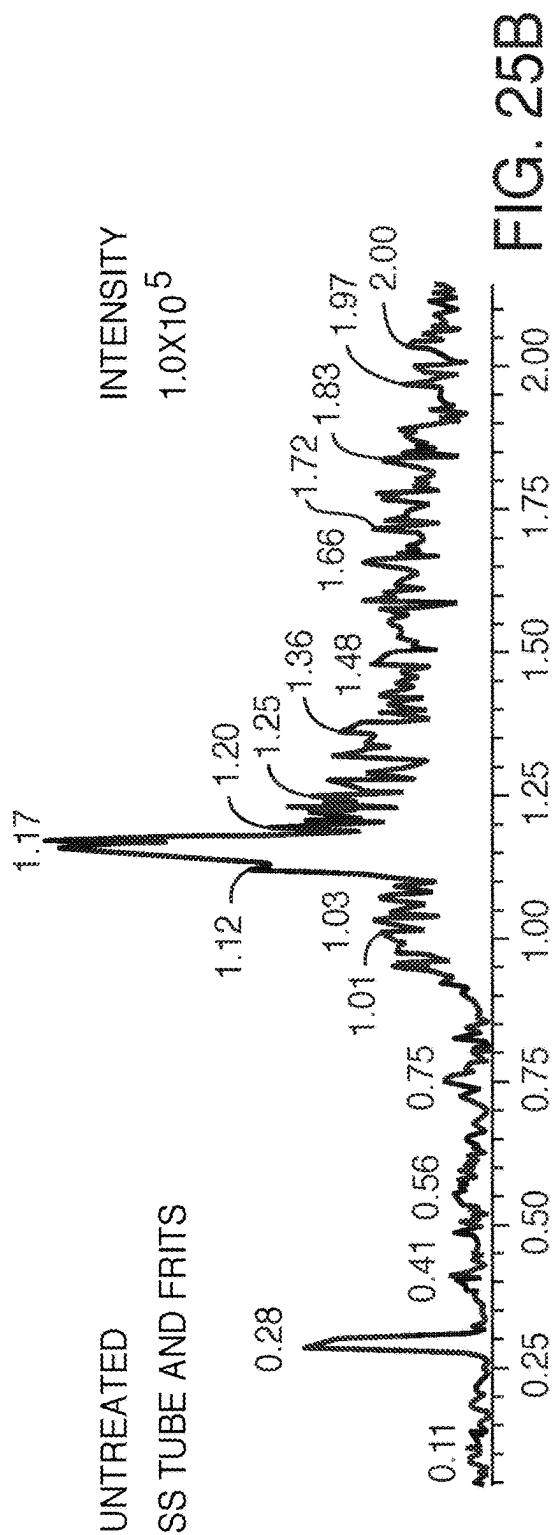
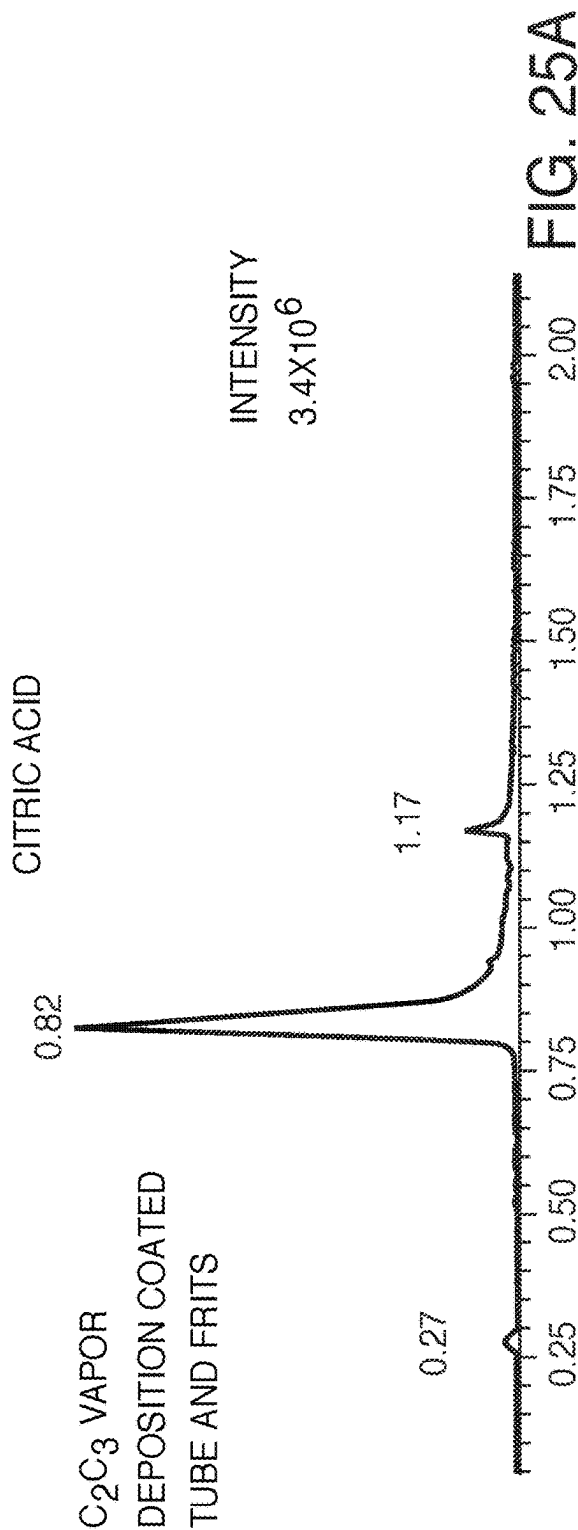
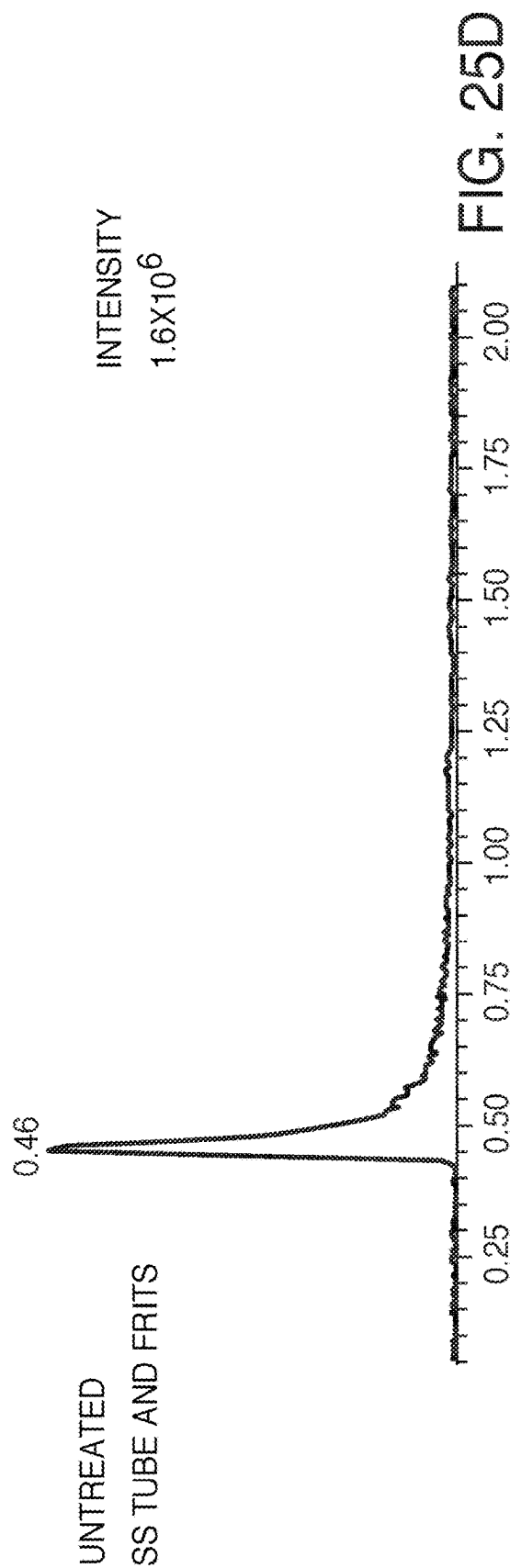
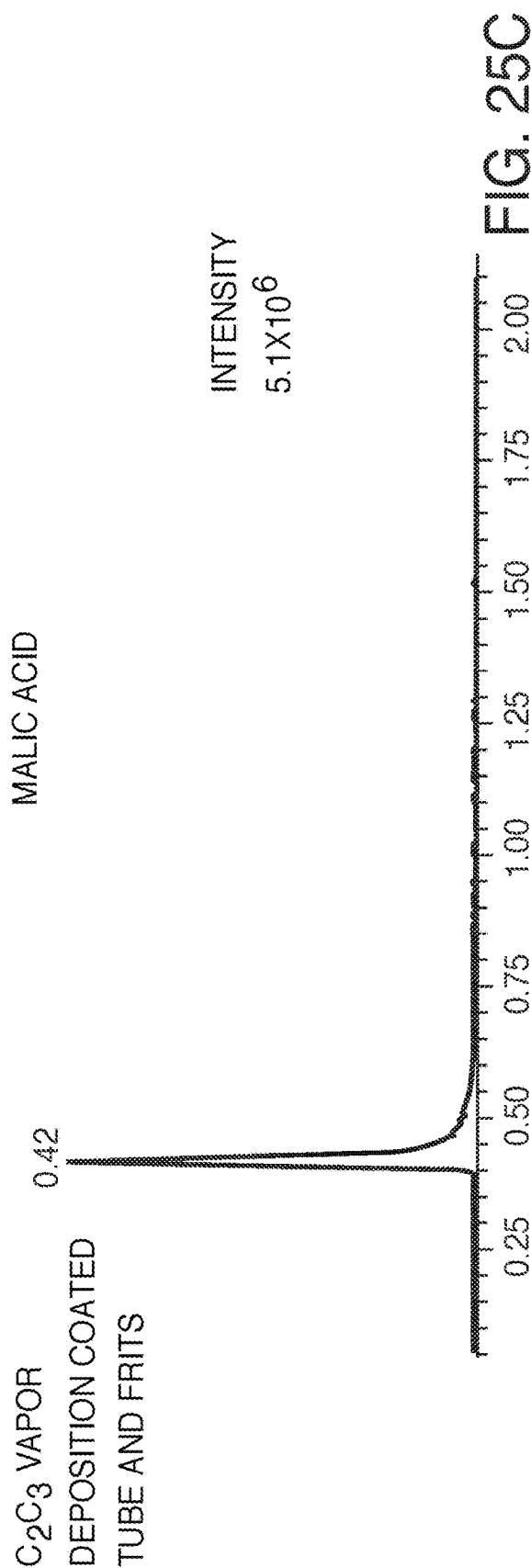
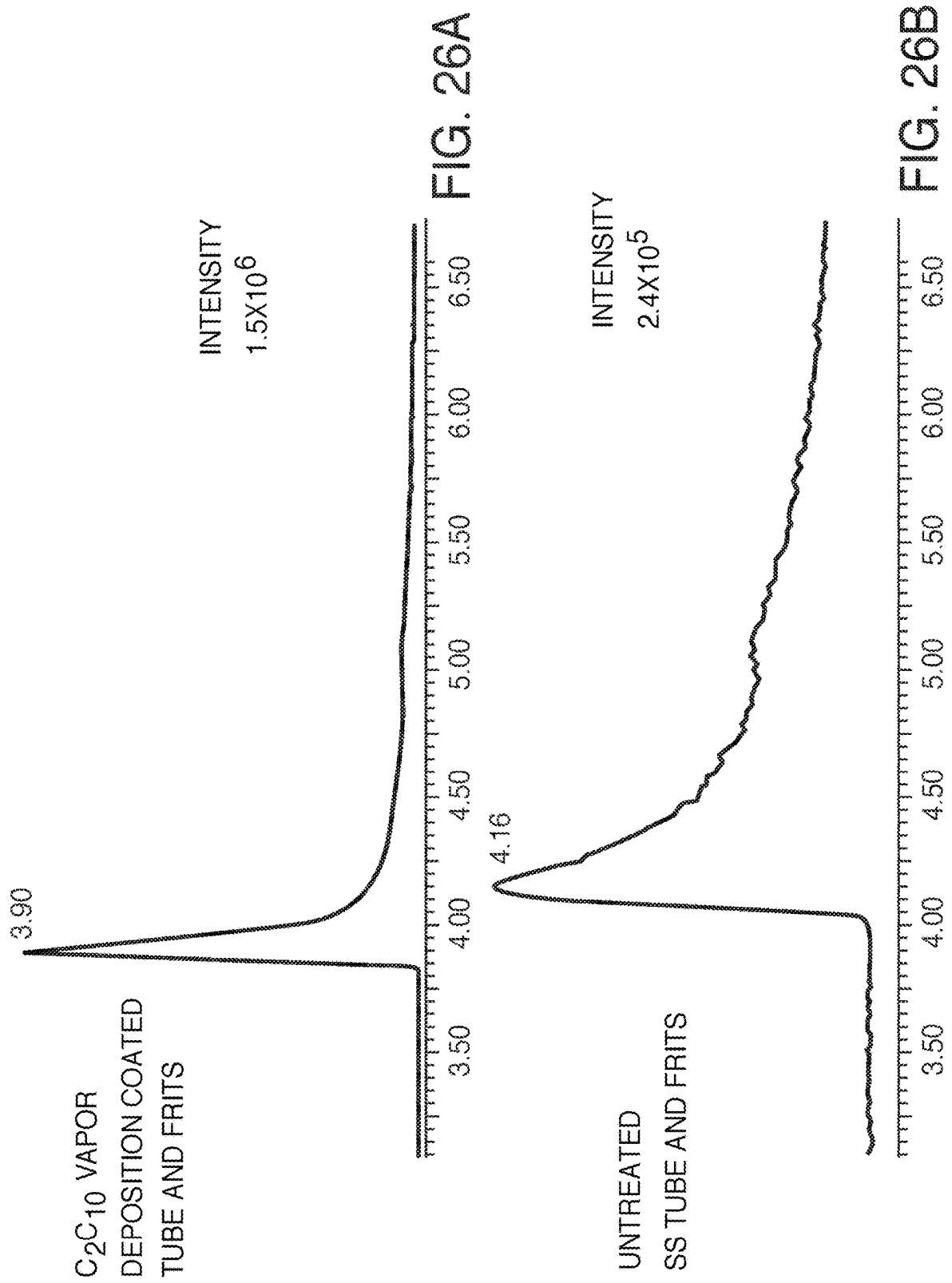


FIG. 24







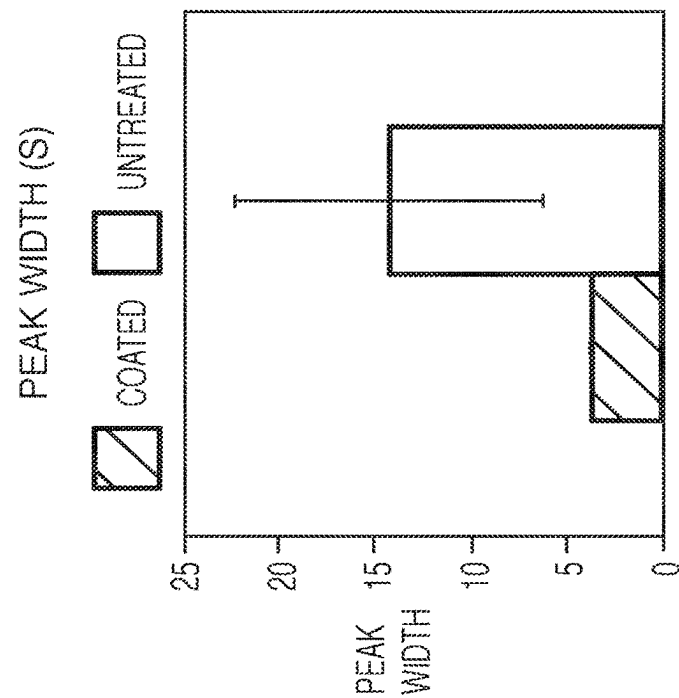


FIG. 27B



FIG. 27A

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USE OF VAPOR DEPOSITION COATED FLOW PATHS FOR IMPROVED CHROMATOGRAPHY OF METAL INTERACTING ANALYTES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/133,089 filed Sep. 17, 2018, which claims priority to and the benefit of U.S. Provisional Application No. 62/559,895 filed Sep. 18, 2017 and entitled "Use of Vapor Deposition Coated Flow Paths for Improved Chromatography of Biomolecules," the contents of which are incorporated herein by reference in its entirety.

FIELD OF THE TECHNOLOGY

This technology relates to the use of vapor deposition coated flow paths for improved chromatography. More specifically, this technology relates to chromatographic devices for separating analytes in a sample having coated flow paths, methods of separating analytes in a sample (for example, glycans, peptides, pesticides, and citric acid cycle metabolites) using a fluidic system that includes coated flow paths, and methods of tailoring a fluidic flow path for separation of a sample.

BACKGROUND OF THE TECHNOLOGY

Analytes that interact with metal have often proven to be very challenging to separate. The desire to have high pressure capable chromatographic systems with minimal dispersion has required that flow paths decrease in diameter and be able to withstand increasingly high pressures at increasingly fast flow rates. As a result, the material of choice for chromatographic flow paths is often metallic in nature. This is despite the fact that characteristics of certain analytes, for example, biomolecules, proteins, glycans, peptides, oligonucleotides, pesticides, bisphosphonic acids, anionic metabolites, and zwitterions like amino acids and neurotransmitters, are known to have unfavorable interactions, so called chromatographic secondary interactions, with metallic surfaces.

The proposed mechanism for metal specific binding interactions requires an understanding of the Lewis theory of acid-base chemistry. Pure metals and metal alloys (along with their corresponding oxide layers) have terminal metal atoms that have characteristics of a Lewis acid. More simply, these metal atoms show a propensity to accept donor electrons. This propensity is even more pronounced with any surface metal ions bearing a positive charge. Analytes with sufficient Lewis base characteristics (any substance that can donate non-bonding electrons) can potentially adsorb to these sites and thus form problematic non-covalent complexes. It is these substances that are defined as metal-interacting analytes.

For example, analytes having phosphate groups are excellent polydentate ligands capable of high affinity metal chelation. This interaction causes phosphorylated species to bind to the flow path metals thus reducing the detected amounts of such species, a particularly troublesome effect given that phosphorylated species are frequently the most important analytes of an assay.

Other characteristics of analytes can likewise pose problems. For example, carboxylate groups also have the ability to chelate to metals, albeit with lower affinities than phos-

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phate groups. Yet, carboxylate functional groups are ubiquitous in, for example, biomolecules, giving the opportunity for cumulative polydentate-based adsorptive losses. These complications can exist not only on peptides and proteins, but also glycans. For example, N-glycan species can at times contain one or more phosphate groups as well as one or more carboxylate containing sialic acid residues. Additionally, smaller biomolecules such as nucleotides and saccharides, like sugar phosphates, can exhibit similar behavior to the previously mentioned N-glycan molecules. Moreover, chromatographic secondary interactions can be especially problematic with biomolecules, particularly larger structures, because they have a capacity (via their size and structural order) to form microenvironments that can adversely interact with separation components and flow path surfaces. In this case, a biomolecule or analyte having larger structures, can present structural regions with chemical properties that amplify a secondary interaction to the material of a flow path. This, combined with the cumulative metal chelation effects curtails the overall effective separation of biomolecules, pesticides, bisphosphonic acids, anionic metabolites, and zwitterions like amino acids and neurotransmitters.

An alternative to using metal flow paths is to use flow paths constructed from polymeric materials, such as polyether ether ketone (PEEK). PEEK tubing, like most polymeric materials, is formed by means of an extrusion process. With polymeric resin, this manufacturing process can lead to highly variable internal diameters. Accordingly, PEEK column hardware yields unfavorable differences in the retention times as can be observed from switching between one column and the next. Often, this variation can be a factor of three higher than a metal constructed column. In addition, the techniques for fabricating polymer based frits are not yet sufficiently optimized to afford suitably rugged components for commercial HPLC columns. For example, commercially available PEEK frits tend to exhibit unacceptably low permeability.

Ongoing efforts to reduce chelation and secondary chromatographic interactions of analytes with metal chromatographic surfaces in an effort to facilitate chromatographic separation having higher resolutions are therefore needed.

SUMMARY OF THE TECHNOLOGY

One advantage of the alkylsilyl coatings of the present technology is that metal chromatographic flow paths can be used while minimizing the interactions between analytes and the metal flow paths. Coating the flow path of instrumentation and chromatographic devices with certain alkylsilyl compositions improves multiple aspects of liquid chromatography separations where the analyte of interest is a metal-interacting analyte. The use of alkylsilyl coatings on metal flow paths allow the use of metal chromatographic flow paths, which are able to withstand high pressures at fast flow rates, while minimizing the secondary chromatographic interactions between the analyte and the metal. Therefore, high pressure components can be manufactured out of stainless steel or other metallic or high pressure material. These components made of high pressure material can then be tailored in that the internal flow paths can be modified with a coating to address the hydrophobicity of the flow path and reduce secondary chromatographic interactions.

Provided herein, therefore, are methods for isolating analytes comprising the use of vapor depositing one or more alkylsilyl derivatives to at least one component of a fluidic system to form a bioinert or low-bind coating, and eluting the analyte through the system. Unlike ambient, liquid phase

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silanization, coatings which are vapor deposited tend to produce, more resilient modifications of substrates with precisely controlled thicknesses. Also, because vapor deposition is a non-line-of-sight process, this leads to a more uniform coating over substrate contours and complex surfaces. This advantage allows for coatings to be applied to flow paths with narrow internal diameters and curved surfaces, therefore addressing the need for increasingly high pressures at increasingly fast flow rates.

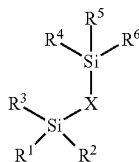
Also provided herein are methods of tailoring a fluidic flow path for separation of a sample comprising an analyte that includes infiltrating a vaporized source of one or more alkylsilyl derivatives through the fluidic flow path to form a bioinert (or low-bind) coating and controlling temperature and pressure to deposit a first coating on wetted surfaces of the flow path.

Also provided are methods of tailoring a fluidic flow path for separation of a sample including an analyte comprising assessing the polarity of the analyte, selecting an appropriate alkylsilyl derivative, and adjusting the hydrophobicity of wetted surfaces of the flow path by vapor depositing the appropriate alkylsilyl derivative to form a bioinert, low-bind coating.

Further provided herein are methods of improving baseline returns in a chromatographic system comprising introducing a sample including an analyte into a fluidic system comprising at least one vapor deposited alkylsilyl derivative to form a bioinert, low-bind coating, and eluting the sample through the system.

The disclosed methods can be applied to stainless steel or other metallic flow path components and provides a manufacturing advantage over alternative non-metallic or non-metallic lined components.

In one aspect, the technology includes a chromatographic device for separating analytes in a sample. The device includes a sample injector having a sample injection needle for injecting the sample into the mobile phase, a sample reservoir container in fluid communication with the sample injector, a chromatography column downstream of the sample injector, wherein the chromatography column having fluid connectors, and fluid conduits connecting the sample injector and the chromatography column. Interior surfaces of the fluid conduits, sample injector, sample reservoir container, and chromatography column form a fluidic flow path having wetted surfaces. At least a portion of the wetted surfaces of the fluidic flow path are coated with an alkylsilyl coating, wherein the alkylsilyl coating is inert to at least one of the analytes in the sample. The alkylsilyl coating has the Formula I:



R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 are each independently selected from $(\text{C}_1\text{-C}_6)\text{alkoxy}$, $-\text{NH}(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{N}((\text{C}_1\text{-C}_6)\text{alkyl})_2$, OH , OR^4 , and halo. R^4 represents a point of attachment to the interior surfaces of the fluidic system. At least one of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is OR^4 . X is $(\text{C}_1\text{-C}_{20})\text{alkyl}$, $-\text{O}[(\text{CH}_2)_2\text{O}]_{1-20}-$, $-(\text{C}_1\text{-C}_{10})[\text{NH}(\text{CO})\text{NH}(\text{C}_1\text{-C}_{10})]_{1-20}-$, or $-(\text{C}_1\text{-C}_{10})[\text{alkylphenyl}(\text{C}_1\text{-C}_{10})\text{alkyl}]_{1-20}-$.

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The device can include one or more of the following embodiments in any combination thereof.

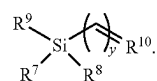
The alkylsilyl coating can have a contact angle of at least 15° . In some embodiments, the alkylsilyl coating has a contact angle less than or equal to 30° or less than or equal to 90° .

The chromatographic device can also include a detector downstream of the chromatography column. The fluidic flow path can also include the detector. In some embodiments the detector is a mass spectrometer and the fluidic flow path includes wetted surfaces of an electrospray needle.

In some embodiments, the fluidic flow path has a length to diameter ratio of at least 20. The alkylsilyl coating can have a thickness of at least 100 Å.

In some embodiments, X is $(\text{C}_2\text{-C}_{10})\text{alkyl}$. In other embodiments, X is ethyl. R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 can each be methoxy or chloro. The alkylsilyl coating of Formula I can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane.

In some embodiments, the device also includes a second alkylsilyl coating in direct contact with the alkylsilyl coating of Formula I. The second alkylsilyl coating has the Formula II:



(II)

R^7 , R^8 , and R^9 are each independently selected from $-\text{NH}(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{N}[(\text{C}_1\text{-C}_6)\text{alkyl}]_2$, $(\text{C}_1\text{-C}_6)\text{alkoxy}$, $(\text{C}_1\text{-C}_6)\text{alkyl}$, $(\text{C}_1\text{-C}_6)\text{alkenyl}$, OH , and halo. R^{10} is selected from $(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{OR}^B$, $-\text{O}[(\text{C}_1\text{-C}_3)\text{alkyl}]_{1-10}\text{O}(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{O}[(\text{C}_1\text{-C}_3)\text{alkyl}]_{1-10}\text{OH}$, and phenyl, wherein said $(\text{C}_1\text{-C}_6)\text{alkyl}$ is optionally substituted with one or more halo and wherein said phenyl is optionally substituted with one or more groups selected from $(\text{C}_1\text{-C}_3)\text{alkyl}$, hydroxyl, fluorine, chlorine, bromine, cyano, $-\text{C}(\text{O})\text{NH}_2$, and carboxyl. R^B is $-(\text{C}_1\text{-C}_3)\text{alkyloxirane}$, $-(\text{C}_1\text{-C}_3)\text{alkyl-3,4-epoxycyclohexyl}$, or $-(\text{C}_1\text{-C}_4)\text{alkylOH}$. The hashed bond to R^{10} represents an optional additional covalent bond between R^{10} and the carbon bridging the silyl group to form an alkene, provided y is not 0. y is an integer from 0 to 20.

In some embodiments, y is an integer from 2 to 9. In some embodiments, y is 9, R^{10} is methyl, and R^7 , R^8 , and R^9 are each ethoxy or chloro.

The alkylsilyl coating of Formula II can be (3-glycidyloxypropyl)trimethoxysilane, n-decyltrichloro silane, trimethylchlorosilane, trimethyldimethyl amino silane, methoxypolyethyleneoxy(1-10) propyl trichlorosilane, or methoxypolyethyleneoxy(1-10) propyl trimethoxysilane. In some embodiments, the alkylsilyl coating of Formula II is (3-glycidyloxypropyl)trimethoxysilane followed by hydrolysis.

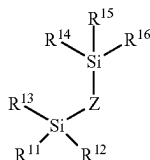
The alkylsilyl coating of Formula I and II can provide a desired contact angle of about 0° to about 105° .

In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is (3-glycidyloxypropyl)trimethoxysilane. The alkylsilyl coating of Formula I can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II can be (3-glycidyloxypropyl)trimethoxysilane followed by hydrolysis. In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is n-decyltrichlorosilane.

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The alkylsilyl coating of Formula I can be bis(trichlorosilyl) ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II can be trimethylchlorosilane or trimethyldimethylaminosilane. In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is methoxy-polyethyleneoxy(3)silane.

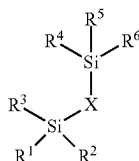
The chromatographic device can also include an alkylsilyl coating having the Formula III in direct contact with the alkylsilyl coating of Formula I,



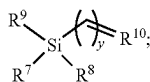
R^{11} , R^{12} , R^{13} , R^{14} , R^{15} , and R^{16} are each independently selected from (C_1-C_6) alkoxy, $-NH(C_1-C_6)alkyl$, $-N((C_1-C_6)alkyl)_2$, OH, and halo. Z is $(C_1-C_{20})alkyl$, $-O[(CH_2)_2O]_{1-20}-$, $-(C_1-C_{10})[NH(CO)NH(C_1-C_{10})]_{1-20}-$, or $-(C_1-C_{10})[alkylphenyl(C_1-C_{10})alkyl]_{1-20}-$.

In some embodiments, the alkylsilyl coating of Formula III is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane. The alkylsilyl coating of Formula I can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula III can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane. In some embodiments, the alkylsilyl coating of I and III has a total thickness of about 400 Å.

In another aspect, the technology relates to a method of separating a sample comprising a glycan, a peptide, or a pesticide. The method includes introducing the sample comprising the glycan, the peptide, or the pesticide to a fluidic system including a flow path disposed in an interior of the fluidic system. The flow path comprising a alkylsilyl coating having the Formula I:



R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 are each independently selected from (C_1-C_6) alkoxy, $-NH(C_1-C_6)alkyl$, $-N((C_1-C_6)alkyl)_2$, OH, OR^4 , and halo. R^4 represents a point of attachment to the interior surfaces of the fluidic system. At least one of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is OR^4 . X is $(C_1-C_{20})alkyl$, $-O[(CH_2)_2O]_{1-20}-$, $-(C_1-C_{10})[NH(CO)NH(C_1-C_{10})]_{1-20}-$, or $-(C_1-C_{10})[alkylphenyl(C_1-C_{10})alkyl]_{1-20}-$. A second alkylsilyl coating is in direct contact with the alkylsilyl coating of Formula I. the second alkylsilyl coating has the Formula II:



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R^7 , R^8 , and R^9 are each independently selected from $-NH(C_1-C_6)alkyl$, $-N[(C_1-C_6)alkyl]_2$, $(C_1-C_6)alkoxy$, $(C_1-C_6)alkyl$, $(C_1-C_6)alkenyl$, OH, and halo. R^{10} is selected from $(C_1-C_6)alkyl$, $-OR^B$, $-[O(C_1-C_3)alkyl]_{1-10}O(C_1-C_6)alkyl$, $-[O(C_1-C_3)alkyl]_{1-10}OH$, and phenyl. $(C_1-C_6)alkyl$ is optionally substituted with one or more halo. The phenyl is optionally substituted with one or more groups selected from $(C_1-C_3)alkyl$, hydroxyl, fluorine, chlorine, bromine, cyano, $-C(O)NH_2$, and carboxy.

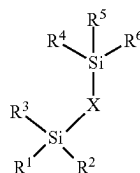
R^B is $-(C_1-C_3)alkyloxirane$, $-(C_1-C_3)alkyl-3,4-epoxycyclohexyl$, or $-(C_1-C_4)alkylOH$. The hashed bond to R^{10} represents an optional additional covalent bond between R^{10} and the carbon bridging the silyl group to form an alkene, provided y is not 0. y is an integer from 0 to 20. The method also includes eluting the sample through the fluidic system, thereby isolating the glycan, the peptide or the pesticide. The method can include one or more of the embodiments described herein in any combination thereof.

In some embodiments, the flow path is defined at least in part by the interior surface of a chromatographic column. The flow path can be further defined at least in part by passageways through a frit of the chromatographic column. In some embodiments, the flow path is further defined at least in part by interior surfaces of tubing. The flow path can be formed of stainless steel.

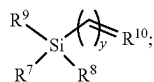
In some embodiments, the glycan is a phosphoglycan. The peptide can be a phosphopeptide. The pesticide can be glyphosate.

In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane. The alkylsilyl coating of Formula II can be n-decyltrichlorosilane. In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is n-decyltrichlorosilane.

In another aspect, the technology includes a method of separating a sample comprising a citric acid cycle metabolite. The method includes introducing the sample comprising the citric acid cycle metabolite to a fluidic system including a flow path disposed in an interior of the fluidic system. The flow path includes a alkylsilyl coating has the Formula I:



R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 are each independently selected from (C_1-C_6) alkoxy, $-NH(C_1-C_6)alkyl$, $-N((C_1-C_6)alkyl)_2$, OH, OR^4 , and halo. R^4 represents a point of attachment to the interior surfaces of the fluidic system. At least one of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is OR^4 . X is $(C_1-C_{20})alkyl$, $-O[(CH_2)_2O]_{1-20}-$, $-(C_1-C_{10})[NH(CO)NH(C_1-C_{10})]_{1-20}-$, or $-(C_1-C_{10})[alkylphenyl(C_1-C_{10})alkyl]_{1-20}-$. A second alkylsilyl coating is in direct contact with the alkylsilyl coating of Formula I. The second alkylsilyl coating has the Formula II:



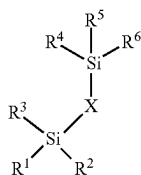
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R^7 , R^8 , and R^9 are each independently selected from $-\text{NH}$ ($\text{C}_1\text{-C}_6$)alkyl, $-\text{N}[(\text{C}_1\text{-C}_6)\text{alkyl}]_2$, ($\text{C}_1\text{-C}_6$)alkoxy, ($\text{C}_1\text{-C}_6$)alkyl, ($\text{C}_1\text{-C}_6$)alkenyl, OH, and halo. R^{10} is selected from ($\text{C}_1\text{-C}_6$)alkyl, $-\text{OR}^B$, $-\text{O}[(\text{C}_1\text{-C}_3)\text{alkyl}]_{1-10}\text{O}(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{O}[(\text{C}_1\text{-C}_3)\text{alkyl}]_{1-10}\text{OH}$, and phenyl. The ($\text{C}_1\text{-C}_6$)alkyl is optionally substituted with one or more halo. The phenyl is optionally substituted with one or more groups selected from ($\text{C}_1\text{-C}_3$)alkyl, hydroxyl, fluorine, chlorine, bromine, cyano, $-\text{C}(\text{O})\text{NH}_2$, and carboxyl. R^B is $-(\text{C}_1\text{-C}_3)\text{alkyloxirane}$, $-(\text{C}_1\text{-C}_3)\text{alkyl-3,4-epoxycyclohexyl}$, or $-(\text{C}_1\text{-C}_4)\text{alkyl-IOH}$. The hashed bond to R^{10} represents an optional additional covalent bond between R^{10} and the carbon bridging the silyl group to form an alkene, provided y is not 0. y is an integer from 0 to 20. The method also includes eluting the sample through the fluidic system, thereby isolating the citric acid cycle metabolite. The method can include one or more of the embodiments described herein in any combination thereof.

In some embodiments, the flow path is defined at least in part by the interior surface of a chromatographic column. The flow path can be further defined at least in part by passageways through a frit of the chromatographic column. The flow path can be further defined at least in part by interior surfaces of tubing. The citric acid cycle metabolite can be citric acid or malic acid.

In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane. The alkylsilyl coating of Formula II can be trimethylchlorosilane or trimethyldimethylaminosilane. In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is trimethylchlorosilane or trimethyldimethylaminosilane.

In another aspect, the technology includes a method of tailoring a fluidic flow path for separation of a sample. The method includes assessing a polarity of an analyte in the sample, selecting a chromatographic media based on the analyte in the sample, and selecting a alkylsilyl coating based on the polarity of the analyte in the sample. The alkylsilyl coating is inert to the analyte in the sample. The alkylsilyl coating has the Formula I:



R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 are each independently selected from ($\text{C}_1\text{-C}_6$)alkoxy, $-\text{NH}(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{N}[(\text{C}_1\text{-C}_6)\text{alkyl}]_2$, OH, OR^A , and halo. R^A represents a point of attachment to wetted surfaces of the fluidic flow path. At least one of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is OR^A . X is ($\text{C}_1\text{-C}_{20}$)alkyl, $-\text{O}[(\text{CH}_2)_2\text{O}]_{1-20}-$, $-(\text{C}_1\text{-C}_{10})[\text{NH}(\text{CO})\text{NH}(\text{C}_1\text{-C}_{10})]_{1-20}-$, or $-(\text{C}_1\text{-C}_{10})[\text{alkylphenyl}(\text{C}_1\text{-C}_{10})\text{alkyl}]_{1-20}-$. The method also includes adjusting a hydrophobicity of the wetted surfaces of the fluidic flow path by vapor depositing the alkylsilyl coating onto the wetted surfaces of the fluidic flow path. The method can include one or more of the embodiments described herein in any combination thereof.

In some embodiments, the analyte in the sample is retained with a retentivity within 10% of the retentivity attributable to the chromatographic media. In some embodi-

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ments, the analyte in the sample is retained with a retentivity within 5% of the retentivity attributable to the chromatographic media. In some embodiments, the analyte in the sample is retained with a retentivity within 1% of the retentivity attributable to the chromatographic media.

The alkylsilyl coating can provide a desired contact angle of about 0° or about 95° .

In some embodiments, the wetted surface of the fluidic flow path are defined at least in part by an interior surface of a column or an interior surface of a sample injection needle. The wetted surface of the fluidic flow path can extend from an interior surface of a sample injection needle through the interior surface of a column. In some embodiments, the wetted surface of the fluidic flow path extends from a sample reservoir container disposed upstream of an in fluidic communication with an interior surface of a sample injection needle throughout the fluidic system to a connector or port of a detector.

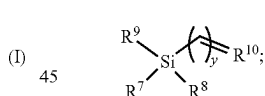
The detector can be a mass spectrometer and the wetted surface of the fluidic flow path includes an interior surface of an electrospray needle.

In some embodiments, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane.

The method can also include annealing the alkylsilyl coating after vapor depositing the alkylsilyl coating onto the wetted surfaces of the fluidic flow path.

In some embodiments, the method also includes modifying the alkylsilyl coating of Formula I with a silanizing reagent to obtain a desired thickness of the alkylsilyl coating. The silanizing reagent can be a non-volatile zwitterion. The non-volatile zwitterion can be sulfobetaine or carboxybetaine. The silanizing reagent can be an acidic or basic silane. In some embodiments, the silanizing agent is methoxy-polyethyleneoxy(6-9)silane.

In some embodiments, the method also includes adjusting the hydrophobicity of the wetted surfaces of the fluidic flow path by vapor depositing a second alkylsilyl coating in direct contact with the vapor deposited alkylsilyl coating of Formula I. The second alkylsilyl coating has the Formula II:



R^7 , R^8 , and R^9 are each independently selected from $-\text{NH}$ ($\text{C}_1\text{-C}_6$)alkyl, $-\text{N}[(\text{C}_1\text{-C}_6)\text{alkyl}]_2$, ($\text{C}_1\text{-C}_6$)alkoxy, ($\text{C}_1\text{-C}_6$)alkyl, ($\text{C}_1\text{-C}_6$)alkenyl, OH, and halo. R^{10} is selected from ($\text{C}_1\text{-C}_6$)alkyl, $-\text{OR}^B$, $-\text{O}[(\text{C}_1\text{-C}_3)\text{alkyl}]_{1-10}\text{O}(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{O}[(\text{C}_1\text{-C}_3)\text{alkyl}]_{1-10}\text{OH}$, and phenyl, wherein said ($\text{C}_1\text{-C}_6$)alkyl is optionally substituted with one or more halo and wherein said phenyl is optionally substituted with one or more groups selected from ($\text{C}_1\text{-C}_3$)alkyl, hydroxyl, fluorine, chlorine, bromine, cyano, $-\text{C}(\text{O})\text{NH}_2$, and carboxyl/ R^B is $-(\text{C}_1\text{-C}_3)\text{alkyloxirane}$, $-(\text{C}_1\text{-C}_3)\text{alkyl-3,4-epoxycyclohexyl}$, or $-(\text{C}_1\text{-C}_4)\text{alkyl-IOH}$. The hashed bond to R^{10} represents an optional additional covalent bond between R^{10} and the carbon bridging the silyl group to form an alkene, provided y is not 0. y is an integer from 0 to 20.

In some embodiments, the alkylsilyl coating of Formula II is (3-glycidyloxypropyl)trimethoxy silane, n-decyltrichlorosilane, trimethylchlorosilane, trimethyldimethylaminosilane, or methoxy-polyethyleneoxy(3)silane. The alkylsilyl coating of Formula II can be (3-glycidyloxypropyl)trimethoxysilane followed by hydrolysis.

In some embodiments, the alkylsilyl coating of Formula I and II provides a desired contact angle of about 0° to about 105°.

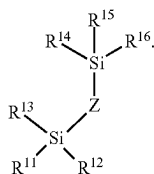
The method can also include modifying the alkylsilyl coating of Formula II with a silanizing reagent to obtain a desired thickness of the alkylsilyl coating. The silanizing reagent can be a non-volatile zwitterion. The non-volatile zwitterion can be sulfobetaine or carboxybetaine. The silanizing reagent can be an acidic or basic silane. In some embodiments, the silanizing agent is methoxy-polyethyleneoxy(6-9)silane.

In some embodiments, the analyte is a biomolecule. The biomolecule is a peptide or peptide fragment, an oligopeptide, a protein, a glycan, a nucleic acid or nucleic acid fragment, a growth factor, a carbohydrate, a fatty acid or a lipid. The analyte can be a citric acid cycle metabolite. In some embodiments, the analyte is a pesticide.

The method can also include assessing a polarity of the chromatographic media and selecting the alkylsilyl coating based on the polarity of the analyte and the chromatographic media.

In some embodiments, the also includes eluting the sample through the fluidic flow path, thereby isolating the analyte.

In another aspect, the technology includes a method of tailoring a fluidic flow path for separation of a sample. The method includes infiltrating a vaporized source of an alkylsilyl of Formula III on wetted surfaces of the fluidic flow path



R^{11} , R^{12} , R^{13} , R^{14} , R^{15} , and R^{16} are each independently selected from $(\text{C}_1\text{-C}_6)\text{alkoxy}$, $\text{—NH}(\text{C}_1\text{-C}_6)\text{alkyl}$, $\text{—N}((\text{C}_1\text{-C}_6)\text{alkyl})_2$, OH , and halo . Z is $(\text{C}_1\text{-C}_{20})\text{alkyl}$, $\text{—O}[(\text{CH}_2)_2\text{O}]_{1-20}\text{—}$, $\text{—}(\text{C}_1\text{-C}_{10})[\text{NH}(\text{CO})\text{NH}(\text{C}_1\text{-C}_{10})]_{1-20}\text{—}$, or $\text{—}(\text{C}_1\text{-C}_{10})[\text{alkylphenyl}(\text{C}_1\text{-C}_{10})\text{alkyl}]_{1-20}\text{—}$. The method also includes controlling temperature and pressure to deposit a first coating of Formula III on the wetted surfaces of the fluidic flow path, the first coating having a thickness of at least 100 Å and a contact angle of at least 15°. The method can include one or more of the embodiments described herein in any combination thereof.

In some embodiments, the method also includes pretreating the wetted surfaces of the fluidic flow path with a plasma prior to depositing the first coating of Formula III.

The method can also include modifying the alkylsilyl coating of Formula III with a silanizing reagent to obtain a desired thickness of the alkylsilyl coating. The silanizing reagent can be a non-volatile zwitterion. The non-volatile zwitterion can be sulfobetaine or carboxybetaine. The silanizing reagent can be an acidic or basic silane. In some embodiments, the silanizing agent is methoxy-polyethyleneoxy(6-9)silane.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic of a chromatographic flow system including a chromatography column and various other com-

ponents, in accordance with an illustrative embodiment of the technology. A fluid is carried through the chromatographic flow system with a fluidic flow path extending from a fluid manager to a detector.

FIG. 2 is a flow chart showing a method of tailoring wetted surfaces of a flow path, in accordance with an illustrative embodiment of the technology.

FIG. 3 is a flow chart showing a method of tailoring a fluidic flow path for separation of a sample including a biomolecule, in accordance with an illustrative embodiment of the technology.

FIG. 4A shows a fluorescence chromatogram obtained using uncoated stainless steel hardware, in accordance with an illustrative embodiment of the technology.

FIG. 4B shows a fluorescence chromatogram obtained using hardware coated with exemplary vapor deposited alkylsilyl, in accordance with an illustrative embodiment of the technology.

FIG. 4C shows a fluorescence chromatogram obtained using hardware coated with exemplary vapor deposited alkylsilyl, in accordance with an illustrative embodiment of the technology.

FIG. 5A is a schematic of exemplified bioinert alkylsilyl coated stainless steel sample flow path components, including column inlet tubing, in accordance with an illustrative embodiment of the technology.

FIG. 5B is a schematic of exemplified bioinert alkylsilyl coated stainless steel sample flow path components, including a sample needle, in accordance with an illustrative embodiment of the technology.

FIG. 6A shows a fluorescence chromatogram obtained using an untreated flow path and untreated tube and frit combination in accordance with an embodiment of the technology.

FIG. 6B shows a fluorescence chromatogram obtained using an untreated flow path and coated tube and frit combination, in accordance with an embodiment of the technology.

FIG. 6C shows a fluorescence chromatogram obtained using a coated flow path and coated tube and frit combination, in accordance with an embodiment of the technology.

FIG. 7A shows a UV chromatogram obtained using an untreated stainless steel tube/frit combination in accordance with an embodiment of the technology.

FIG. 7B shows a UV chromatogram obtained using a C_2 vapor deposition coated tube/frit combination, in accordance with an embodiment of the technology.

FIG. 7C shows a UV chromatogram obtained using a C_{20} vapor deposition coated tube/frit combination, in accordance with an embodiment of the technology.

FIG. 8A is a chromatogram showing the effects of employing vapor deposition coated column hardware for the reversed phase LC analyses of glucose-6-phosphate, in accordance with an illustrative embodiment of the technology.

FIG. 8B is a chromatogram showing the effects of employing untreated column hardware for the reversed phase LC analyses of glucose-6-phosphate, in accordance with an illustrative embodiment of the technology.

FIG. 9A is a chromatogram showing the effects of employing vapor deposition coated column hardware for the reversed phase LC analyses of fructose-6-phosphate, in accordance with an illustrative embodiment of the technology.

FIG. 9B is a chromatogram showing the effects of employing untreated column hardware for the reversed

phase LC analyses of fructose-6-phosphate, in accordance with an illustrative embodiment of the technology.

FIG. 10A is a chromatogram showing the effects of employing vapor deposition coated column hardware for the reversed phase LC analyses of adenosine triphosphate, in accordance with an illustrative embodiment of the technology.

FIG. 10B is a chromatogram showing the effects of employing untreated column hardware for the reversed phase LC analyses of adenosine triphosphate, in accordance with an illustrative embodiment of the technology.

FIG. 11A is a chromatogram showing the effects of employing vapor deposition coated column hardware for the reversed phase LC analyses of adenosine monophosphate, in accordance with an illustrative embodiment of the technology.

FIG. 11B is a chromatogram showing the effects of employing untreated column hardware for the reversed phase LC analyses of adenosine monophosphate, in accordance with an illustrative embodiment of the technology.

FIG. 12A is a fluorescence chromatogram for fetuin N-glycans obtained with untreated stainless steel, in accordance with an illustrative embodiment of the technology.

FIG. 12B is a fluorescence chromatogram for fetuin N-glycans obtained with vapor deposition coated hardware, in accordance with an illustrative embodiment of the technology.

FIG. 13A is a graph showing fluorescence peak areas for disialylated glycans obtained with untreated stainless steel hardware compared to stainless steel hardware coated different types of vapor deposited coatings, in accordance with an illustrative embodiment of the technology.

FIG. 13B is a graph showing fluorescence peak areas for trisialylated glycans obtained with untreated stainless steel hardware compared to stainless steel hardware coated different types of vapor deposited coatings, in accordance with an illustrative embodiment of the technology.

FIG. 13C is a graph showing fluorescence peak areas for tetrasialylated glycans obtained with untreated stainless steel hardware compared to stainless steel hardware coated different types of vapor deposited coatings, in accordance with an illustrative embodiment of the technology.

FIG. 13D is a graph showing fluorescence peak areas for pentasialylated glycans obtained with untreated stainless steel hardware compared to stainless steel hardware coated different types of vapor deposited coatings, in accordance with an illustrative embodiment of the technology.

FIG. 14A is a reversed phase fluorescence chromatogram of reduced, IdeS-digested NIST Reference Material 8671 obtained with column hardware components treated with coatings in accordance with illustrative embodiments of the technology.

FIG. 14B is a reversed phase fluorescence chromatogram of reduced, IdeS-digested NIST Reference Material 8671 obtained with column hardware components treated with coatings in accordance with illustrative embodiments of the technology.

FIG. 15A is a reversed phase total ion chromatogram for columns constructed with stainless steel alternatives, namely polyether ether ketone (PEEK) and a low titanium, nickel cobalt alloy (MP35NLT) with various components coated, in accordance with an illustrative embodiment of the technology.

FIG. 15B is a reversed phase total ion chromatogram for column components constructed with stainless steel, C₂ coatings and C₂C₁₀ coatings, in accordance with an illustrative embodiment of the technology.

FIG. 16A shows fluorescence chromatograms of reduced, IdeS-digested NIST Reference Material 8671 and the effect on baseline return when various components of the system are coated, in accordance with an illustrative embodiment of the technology.

FIG. 16B shows reversed phase total ion chromatograms (TICs) reduced, IdeS-digested NIST Reference Material 8671 and the effect on baseline return when various components of the system are coated, in accordance with an illustrative embodiment of the technology.

FIG. 16C is a schematic of the column tube and frits that were coated and used to obtain the chromatograms of FIGS. 16A and 16B, in accordance with an illustrative embodiment of the technology.

FIG. 17A shows fluorescence chromatograms of reduced, IdeS-digested NIST Reference Material 8671 and the effect on baseline return when various components of the system are coated, in accordance with an illustrative embodiment of the technology.

FIG. 17B shows reversed phase total ion chromatograms (TICs) of reduced, IdeS-digested NIST Reference Material 8671 and the effect on baseline return when various components of the system are coated, in accordance with an illustrative embodiment of the technology.

FIG. 17C is a schematic of the column tube and frits that were coated and used to obtain the chromatograms of FIGS. 17A and 17B, in accordance with an illustrative embodiment of the technology.

FIG. 18 is a bar graph showing bubble point pressure in each of water and IPA for a non-coated stainless steel frit and stainless steel frits coated in accordance with one or more illustrative embodiments of the technology. The bubble point in water is provided as the left bar, and the bubble point in IPA is provided as the right bar for each type of frit.

FIG. 19 is a bar graph showing a comparison of frit porosity contact angle with water for a non-coated stainless steel frit versus stainless steel frits coated in accordance with one or more embodiments of the technology.

FIG. 20 is a bar graph showing a comparison of mass loss test according to ASTM G48 Method A for a bare or uncoated stainless steel frit versus stainless steel frits coated in accordance with one or more embodiments of the technology.

FIG. 21A is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 1 using hardware A, uncoated, in accordance with an illustrative embodiment of the technology.

FIG. 21B is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 2 using hardware A, uncoated, in accordance with an illustrative embodiment of the technology.

FIG. 21C is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 3 using hardware A, uncoated, in accordance with an illustrative embodiment of the technology.

FIG. 21D is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 1 using hardware A, C₂ coating, in accordance with an illustrative embodiment of the technology.

FIG. 21E is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 2 using hardware A, C₂ coating, in accordance with an illustrative embodiment of the technology.

FIG. 21F is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 3 using hardware A, C₂ coating, in accordance with an illustrative embodiment of the technology.

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FIG. 21G is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 1 using hardware B, uncoated, in accordance with an illustrative embodiment of the technology.

FIG. 21H is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 2 using hardware B, uncoated, in accordance with an illustrative embodiment of the technology.

FIG. 21I is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 3 using hardware B, uncoated, in accordance with an illustrative embodiment of the technology.

FIG. 21J is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 1 using hardware B, C₂-GPTMS-OH coating, in accordance with an illustrative embodiment of the technology.

FIG. 21K is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 2 using hardware B, C₂-GPTMS-OH coating, in accordance with an illustrative embodiment of the technology.

FIG. 21L is a chromatogram of NIST reference material 8671, an IgG1κ mAb, as obtained from injection 3 using hardware B, C₂-GPTMS-OH coating, in accordance with an illustrative embodiment of the technology.

FIG. 22 presents a bar graph showing peak areas of NIST reference materials 8671 obtained from sequential cation exchange separations over three injections of the sample, in accordance with an illustrative embodiment of the technology. This bar graph compares the peak areas for four different constructions in which the left most bar in each injection is an uncoated hardware A construction. The second from the left is a coated version of hardware A. The third bar from the left is an uncoated hardware B construction and the fourth or last bar per injection is a coated hardware B construction.

FIG. 23A is a reversed-phase chromatogram of the first injection of 5 picomoles of deoxythymidine oligomers (15, 20, 25, 30, and 35-mer) obtained from a 2.1×50 mm 1.7 μm organosilica 130 Å C₁₈ column constructed with an untreated stainless steel (SS) tube and frits, in accordance with an illustrative embodiment of the technology.

FIG. 23B is a reversed-phase chromatogram of the second injection of 5 picomoles of deoxythymidine oligomers (15, 20, 25, 30, and 35-mer) obtained from a 2.1×50 mm 1.7 μm organosilica 130 Å C₁₈ column constructed with an untreated stainless steel (SS) tube and frits, in accordance with an illustrative embodiment of the technology.

FIG. 23C is a reversed-phase chromatogram of the third injection of 5 picomoles of deoxythymidine oligomers (15, 20, 25, 30, and 35-mer) obtained from a 2.1×50 mm 1.7 μm organosilica 130 Å C₁₈ column constructed with an untreated stainless steel (SS) tube and frits, in accordance with an illustrative embodiment of the technology.

FIG. 23D is a reversed-phase chromatogram of the first injection of 5 picomoles of deoxythymidine oligomers (15, 20, 25, 30, and 35-mer) obtained from a column constructed with a C₂C₁₀ vapor deposition coated tube and frits, in accordance with an illustrative embodiment of the technology.

FIG. 23E is a reversed-phase chromatogram of the second injection of 5 picomoles of deoxythymidine oligomers (15, 20, 25, 30, and 35-mer) obtained from a column constructed with a C₂C₁₀ vapor deposition coated tube and frits, in accordance with an illustrative embodiment of the technology.

FIG. 23F is a reversed-phase chromatogram of the third injection of 5 picomoles of deoxythymidine oligomers (15,

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20, 25, 30, and 35-mer) obtained from a column constructed with a C₂C₁₀ vapor deposition coated tube and frits, in accordance with an illustrative embodiment of the technology.

FIG. 24 is a graph showing the average UV peak areas of a 15-mer deoxythymidine analyte as observed during reversed phase chromatography and initial injections onto either a 2.1×50 mm 1.7 μm organosilica 130 Å C₁₈ column constructed with untreated stainless steel (SS) or C₂C₁₀ vapor deposition coated components, in accordance with an illustrative embodiment of the technology. Analyses were performed in duplicate using two untreated columns and two C₂C₁₀ vapor deposition coated columns.

FIG. 25A is a reversed phase MRM (multiple reaction monitoring) chromatogram obtained for citric acid with the use of a 2.1×50 mm 1.8 μm silica 100 Å C₁₈ 1.8 μm column constructed with C₂C₃ vapor deposition coated components, in accordance with an illustrative embodiment of the technology.

FIG. 25B is a reversed phase MRM (multiple reaction monitoring) chromatogram obtained for citric acid with the use of a 2.1×50 mm 1.8 μm silica 100 Å C₁₈ 1.8 μm column constructed with untreated components, in accordance with an illustrative embodiment of the technology.

FIG. 25C is a reversed phase MRM (multiple reaction monitoring) chromatogram obtained for malic acid with the use of a 2.1×50 mm 1.8 μm silica 100 Å C₁₈ 1.8 μm column constructed with C₂C₃ vapor deposition coated components, in accordance with an illustrative embodiment of the technology.

FIG. 25D is a reversed phase MRM (multiple reaction monitoring) chromatogram obtained for malic acid with the use of a 2.1×50 mm 1.8 μm silica 100 Å C₁₈ 1.8 μm column constructed with untreated components, in accordance with an illustrative embodiment of the technology.

FIG. 26A is a mixed mode hydrophilic interaction chromatography (HILIC) MRM (multiple reaction monitoring) chromatogram of glyphosate showing MRM peak intensities obtained for glyphosate with the use of a 2.1×100 mm 1.7 μm diethylamine bonded organosilica 130 Å column constructed with C₂C₁₀ vapor deposition coated components, in accordance with an illustrative embodiment of the technology.

FIG. 26B is a mixed mode hydrophilic interaction chromatography (HILIC) MRM (multiple reaction monitoring) chromatogram of glyphosate showing MRM peak intensities obtained for glyphosate with the use of a 2.1×100 mm 1.7 μm diethylamine bonded organosilica 130 Å column with uncoated components, in accordance with an illustrative embodiment of the technology.

FIG. 27A is a graph showing the average peak areas of glyphosate as observed during mixed mode HILIC using either a 2.1×100 mm 1.7 μm diethylamine bonded organosilica 130 Å column constructed with either untreated or C₂C₁₀ vapor deposition coated components in accordance with an illustrative embodiment of the technology. The analyses were performed with six replicate injections.

FIG. 27B is a graph showing the average peak widths of glyphosate as observed during mixed mode HILIC using either a 2.1×100 mm 1.7 μm diethylamine bonded organosilica 130 Å column constructed with either untreated or C₂C₁₀ vapor deposition coated components, in accordance with an illustrative embodiment of the technology. The analyses were performed with six replicate injections.

DETAILED DESCRIPTION

In general, a number of aspects of the technology are directed to (1) devices having an alkylsilyl coating; (2)

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methods of tailoring or tuning a flow path for isolation of an analyte, in particular a metal-interacting analyte; (3) method of isolating an analyte in a sample, in particular a metal-interacting analyte; and (4) kits comprising various chromatographic components coated with an alkylsilyl coating and instructions for use. In some aspects, a bioinert, low-bind coating is used to modify a flow path to address flow path interactions with an analyte. That is, the bioinert, low-bind coating minimizes surface reactions with the metal interacting analyte and allows the analyte to pass along a flow path without clogging, attaching to surfaces, or change in analyte properties. The reduction/elimination of these interactions is advantageous because it allows for accurate quantification and analysis of a sample containing a metal-interacting analyte, for example biomolecules, proteins, glycans, peptides, oligonucleotides, pesticides, bisphosphonic acids, anionic metabolites, and zwitterions like amino acids and neurotransmitters. The biomolecule can be selected from a peptide or peptide fragment, an oligopeptide, a protein, a glycan, a nucleic acid or nucleic acid fragment, a growth factor, a carbohydrate, a fatty acid, and a lipid. In one aspect, the biomolecule is a protein, a peptide, or a glycan. The biomolecule can be a phosphoglycan or a phosphopeptide.

In the present technology, vapor deposited alkylsilyl coatings on wetted surfaces of fluidic systems (e.g., liquid chromatography systems) modify the fluidic path and decrease secondary interactions. As such, they are bioinert or low-bind (meaning that analytes of a sample do not interact, or have minimal interaction, with the alkylsilyl coating). In addition, the deposited coatings are highly tunable to provide a range of desirable contact angles (e.g., make the wetted surfaces hydrophilic or hydrophobic), chemistries, and properties to achieve a desired effect on the flow path and ultimately the sample passing through the flow path.

Devices

FIG. 1 is a representative schematic of a chromatographic flow system/device **100** that can be used to separate analytes in a sample. Chromatographic flow system **100** includes several components including a fluid manager system **105** (e.g., controls mobile phase flow through the system), tubing **110** (which could also be replaced or used together with microfabricated fluid conduits), fluid connectors **115** (e.g., fluidic caps), frits **120**, a chromatography column **125**, a sample injector **135** including a needle (not shown) to insert or inject the sample into the mobile phase, a vial, sinker, or sample reservoir **130** for holding the sample prior to injection, a detector **150** and a pressure regulator **140** for controlling pressure of the flow. Interior surfaces of the components of the chromatographic system/device form a fluidic flow path that has wetted surfaces. The fluidic flow path can have a length to diameter ratio of at least 20, at least 25, at least 30, at least 35 or at least 40.

The detector **150**, can be a mass spectrometer. The fluidic flow path can include wetted surfaces of an electrospray needle (not shown).

At least a portion of the wetted surfaces can be coated with an alkyl silyl coating, described in detail herein, to tailor its hydrophobicity. The coating can be applied by vapor deposition. As such, methods and devices of the present technology provide the advantage of being able to use high pressure resistant materials (e.g., stainless steel) for the creation of the flow system, but also being able to tailor the wetted surfaces of the fluidic flow path to provide the

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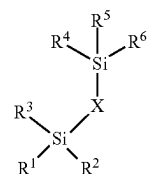
appropriate hydrophobicity so deleterious interactions or undesirable chemical effects on the sample can be minimized.

The alkylsilyl coating can be provided throughout the system from the tubing or fluid conduits **110** extending from the fluid manager system **105** all the way through to the detector **150**. The coatings can also be applied to portions of the fluidic fluid path. That is, one may choose to coat one or more components or portions of a component and not the entire fluidic path. For example, the internal portions of the column **125** and its frits **120** and end caps **115** can be coated whereas the remainder of the flow path can be left unmodified. Further, removable/replaceable components can be coated. For example, the vial or sinker **130** containing the sample reservoir can be coated as well as frits **120**.

In one aspect, the flow path of the fluidic systems described herein is defined at least in part by an interior surface of tubing. In another aspect, the flow path of the fluidic systems described herein is defined at least in part by an interior surface of microfabricated fluid conduits. In another aspect, the flow path of the fluidic systems described herein is defined at least in part by an interior surface of a column. In another aspect, the flow path of the fluidic systems described herein is defined at least in part by passageways through a frit. In another aspect, the flow path of the fluidic systems described herein is defined at least in part by an interior surface of a sample injection needle. In another aspect, the flow path of the fluidic systems described herein extends from the interior surface of a sample injection needle throughout the interior surface of a column. In another aspect, the flow path extends from a sample reservoir container (e.g. sinker) disposed upstream of and in fluidic communication with the interior surface of a sample injection needle throughout the fluidic system to a connector/port to a detector.

In some embodiments, only the wetted surfaces of the chromatographic column and the components located upstream of the chromatographic column are coated with the alkylsilyl coatings described herein while wetted surfaces located downstream of the column are not coated. The coating can be applied to the wetted surfaces via vapor deposition.

At least a portion of the wetted surfaces of the fluidic flow path are coated with an alkylsilyl coating. The alkylsilyl coating is inert to at least one of the analytes in the sample. The alkylsilyl coating can have the Formula I:



(I)

R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 are each independently selected from $(\text{C}_1\text{-C}_6)\text{alkoxy}$, $-\text{NH}(\text{C}_1\text{-C}_6)\text{alkyl}$, $-\text{N}((\text{C}_1\text{-C}_6)\text{alkyl})_2$, OH, OR⁴, and halo (i.e., a halogen, for example chloro). R⁴ represents a point of attachment to the interior surfaces of the fluidic system. At least one of R¹, R², R³, R⁴, R⁵, and R⁶ is OR⁴. X is $(\text{C}_1\text{-C}_{20})\text{alkyl}$, $-\text{O}[(\text{CH}_2)_2\text{O}]_{1-20}-$, $-(\text{C}_1\text{-C}_{10})[\text{NH}(\text{CO})\text{NH}(\text{C}_1\text{-C}_{10})]_{1-20}-$, or $-(\text{C}_1\text{-C}_{10})[\text{alkylphenyl}(\text{C}_1\text{-C}_{10})\text{alkyl}]_{1-20}-$.

When used in the context of a chemical formula, a hyphen (“-”) indicates the point of attachment. For example, when

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X is $[(C_1-C_{10})\text{alkylphenyl}(C_1-C_{10})\text{alkyl}]_{1-20}$, that means that X is connected to $\text{SiR}^1\text{R}^2\text{R}^3$ via the $(C_1-C_{10})\text{alkyl}$ and connected to $\text{SiR}^4\text{R}^5\text{R}^6$ via the other $(C_1-C_{10})\text{alkyl}$. This applies to the remaining variables.

In one aspect, X in Formula I is $(C_1-C_{15})\text{alkyl}$, $(C_1-C_{12})\text{alkyl}$, or $(C_1-C_{10})\text{alkyl}$. In some aspects, X in Formula I is methyl, ethyl, propyl, isopropyl, butyl, sec-butyl, iso-butyl, t-butyl, pentyl, hexyl, heptyl, nonyl, or decanyl. In other aspect, X in Formula I is ethyl or decanyl.

In one aspect, at least one of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is $(C_1-C_6)\text{alkoxy}$, e.g., ethoxy, wherein the values for X are described in Formula I or the preceding paragraph. In another aspect, at least two of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is $(C_1-C_6)\text{alkoxy}$, e.g., ethoxy, wherein the values for X are described in Formula I or the preceding paragraph. In another aspect, at least three of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is $(C_1-C_6)\text{alkoxy}$, e.g., ethoxy, wherein the values for X are described in Formula I or the preceding paragraph. In another aspect, at least four of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is $(C_1-C_6)\text{alkoxy}$, e.g., ethoxy, wherein the values for X are described in Formula I or the preceding paragraph. In another aspect, at least five of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is $(C_1-C_6)\text{alkoxy}$, e.g., ethoxy, wherein the values for X are described in Formula I or the preceding paragraph.

In one aspect, at least one of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is halo, e.g., chloro, wherein the values for X are described in Formula I or the preceding paragraphs above. In another aspect, at least two of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is halo, e.g., chloro, wherein the values for X are described in Formula I or the preceding paragraphs above. In another aspect, at least three of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is halo, e.g., chloro, wherein the values for X are described in Formula I or the preceding paragraphs above. In another aspect, at least four of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is halo, e.g., chloro, wherein the values for X are described in Formula I or the preceding paragraphs above. In another aspect, at least five of R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 is halo, e.g., chloro, wherein the values for X are described in Formula I or the preceding paragraphs above.

In another aspect, R^1 , R^2 , R^3 , R^4 , R^5 , and R^6 are each methoxy or chloro.

The alkylsilyl coating of Formula I can have a contact angle of at least about 15° . In some embodiments, the alkylsilyl coating of Formula I can have a contact angle of less than or equal to 30° . The contact angle can be less than or equal to about 90° . In some embodiments, the contact angle of the alkylsilyl coating of Formula I is between about 15° to about 105° . For example, the contact angle of the alkylsilyl coating of Formula I can be about 0° , 5° , 10° , 15° , 20° , 25° , 30° , 35° , 40° , 45° , 50° , 60° , 65° , 70° , 75° , 80° , 85° , 90° , 95° , 100° , or 105° .

The thickness of the alkylsilyl coating can be at least about $100 \text{ \AA}$. For example the thickness can be between about $100 \text{ \AA}$ to about $1600 \text{ \AA}$. The thickness of the alkylsilyl coating for Formula I can be about $100 \text{ \AA}$, $200 \text{ \AA}$, $300 \text{ \AA}$, $400 \text{ \AA}$, $500 \text{ \AA}$, $600 \text{ \AA}$, $700 \text{ \AA}$, $800 \text{ \AA}$, $900 \text{ \AA}$, $1000 \text{ \AA}$, $1100 \text{ \AA}$, $1200 \text{ \AA}$, $1300 \text{ \AA}$, $1400 \text{ \AA}$, $1500 \text{ \AA}$ or $1600 \text{ \AA}$. The thickness of the alkylsilyl coating (e.g., a vapor deposited alkylsilyl coating) can be detected optically by the naked eye. For example, more opaqueness and coloration is indicative of a thicker coating. Thus, coatings with pronounced visual distinction are an embodiment of this technology. From thin to thick, the color changes from yellow, to violet, to blue, to slightly greenish and then back to yellow when coated parts are observed under full-spectrum light, such as sunlight. For example, when the alkylsilyl coating is $300 \text{ \AA}$ thick, the coating can appear yellow and reflect light with a peak

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wavelength between 560 and 590 nm . When the alkylsilyl coating is $600 \text{ \AA}$ thick, the coating can appear violet and reflect light with a peak wavelength between 400 and 450 nm . When the alkylsilyl coating is $1000 \text{ \AA}$ thick, the coating can appear blue and reflect light with a peak wavelength between 450 and 490 nm . See, e.g., Faucheu et al., *Relating Gloss Loss to Topographical Features of a PVDF Coating*, Published Oct. 6, 2004; Bohlin, Erik, *Surface and Porous Structure of Pigment Coatings, Interactions with flexographic ink and effects of print quality*, Dissertation, Karlstad University Studies, 2013:49.

In one aspect, the vapor deposited coating of Formula I is the product of vapor deposited bis(trichlorosilyl)ethane, bis(trimethoxysilyl)ethane, bis(trichlorosilyl)octane, bis(trimethoxysilyl)octane, bis(trimethoxysilyl)hexane, and bis(trichlorosilyl)hexane.

In some aspects, at least a portion of the wetted surfaces of the fluidic flow path are coated with multiple layers of the same or different alkylsilyls, where the thickness of the alkylsilyl coatings correlate with the number of layering steps performed (e.g., the number of deposited layers of alkylsilyl coating on wetted surface of the fluidic flow path of the chromatographic system/device). In this manner, increasingly thick bioinert coatings can be produced and tailored to achieve desirable separations.

The chromatographic device can have a second alkylsilyl coating in direct contact with the alkylsilyl coating of Formula I. The second alkylsilyl coating has the Formula II



wherein R^7 , R^8 , and R^9 are each independently selected from $-\text{NH}(C_1-C_6)\text{alkyl}$, $-\text{N}[(C_1-C_6)\text{alkyl}]_2$, $(C_1-C_6)\text{alkoxy}$, $(C_1-C_6)\text{alkyl}$, $(C_1-C_6)\text{alkenyl}$, OH, and halo; R^{10} is selected from $(C_1-C_6)\text{alkyl}$, $-\text{OR}^B$, $-\text{[O}(C_1-C_3)\text{alkyl}]_{1-10}\text{O}(C_1-C_6)\text{alkyl}$, $-\text{[O}(C_1-C_3)\text{alkyl}]_{1-10}\text{OH}$ and phenyl. $(C_1-C_6)\text{alkyl}$ is optionally substituted with one or more halo. The phenyl is optionally substituted with one or more groups selected from $(C_1-C_3)\text{alkyl}$, hydroxyl, fluorine, chlorine, bromine, cyano, $-\text{C}(\text{O})\text{NH}_2$, and carboxyl. R^B is $-(C_1-C_3)\text{alkyloxirane}$, $-(C_1-C_3)\text{alkyl-3,4-epoxycyclohexyl}$, or $-(C_1-C_4)\text{alkyl-OH}$. The hashed bond to R^{10} represents an optional additional covalent bond between R^{10} and the carbon bridging the silyl group to form an alkene, provided y is not 0. y is an integer from 0 to 20.

In one aspect, y in Formula II is an integer from 1 to 15. In another aspect, y in Formula II is an integer from 1 to 12. In another aspect, y in Formula II is an integer from 1 to 10. In another aspect, y in Formula II is an integer from 2 to 9.

In one aspect R^{10} in Formula II is methyl and y is as described above for Formula II or the preceding paragraph.

In one aspect, R^7 , R^8 , and R^9 in Formula II are each the same, wherein R^{10} and y are as described above. In one aspect, R^7 , R^8 , and R^9 are each halo (e.g., chloro) or $(C_1-C_6)\text{alkoxy}$ such as methoxy, wherein R^{10} and y are as described above.

In one aspect, y in Formula II is 9, R^{10} is methyl, and R^7 , R^8 , and R^9 are each ethoxy or chloro.

In one aspect, the coating of the formula II is n-decyl-trichlorosilane, (3-glycidyloxypropyl)trimethoxysilane (GPTMS), (3-glycidyloxypropyl)trimethoxysilane (GPTMS) followed by hydrolysis, 2-(3,4-epoxycyclohexyl)

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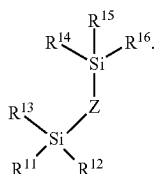
ethyltrimethoxysilane, trimethylchlorosilane, trimethyldimethylaminosilane, methoxy-polyethyleneoxy(3)silane propyltrichlorosilane, propyltrimethoxysilane, (heptadecafluoro-1,1,2,2-tetrahydrodecyl)tris(dimethylamino)silane, (heptadecafluoro-1,1,2,2-tetrahydrodecyl)trichlorosilane, (heptadecafluoro-1,1,2,2-tetrahydrodecyl)trimethoxysilane vinyltrichlorosilane, vinyltrimethoxysilane, allyltrichlorosilane, 2-[methoxy (polyethyleneoxy)3propyl]trichlorosilane, 2-[methoxy(polyethyleneoxy)3propyl]trimethoxysilane, or 2-[methoxy(polyethyleneoxy)3propyl]tris(dimethylamino)silane.

The alkylsilyl coating of Formula I and II can have a contact angle of at least about 15°. In some embodiments, the alkylsilyl coating of Formula I and II can have a contact angle of less than or equal to 105°. The contact angle can be less than or equal to about 90°. In some embodiments, the contact angle of the alkylsilyl coating of Formula I and II is between about 15° to about 105°. For example, the contact angle of the alkylsilyl coating of Formula I and II can be about 0°, 5°, 10°, 15°, 20°, 25°, 30°, 35°, 40°, 45°, 50°, 55°, 60°, 65°, 70°, 75°, 80°, 85°, 90°, 95°, 100°, or 105°.

The thickness of the multi-layered alkylsilyl coating can be at least about 100 Å. For example the thickness can be between about 100 Å to about 1600 Å. The thickness of the multi-layered alkylsilyl coating for Formula I can be about 100 Å, 200 Å, 300 Å, 400 Å, 500 Å, 600 Å, 700 Å, 800 Å, 900 Å, 1000 Å, 1100 Å, 1200 Å, 1300 Å, 1400 Å, 1500 Å or 1600 Å.

In one aspect, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is (3-glycidyloxypropyl)trimethoxysilane. In another aspect, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is (3-glycidyloxypropyl)trimethoxysilane followed by hydrolysis. In one aspect, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is n-decyltrichlorosilane. The alkylsilyl coating of Formula I can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II can be trimethylchlorosilane or trimethyldimethylaminosilane. In one aspect, the alkylsilyl coating of Formula I is bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II is methoxy-polyethyleneoxy(3) propyl trichlorosilane or methoxy-polyethyleneoxy(3) propyl trimethoxysilane.

The chromatographic device can have an alkylsilyl coating in direct contact with the alkylsilyl coating of Formula I. III in direct contact with the alkylsilyl coating of Formula I.



R¹¹, R¹², R¹³, R¹⁴, R¹⁵, and R¹⁶ are each independently selected from (C₁-C₆)alkoxy, —NH(C₁-C₆)alkyl, —N((C₁-C₆)alkyl)₂, OH, and halo (i.e., a halogen, for example, chloro). Z is (C₁-C₂₀)alkyl, —O[(CH₂)₂O]₁₋₂₀—, —(C₁-C₁₀) [NH(CO)NH(C₁-C₁₀)]₁₋₂₀—, or —(C₁-C₁₀)[alkylphenyl(C₁-C₁₀)alkyl]₁₋₂₀—.

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In some aspects, Z in Formula III is (C₁-C₁₀)alkyl; and R¹, R², R³, R⁴, R⁵, and R⁶ are each methoxy or chloro. In other aspects, Z in Formula III is (C₂-C₁₀)alkyl. In other aspects, Z in Formula III is ethyl.

In the layered alkylsilyl coating of Formula I and Formula III, Formula I and Formula III can be the same (for example, C₂C₂) or Formula I and Formula III can be different. Formula III is attached directly to the coating of Formula I, i.e., in Formula III, there is no point of attachment to the interior of the fluidic system; instead Formula III is deposited directly on Formula I.

The alkylsilyl coating of Formula III can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane. The alkylsilyl coating of Formula I can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula III can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane.

The alkylsilyl coating of Formula I and III can have a contact angle of at least about 15°. In some embodiments, the alkylsilyl coating of Formula I and III can have a contact angle of less than or equal to 105°. The contact angle can be less than or equal to about 90°. In some embodiments, the contact angle of the alkylsilyl coating of Formula I and III is between about 15° to about 105°. For example, the contact angle of the alkylsilyl coating of Formula I and III can be about 0°, 5°, 10°, 20°, 25°, 30°, 35°, 40°, 45°, 50°, 55°, 60°, 65°, 70°, 75°, 80°, 85°, 90°, 95°, 100°, or 105°.

The thickness of the multi-layered alkylsilyl coating can be at least about 100 Å. For example the thickness can be between about 100 Å to about 1600 Å. The thickness of the multi-layered alkylsilyl coating for Formula I can be about 100 Å, 200 Å, 300 Å, 400 Å, 500 Å, 600 Å, 700 Å, 800 Å, 900 Å, 1000 Å, 1100 Å, 1200 Å, 1300 Å, 1400 Å, 1500 Å or 1600 Å.

In one aspect, the alkylsilyl coating of Formula II is applied directly to wetted surfaces of the fluidic flow path. Therefore, in some embodiments, one of R⁷, R⁸, and R⁹ of Formula II can also include OR⁴, where R⁴ represents a point of attachment to the interior surfaces (e.g., wetted surfaces) of the fluidic system. In other embodiments, R⁷, R⁸, and R⁹ of the alkylsilyl coating of Formula II does not include OR⁴, by the alkylsilyl coating of Formula II is deposited directly onto wetted surfaces of the fluidic flow path that have been pre-treated with, for example, a plasma.

In one aspect, stainless steel flow path components, including but not limited to tubing, microfabricated fluid conduits, column frits, column inlet tubing, and sample injection needles, are coated via vapor deposition with one or more of the disclosed alkylsilyls. In one aspect, these coated components are annealed to alter their chemical or physical properties.

In another aspect, flow path components that are made of other materials than stainless steel or other metallics, (e.g., polymers, glass, etc.) are coated via vapor deposition with one or more of the disclosed alkylsilyls. In particular, frits for use within the system or sample vials connectable to the injection needle are coated.

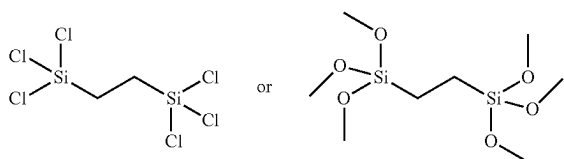
Exemplary coatings with their respective approximate thickness and contact angle are provided in Table 1.

TABLE 1

VPD #	Vapor Deposited Material	Alternative Coating Abbreviation	Approximate Thickness of Product	Approximate Contact Angle
1	bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by GPTMS followed by hydrolysis to form GPTMS-OH	C ₂ -GPTMS-OH	500 Å	15°
2	bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane	C ₂	500 Å	35°
3	bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a second layer.	C ₂ -C ₂	1600 Å	35°
4	bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by GPTMS as a second layer	C ₂ -GPTMS	500 Å	50°
5	Annealed bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane	Annealed C ₂	500 Å	95°
6	Annealed bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by annealed bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a second layer	Annealed C ₂ -C ₂	1600 Å	95°
7	bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by n-decyltrichlorosilane as a second layer	C ₂ C ₁₀	500 Å	105°
8	Annealed bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by annealed n-decyltrichlorosilane as a second layer	Annealed C ₂ C ₁₀	500 Å	105°
9	GPTMS	GPTMS	100 to 200 Å	~50°
10	GPTMS followed by hydrolysis to form GPTMS-OH	GPTMS-OH	100 to 200 Å	~20°
11	bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by trimethylchlorosilane or trimethyldimethylaminosilane	C ₂ C ₃	500 Å	40-90°
12	annealed bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by trimethylchlorosilane or trimethyldimethylaminosilane	Annealed C ₂ C ₃	500 Å	95°
13	bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by a methoxy-polyethyleneoxy(3) propyl trichlorosilane or methoxy-polyethyleneoxy(3) propyl trimethoxysilane	C ₂ PEO	500 Å	15°

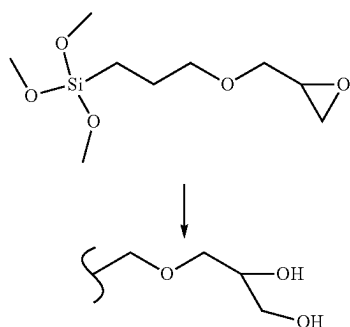
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Referring to VPD #1 (C_2 -GPTMS-OH), the first coating layer, C_2 shown below, is a layer according to Formula I, described above.



structure of bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane (C_2)

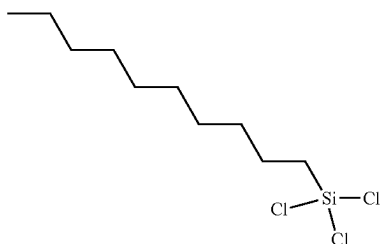
The second layer of VPD #1, GPTMS-OH, shown below, is a layer according to Formula II.



Structure of GPTMS-OH

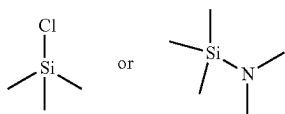
VPD #3 (C_2 - C_2) is an example of a coating of Formula I and then a coating for Formula III.

VPD #7 (C_2C_{10}) is another example of a coating of Formula I and a second layer of Formula II. The structure of bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane (C_2) is shown above. The structure of C_{10} is shown below.



Structure of n-decyltrichlorosilane (C_{10})

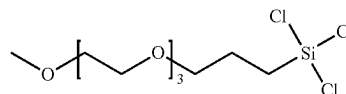
VPD #11 (C_2C_3) is another example of a coating of Formula I and a second layer of Formula II. The structure of bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane (C_2) is shown above. The structure of C_3 is shown below.



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Structure of trimethylchlorosilane or trimethyldimethylaminosilane (C_3)

VPD #13 is another example of a coating of Formula I and a second layer of Formula II. The structure of bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane (C_2) is shown above. The structure of methoxy-polyethyleneoxy(3)propyl trichlorosilane (PEO) is shown below.



Structure of methoxy-polyethyleneoxy(3)propyl trichlorosilane (PEO)

Alternatively, commercially available vapor deposition coatings can be used in the disclosed systems, devices, and methods, including but not limited to Dursan® (commercially available from SikcoTek Corporation, Bellefonte, PA).

In one aspect, the alkylsilyl coatings described herein enhance the corrosion performance of metals, e.g., as in metallic chromatography columns. Depending on the denseness and thickness, the coatings act as a barrier, thereby preventing water and corrosive molecules from reacting with the base metal. While increasing the hydrophobicity and density improves the corrosion performance, even coatings derived from C_2 and GPTMS (C_2 -GPTMS) followed by hydrolysis to form C_2 -GPTMS-OH shows a 10x improvement in the ASTM G48 Method A pitting corrosion, see e.g., Example 4 below. In terms of most corrosion resistant to least, the ranking is the material formed from VPD #7>2>1 (bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane as a first layer followed by GPTMS then hydrolysis to form GPTMS-OH as a second layer). This also correlates to hydrophobicity rankings.

Methods of Tailoring a Fluidic Flow Path

The coatings described above can be used to tailor a fluidic flow path of a chromatography system for the separation of a sample. The coatings can be vapor deposited. In general, the deposited coatings can be used to adjust the hydrophobicity of internal surfaces of the fluidic flow path that come into contact with a fluid (i.e. wetted surfaces or surfaces coming into contact with the mobile phase and/or sample/analyte). By coating wetted surfaces of one or more components of a flow path within a chromatography system, a user can tailor the wetted surfaces to provide a desired interaction (or lack of interaction) between the flow path and fluids therein (including any sample, such as biomolecules, proteins, glycans, peptides, oligonucleotides, pesticides, bisphosphonic acids, anionic metabolites, and zwitterions like amino acids and neurotransmitters, within the fluid).

In one aspect, an effective coating is produced from a low temperature, vacuum-assisted vapor deposition process. In one aspect, an oxygen plasma pretreatment step precedes the coating deposition. The oxygen plasma removes organic compounds and improves surface wettability for the coatings. Time, temperature, and pressure are controlled for each processing step. Each coating run can use a silicon wafer to monitor the thickness and contact angle of the resultant coating. Ellipsometry can be used to measure the coating thickness, and an optical goniometer can be used to measure the contact angle of the coating. A post coating annealing

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step can be utilized to increase coating cross-linking and increase coating hydrophobicity.

FIG. 2 is a flow chart illustrating method **200** for tailoring a fluidic flow path for separation of a sample including biomolecules, proteins, glycans, peptides, oligonucleotides, pesticides, bisphosphonic acids, anionic metabolites, and zwitterions like amino acids and neurotransmitters. The method has certain steps which are optional as indicated by the dashed outline surrounding a particular step. Method **200** can start with a pretreatment step (**205**) for cleaning and/or preparing a flow path within a component for tailoring. Pretreatment step **205** can include cleaning the flow path with plasma, such as oxygen plasma. This pretreatment step is optional.

Next, an infiltration step (**210**) is initiated. A vaporized source of an alkylsilyl compound (e.g., the alkylsilyl compounds of Formulas I, II and/or III) is infiltrated into the flow path. The vaporized source is free to travel throughout and along the internal surfaces of the flow path. Temperature and/or pressure is controlled during infiltration such that the vaporized source is allowed to permeate throughout the internal flow path and to deposit a coating from the vaporized source on the exposed surface (e.g., wetted surfaces) of the flow path as shown in step **215**. Additional steps can be taken to further tailor the flow path. For example, after the coating is deposited, it can be heat treated or annealed (step **220**) to create cross linking within the deposited coating and/or to adjust the contact angle or hydrophobicity of the coating. Additionally or alternatively, a second coating of alkylsilyl compound (having the same or different form) can be deposited by infiltrating a vaporized source into the flow path and depositing a second or additional layers in contact with the first deposited layer as shown in step **225**. After the deposition of each coating layer, an annealing step can occur. Numerous infiltration and annealing steps can be provided to tailor the flow path accordingly (step **230**).

FIG. 3 provides a flow chart illustrating a method (**300**) of tailoring a fluidic flow path for separation of a sample including a biomolecule or a metal interacting analyte. The method can be used to tailor a flow system for use in isolating, separating, and/or analyzing the biomolecule or metal interacting analyte. In step **305**, the analyte is assessed to determine its polarity. Understanding the polarity will allow an operator to select (by either look up table or make a determination) a desired coating chemistry and, optionally, contact angle as shown in step **310**. In some embodiments, in addition to assessing the polarity of the biomolecule or metal interacting analyte, the polarity of a stationary phase to be used to separate the biomolecule or metal interacting analyte (e.g., stationary phase to be included in at least a portion of the fluidic flow path) is also assessed. A chromatographic media can be selected based on the analyte in the sample. Understanding the polarity of both the analyte and the stationary phase is used in certain embodiments, by the operator to select the desired coating chemistry and contact angle in step **310**. The components to be tailored can then be positioned within a chemical infiltration system with environmental control (e.g., pressure, atmosphere, temperature, etc.) and precursor materials are infiltrated into the flow path of the component to deposit one or more coatings along the wetted surfaces to adjust the hydrophobicity as shown in step **315**. During any one of infiltration, deposition, and condition steps (e.g. annealing), coatings deposited from the infiltration system can be monitored and if necessary precursors and or depositing conditions can be adjusted if required allowing for fine tuning of coating properties. The

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alkylsilyl coating material selected in step **310** can be the alkylsilyl compounds of Formulas I, II and/or III.

A method of tailoring a fluidic flow path for separation of a sample is provided that includes assessing a polarity of an analyte in the sample and selecting a chromatographic media based on the analyte in the sample. An alkylsilyl coating is selected based on the polarity of the analyte in the sample. The alkylsilyl coating is selected so that the coating is inert to the analyte(s) being separated. In other words, the alkylsilyl coating does not produce any secondary chromatographic effects that are attributable to the alkylsilyl coating. In some embodiments, the analyte is a biomolecule. The biomolecule can be a peptide or peptide fragment, an oligopeptide, a protein, a glycan, a nucleic acid or nucleic acid fragment, a growth factor, a carbohydrate, a fatty acid or a lipid. The analyte can be a citric acid cycle metabolite. The analyte can be a pesticide.

The alkylsilyl coating can have the Formula I, II, or III as described above. In one embodiment, the alkylsilyl coating has the Formula I as a first layer and Formula II as a second layer. In some embodiments, there is only a single layer coating having Formula I (e.g., bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane). In some embodiments, there is only a single layer coating having Formula II (e.g., (3-glycidyloxypropyl)trimethoxysilane, n-decyltrichlorosilane, trimethylchlorosilane, trimethyldimethylaminosilane, or methoxy-polyethyleneoxy(3)silane). In some embodiments, there is only a single layer coating having Formula III (e.g., bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane).

The method also includes adjusting a hydrophobicity of the wetted surfaces of the fluidic flow path by vapor depositing the alkylsilyl coating onto the wetted surfaces of the fluidic flow path. In some embodiments, the hydrophobicity of the wetted surfaces is adjusted by adjusting the contact angle of the alkylsilyl coating. For example, the contact angle of the alkylsilyl coating can be between about 0° to about 105°.

The analyte in the sample can be retained with a retentivity within 10% of the retentivity attributable to the chromatography media. In some embodiments, the sample can be retained with a retentivity within 5% or within 1% of the retentivity attributable to the chromatography media. Therefore, the alkylsilyl coating solves the problem of metal interaction between the analyte and the metal chromatographic materials without introducing any secondary reactions that would have a negative effect on the quality of the separation. The alkylsilyl coating does not impart any retention mechanism on the analyte of interest, making the coating inert to the analyte of interest and low-bind.

In addition, the alkylsilyl coating does not produce any changes to peak width. The analyte in the sample has a peak width that is within 10%, 5%, or 1% of the peak width attributable to the chromatographic media.

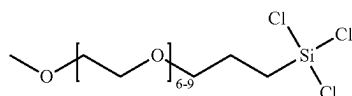
The wetted surfaces of the fluidic flow path can be any of those described above with respect to aspects and embodiments of the chromatographic device.

The method can also include annealing the alkylsilyl coating after vapor depositing the alkylsilyl coating on the wetted surfaces of the fluidic flow path. Typically, the annealing cycle involves subjecting the coating to 200° C. for 3 hours under vacuum.

The method can also include assessing the polarity of the chromatographic media and selecting the alkylsilyl coating based on the polarity of the analyte and the chromatographic media. The method can also include eluting the sample through the fluidic flow path, thereby isolating the analyte.

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In some embodiments, the alkylsilyl coating is modified with a silanizing reagent to obtain a desired thickness of the alkylsilyl coating. The silanizing reagent can be a non-volatile zwitterion. The non-volatile zwitterion can be sulfo-



Structure of methoxy-polyethyleneoxy(6-9)silane

In some aspects, the method of tailoring a fluidic flow path for separation of a sample including a biomolecule further comprises: pretreating the wetted surfaces of the flow path with a plasma prior to depositing the first coating. In other aspects, the method of tailoring a fluidic flow path for separation of a sample including a metal interacting analyte further comprises annealing the first coating at a temperature to increase cross-linking in the first coating. In yet another aspect, the method of tailoring a fluidic flow path for separation of a sample including a metal interacting analyte further comprises annealing the first coating at a temperature to alter hydrophobicity.

In one aspect, the method of tailoring a fluidic flow path for separation of a sample including a metal interacting analyte further comprises performing a second infiltration with a vaporized source having the Formula II, wherein the features for Formula II are as described above; along and throughout the interior flow path of the fluidic system to form a second coating deposited in direct contact with the first coating. In one aspect, the step of performing a second infiltration in the preceding method further comprises performing an annealing step after depositing the second coating. In another aspect, the preceding method further comprises connecting in fluid communication with the flow path at least one coated component selected from the group consisting of a sample reservoir container and a frit.

Also provided herein is a method of tailoring a fluidic flow path for separation of a sample including a metal interacting analyte, the method comprising: assessing polarity of the analyte in the sample; selecting an alkylsilyl coating having the Formula I, wherein the features for Formula I are as described above, and desired contact angle based on polarity assessment; and adjusting the hydrophobicity of wetted surfaces of the flow path by vapor depositing an alkylsilyl having the Formula III, wherein the features for Formula III are as described above, and providing the desired contact angle. In some embodiments of the above method, in addition to assessing polarity of the analyte in the sample, the polarity of a stationary phase disposed within at least a portion of the flow path is also assessed and the polarity assessment is obtained from both the polarity of the biomolecule in the sample and the stationary phase.

Methods of Isolating an Analyte

In one aspect, provided herein are methods of isolating an analyte. The method includes introducing a sample including a glycan, a peptide, a pesticide, or a citric acid cycle metabolite into a fluidic system including a flow path

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disposed in an interior of the fluidic system. The flow path includes a first vapor deposited alkylsilyl inert coating having the Formula I described above and a second vapor deposited coating of the Formula II described above. The sample is eluted through the fluidic system, thereby isolating the glycan, peptide, pesticide, or citric acid cycle metabolite.

The glycan can be a phosphoglycan. The peptide can be a phosphopeptide and the pesticide can be glyphosate. The citric acid cycle metabolite can be citric acid or malic acid.

When the analyte is a glycan, peptide or pesticide, the alkylsilyl coating of Formula I can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II can be n-decyltrichlorosilane. When the analyte is a citric acid cycle metabolite, the alkylsilyl coating of Formula I can be bis(trichlorosilyl)ethane or bis(trimethoxysilyl)ethane and the alkylsilyl coating of Formula II can be trimethylchlorosilane or trimethyldimethylaminosilane.

The flow path can be defined at least in part by the interior surface of a chromatographic system. The flow path can be further defined at least in part by passageways through a frit of the chromatographic column. The flow path can be defined at least in part by interior surfaces of tubing. The flow path can be any flow path described herein, for example, the flow paths described with respect to the chromatographic device.

Methods of Improving Baseline Returns

Also provided herein is a method of improving baseline returns in a chromatographic system, the method comprising: introducing a sample including an analyte into a fluidic system including a flow path disposed in an interior of the fluidic system, the flow path having a length to diameter ratio of at least 20 and comprising a vapor deposited alkylsilyl coating having the Formula I, wherein the features for Formula I are as described above, a thickness of at least 100 angstroms and a contact angle of about 30 degrees to 110 degrees; and eluting the sample through the fluidic system, thereby isolating the biomolecule. In some embodiments, the method includes a second layer of Formula II or Formula III, wherein the features of Formula II and II are described above.

Kits

Also provided here are kits. The kits include a chromatographic components, for example, a chromatographic column, that has been coated with an alkylsilyl coating of Formulas I, II, and/or III, as described above. Other components can be provided in the kit that can also include the coatings described herein, for example, the tubing, frits, and/or connectors. The kit can also include instructions for separating analytes, for example, biomolecules, proteins, glycans, peptides, oligonucleotides, pesticides, bisphosphonic acids, anionic metabolites, and zwitterions like amino acids and neurotransmitters.

Exemplary Separations

Separation of Phosphoglycans

The disclosed coatings, which can be vapor deposited, have been found to dramatically improve separations of phosphoglycans by hydrophilic interaction chromatography (HILIC). To demonstrate the significance of this, the released N-glycans from a recombinant alpha-galactosidase which can be used as an enzyme replacement therapy for Fabry's disease were evaluated. This particular type of enzyme is taken up from circulation and delivered intercellularly to lysosomes through the mannose-6-phosphate pathway, making it important to identify and monitor the levels

of phosphorylated glycans that are present on its surface. Vapor deposition coated stainless steel column tubes along with matching coated stainless steel frits were first tested against corresponding untreated stainless steel hardware. In this instance, two different types of coating chemistries were used. The coating chemistries used to coat the frits and tubing were VPD #2 and VPD #7. FIG. 4A-4C show fluorescence chromatograms obtained with these types of column hardware. From these data, it was found that use of coated column hardware significantly improved the recovery of each phosphorylated N-glycan species. For example, there was a marked increase in the peak area of Man7-PP, a high mannose glycan with two mannose-6-phosphate residues, mannose-7-bisphosphate. Where Man7-PP could not be detected with stainless steel column hardware, it was easily detected with vapor deposition coated column hardware. This indicated that this species of N-glycan was interacting with the metallic surfaces of the column hardware in such a way that prevented it from reaching the detector. When using vapor deposition coated hardware, the peak area ratio of Man-7-PP to Man5 (a high mannose glycan without phosphorylation) was 0.24:1 (FIG. 4A-4C).

The increased recovery of phosphorylated glycans using coated column hardware and frits shows that adsorption to metallic column hardware surfaces is detrimental to recovery. With this in mind, separations were also performed with vapor deposition coated stainless steel sample flow path components (FIGS. 5A and 5B). FIG. 6A-6C show fluorescence chromatograms obtained using coated LC system components in conjunction with coated stainless steel column hardware. Phosphoglycan recovery improved even more with the use of coated column hardware and C₂C₁₀ vapor deposition coated flow path components. Most notably, the peak area ratio of Man7-PP to Man5 increased to 0.8:1 from the ratio of 0.24:1 that was obtained by using coated column hardware alone. The observed relative abundance for Man7-PP with the coated system and coated column hardware is indicative of full recovery for the phosphorylated glycans, as can be determined by orthogonal assays to HILIC of RapiFluor-MS labeled released glycans. In sum, these results confirm that the loss of phosphorylated N-glycan species to sample flow path surfaces can be alleviated with the use of vapor deposition coatings.

Separation of Other Phosphorylated Molecules

The principles learned from using vapor deposition coatings for phosphoglycan analysis were extended to facilitate the analysis of other types of phosphorylated biomolecules. In which case, the coatings have been found to be beneficial to improving the recovery of phosphorylated peptides under reversed phase chromatography conditions. To demonstrate these recovery advantages, we evaluated a mixture containing phosphopeptides. This particular sample contains three peptides that are singly phosphorylated and one that is doubly phosphorylated. Vapor deposition coated stainless steel column tubes along with matching coated stainless steel frits were first tested against corresponding untreated stainless steel hardware. FIG. 7A-7C show UV chromatograms obtained with these types of column hardware. In each case, the addition of the VPD #2, and the VPD #7 coatings increased the recovery of the singly phosphorylated peptides by at least 13% over the stainless steel alone (FIG. 4A-4C). The impact of coating the chromatographic flow path was much more pronounced with the doubly phosphorylated peptide. When using the stainless steel column hardware, there was no detectable recovery of the doubly phosphorylated peptide. However, when using either type of coated column hardware (VPD #2, and the VPD #7), this

peptide became clearly visible in the obtained chromatograms. This result indicates, once again, that vapor deposition coatings can be used to minimize undesirable interactions with the metallic surfaces of chromatographic flow paths and in doing so allow for improved analyses of phosphorylated biomolecules.

As such, in one aspect, the vapor deposition coated column hardware is used to improve the recovery of phosphorylated biomolecules during analyses by liquid chromatography. In yet another embodiment of this invention, vapor deposition coated flow path components are used in conjunction with vapor deposition coated column hardware to improve the recovery of phosphorylated biomolecules during analyses by liquid chromatography.

The effects of this finding have been demonstrated for two examples of phosphorylated biomolecules, phosphorylated glycans and phosphorylated peptides. Phosphorylated biomolecules refers to any molecule naturally produced by an organism that contains a phospho group, including but not limited to phosphorylated proteins and polynucleotides. Furthermore, it is reasonable to envision this disclosure being used to improve liquid chromatographic analyses of smaller biomolecules, including but not limited to phospholipids, nucleotides and sugar phosphates. Indeed, vapor deposition coated column hardware has been found to be useful in improving the recovery and peak shape of sugar phosphates and nucleotides. The effects of employing vapor deposition coated versus untreated column hardware for the reversed phase LC analyses of glucose-6-phosphate, fructose-6-phosphate, adenosine triphosphate, and adenosine monophosphate are captured in FIGS. 8-11. Interestingly, these data indicate that the use of the vapor deposition coated column hardware can yield a significant improvement in both the overall recovery and peak shape of these phosphate containing small biomolecules. Thus, it is foreseeable that this disclosure could also be used to improve the chromatography of non-biomolecules, such as small-molecule pharmaceuticals containing either phospho or phosphonate functional groups.

Separation of Sialylated Glycans and Molecules Having Carboxylic Acid Moieties

It has additionally been discovered that vapor deposition coated hardware can be of benefit to mixed mode separations of sialylated glycans. In such a technique, sialylated glycans can be resolved using a stationary phase that exhibits its anion exchange and reversed phase retention mechanisms. It was just recently discovered that a unique class of stationary phase, referred to as charged surface reversed phase chromatographic materials and described in International Application No. PCT/US2017/028856, entitled "CHARGED SURFACE REVERSED PHASE CHROMATOGRAPHIC MATERIALS METHOD FOR ANALYSIS OF GLYCANS MODIFIED WITH AMPHIPATHIC, STRONGLY BASED MOIETIES" and published as WO2017/189357 (and incorporated herein by reference in its entirety), is ideally suited to producing these types of separations. The use of a high purity chromatographic material (HPCM) with a chromatographic surface comprised of a diethylaminopropyl (DEAP) ionizable modifier, a C18 hydrophobic group and endcapping on a bridged ethylene hybrid particle has proven to be an exemplary embodiment for the separation of glycans labeled with amphipathic, strongly basic moieties, like that imparted by the novel labeling reagent described in International Application No. PCT/US2017/028856 (WO2017/189357). This so-called diethylaminopropyl high purity chromatographic material (DEAP HPCM) stationary phase is effective in

separating acidic glycans as a result of it being modified with a relatively high pKa (~10) ionizable modifier that yields uniquely pronounced anionic retention.

In an application to DEAP HPCM mixed mode separations of sialylated glycans, vapor deposition coated hardware has been shown to yield improved chromatographic recoveries and peak shapes of glycans containing greater than three sialic acid residues. A comparison of fluorescence chromatograms for fetuin N-glycans obtained with untreated stainless steel versus VPD #7 coated hardware is provided in FIG. 12, wherein the effect on peak shape and recovery of tetra- and penta-sialylated glycans is easily visualized. The observed chromatographic differences are likewise easily quantified. In particular, fluorescence peak areas for the most abundant di-, tri-, tetra- and penta-sialylated glycans showed there were indeed very distinct differences in recoveries (FIG. 13). This testing was also used to demonstrate that other, chemically unique vapor deposition coating could be used with equally good effect. Much like the VPD #7 coated hardware, VPD #2 and SilcoTek Dursan® coated hardware showed equivalent capabilities in improving peak shape and recovery of the tetra- and penta-sialylated N-glycans. Interestingly though, it was not found to be necessary to use a coated flow through needle or column inlet in order to optimize peak shape and recovery.

As with phosphorylated species, this effect on the chromatography of sialylated glycans is believed to result from masking the metallic surface of the hardware and minimizing adsorptive sample losses that can occur with analytes that exhibit a propensity for metal chelation. However, the origin of the metal chelation is different in that the effect is a consequence of a glycan carrying multiple carboxylate residues versus one or two phosphorylated residues. Carboxylate containing compounds generally have a weak affinity for metals. Yet, when there are multiple carboxylate moieties present in one molecule, an opportunity for polydentate chelation is created, as is the case with tetra- and penta-sialylated glycans.

Accordingly, in an embodiment of this invention, vapor deposition coated column hardware is used during liquid chromatography of biomolecules containing greater than three carboxylic acid residues as a means to improve their peak shape and recovery. In yet another embodiment of this invention, vapor deposition coated flow path components are used in conjunction with vapor deposition coated column hardware to improve the peak shape and recovery of biomolecules containing greater than three carboxylic acid residues.

Separation of Proteins

Certain vapor deposition coatings have also been found to beneficially impact protein reversed phase chromatography. To demonstrate such, we evaluated a paradigmatic protein separation that is very important to the analysis of biopharmaceuticals, a monoclonal (mAb) subunit separation with MS-friendly, formic acid modified mobile phase. Using such a test, numerous combinations of column hardware materials have been examined. Vapor deposition coated stainless steel column tubes along with matching coated stainless steel frits were first tested against corresponding untreated stainless steel hardware. FIGS. 14A and 14B show fluorescence chromatograms obtained with these column hardware materials. From these data, it was found that hardware coated with VPD #7, but not hardware coated with VPD #2, was uniquely able to improve the baseline quality of the model separation, particularly in providing quicker returns to baseline. This improvement to the chromatographic performance of the separation is underscored by the fact that the

chromatogram produced with the VPD #7 coated column also shows higher peak intensities for some of the subunits. The nature of this baseline issue, as it exists with stainless steel hardware, can be reasoned to be a result of the protein analytes undergoing problematic secondary interactions and not homogeneously eluting at one particular eluotropic strength. Interestingly, in this example, the VPD #7 MVD hardware did not appear to significantly improve half height peak capacity nor the carryover of the columns, which was universally found to be ~0.9%. That is to say, for protein reversed phase chromatography, it would seem that vapor deposition coatings improve the quality of separation predominantly through affecting baseline properties.

An effect such as this can be very significant to protein reversed phase separations, particularly those intended to facilitate detection by online electrospray ionization (ESI)—mass spectrometry (MS). Often, it is critical to have quick returns to baseline in ESI-MS data given that it will make the assignment of chromatographic peaks less ambiguous. Signal from previously eluted species will be less abundant and therefore less confounding in data accumulated for later eluting peaks. With this in mind, 11 additional combinations of column hardware materials were screened, using ESI-MS detection as the means to assessing the quality of the data. FIG. 15A presents total ion chromatograms (TICs) for some of these materials, including columns constructed with stainless steel alternatives, namely polyether ether ketone (PEEK) and a low titanium, nickel cobalt alloy (MP35NLT). Surprisingly, columns constructed of VPD #7 coated hardware were the only found to give uniquely quick returns to baseline. Stainless steel, PEEK, and VPD #2 coated hardware showed comparatively slower returns to baseline. In addition, control experiments showed that the improvement to baseline quality can be achieved through the use of a VPD #7 coated frit alone and that coated tubing is not required to achieve an effect. Further experimentation culminating in the chromatograms of FIG. 15B has made it possible to glean additional insights. One of which is that it does not matter if the frit has a 0.2 or 0.5 μm porosity or if the VPD #7 coating has been thermally cured in the form of an annealing process (resulting in a VPD #8 coating). In contrast, neither a thicker VPD #3 coating (~1800 Å thickness) nor a cured coating (VPD #5) with an increased contact angle of 90° (up from ~35°) were able to produce the effect. Accordingly, VPD #7 coated frits are very unique in their ability being to affect the baseline of the example protein separation. While not limited to theory, it would seem reasonable to suggest that this effect derives from the hydrophobicity/contact angle of this coating. It could be that these coated frits closely mimic the surface chemistry of the reversed phase stationary phase. Consequently, a column with VPD #7 coated frits might exhibit adsorption sites (particularly those near the frit surface) that are more uniform in their chemical properties. Testing has shown that this effect on the protein reversed phase separation can be localized to the inlet frit of the column, lending credence to this hypothesis (FIG. 16). Indeed, one hydrophobic VPD #7 vapor deposition coated frit at the column inlet is sufficient to produce uniquely quick returns to baseline for the example mAb subunit separations. Proteins undergo reversed phase chromatography via fairly discrete adsorption/desorption events. Consequently, upon loading, protein analytes will be most concentrated at and likewise spend a significant amount of time at the head of the column, where an interface exists between the inlet frit and the packed bed of the stationary phase. At this interface, a protein analyte would have an opportunity to establish undesired secondary

interactions that would be cumulative to and energetically different than the desired hydrophobic interaction with the stationary phase. It is plausible that using a frit with surface properties similar to the stationary phase mitigates any chromatographic problems related to there being energetically and chemically diverse adsorption sites present at this packed bed interface. While not limited to theory, it may also be possible that a frit, such as the C_2C_{10} vapor deposition coated inlet frit (e.g., frit coated with VPD #7), imparts an entirely novel focusing effect to protein reversed phase separations that cannot be explained by the understanding and descriptions noted above. In addition, it is possible that a frit, such as the VPD #7 vapor deposition coated inlet frit, makes a unique contribution to how a stationary phase packs into a column. Use of a vapor deposition coated frit as the substrate for building a packed column bed may advantageously impact the properties of a stationary phase and resultant chromatography.

As such, in an embodiment of this invention, vapor deposition coated column hardware is used to improve the chromatographic performance of protein reversed phase separations. In yet another embodiment of this invention, a vapor deposition coating with a contact angle of $>90^\circ$, more preferably greater than 100° , is used to coat the tubing and frits of a column, or chromatographic device, as a means to improve the baseline and/or tailing factors of protein separations.

In a separate embodiment, this invention may utilize a frit material that is constructed of a specific polymer, such that an equivalently hydrophobic surface is achieved, specifically one with a contact angle greater than 90° , more preferably greater than 100° . Polytetrafluoroethylene (PTFE), polymethylpentene (PMP), high density polyethylene (HDPE), low density polyethylene (LDPE) and ultra high molecular weight polyethylene (UHMWPE) are examples of hydrophobic polymers that could be suitable for use as the frit or column material in other embodiments of this invention. In fact, an inlet frit constructed of porous PTFE (1.5 mm thick, Porex PM0515) was found to favorably affect protein reversed phase baselines, in a manner similar to that of the previously mentioned VPD #7 vapor deposition coated inlet frit (FIG. 17). Frits of alternative compositions are also relevant to this invention. In yet another embodiment, parylene, that is poly p-xylylene polymer, coatings could be used to treat column frits and to thereby improve the properties of a protein reversed phase separation. In addition, glass membranes could be used as the basis of a frit material. Onto the glass membrane substrate, silanes could be bonded to advantageously manipulate the hydrophobicity and contact angle of the material. These and other such membranes could also be used in conjunction with a backing material, like a porous polymer sheet, to lend physical rigidity to the apparatus.

Finally, vapor deposition coated hardware has been found to be of benefit to aqueous biomolecule separations, such as protein ion exchange chromatography. When looking to understand the charge heterogeneity of a sample, an analyst will often choose to resolve the components of a sample by ion exchange. In the case of protein therapeutics, this type of analysis is performed as a means to interrogate so-called charge variants, such as deamidation variants, that can have a detrimental effect on the efficacy of the corresponding drug product. Charge variant separations by way of ion exchange can therefore be critical to the effectiveness of a characterization approach for a protein therapeutic, most particularly a monoclonal antibody. Being such an important analytical

approach, protein ion exchange must be robust and able to quickly and reliably yield accurate information.

To this end, ion exchange separations of a monoclonal antibody were evaluated, and the effects of using uncoated versus vapor deposition coated column hardware were contrasted. FIGS. 21A-21L presents chromatograms of NIST reference material 8671, an IgG1 κ mAb, as obtained from sequential cation exchange separations and repeat injections of sample. In this evaluation, columns derived from four different constructions were tested. These columns varied with respect to both hardware design and vapor deposition coating. From the observed results, it was most apparent that uncoated hardware showed a prominent conditioning effect, as manifest in there having been low peak areas on initial injections. While not limited to theory, it is believed that the metallic surfaces of the uncoated column hardware imposed adsorptive losses on these separations and thereby hindered recovery of the sample. In contrast, vapor deposition coated hardware, both C_2 or C_2 -GPTMS-OH chemistries, yielded comparatively high peak areas even on the very first runs of the columns (FIG. 22). That is, coated hardware showed no evidence of requiring a passivation step, giving it the unique advantage of more quickly providing accurate chromatographic data. Here, it is clear that the noted vapor deposition coatings enhance the chromatographic properties of metallic hardware. Little can be seen in the way of distinguishing the chromatographic performance of the two tested vapor deposition coatings, namely the C_2 and C_2 -GPTMS-OH chemistries. However, the C_2 -GPTMS-OH coating has an inordinately low contact angle (as does C_2 PEO). It is foreseeable that certain types and classes of biomolecules will require a highly hydrophilic flow path. One such example could indeed be aqueous protein separations in which hydrophobic interactions could lead to poor recovery or peak tailing. As a whole, it is believed that vapor deposition coated hardware will show advantages for numerous forms of aqueous separations, including but not limited to ion exchange, size exclusion and hydrophobic interaction chromatography, and that the most ideal vapor deposition coating would be one that is very hydrophilic. Accordingly, in an embodiment of this invention, a vapor deposition coated column is used to improve the recovery of samples from aqueous chromatographic separations. In a more specific embodiment, a vapor deposition coating with a contact angle less than 20° is used to improve the recovery of biomolecules in ion exchange, size exclusion or hydrophobic interaction chromatography.

EXAMPLES

Example 1

C_2 and C_2C_{10} Vapor Deposition Coatings

Prior to coating, all metal components are passivated according to a nitric acid passivation. Passivated parts and a silicon wafer are then introduced to the vapor deposition chamber and vacuum is established. The first step is a 15 minute, 200 Watt, 200 cc/min oxygen plasma cleaning step. Next is the first vapor deposition cycle. Each vapor deposition cycle contains a silane vapor deposition, followed by the introduction of water vapor for silane hydrolysis. The silane vapor is delivered at a pressure of 2.0 Torr for 5 seconds, and then the water vapor is delivered at a pressure of 50 Torr for 5 seconds. Following delivery, the silane and water is left to react with the substrate for 15 minutes. This cycle is repeated to produce the desired number of layers and coating thickness. An additional processing cycle can be implemented to functionalize the coating with yet another

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silane. Moreover, a post coating annealing step can be used to further cross-link and increase the hydrophobicity of the coating. Typically, the annealing cycle involves subjecting the coating to 200° C. for 3 hours under vacuum.

A silicon wafer is used as a coupon to measure the thickness and contact angle of the coating. To measure the thickness, a Gaertner Scientific Corporation stokes ellipsometer model LSE is used. By analyzing the change in polarization of light, and comparing to a model, the film thickness can be established. To measure the contact angle, a Ramé-Hart goniometer model 190 is used. After dropping a controlled amount of water onto a perfectly level silicon wafer, optical techniques are used to measure the contact angle.

Example 2

C₂-GPTMS-OH Vapor Deposition Coatings

Prior to coating, all metal components are passivated according to a nitric acid passivation. Passivated parts and a silicon wafer are then introduced to the vapor deposition chamber and vacuum is established. The first step is a 15 minute, 200 Watt, 200 cc/min oxygen plasma cleaning step. Next is the first vapor deposition cycle. Each vapor deposition cycle contains a silane vapor deposition, followed by the introduction of water vapor for silane hydrolysis. The silane vapor is delivered at a pressure of 2.0 Torr for 5 seconds, and then the water vapor is delivered at a pressure of 50 Torr for 5 seconds. Following delivery, the silane and water is left to react with the substrate for 15 minutes. This cycle is repeated to produce the desired number of layers and coating thickness. In this example, the bis(trichlorosilyl) ethane silane is used to build up an adhesion or primer layer of approximately 800 Å. After C₂ deposition, the 3-(glycidoxypentyl)trimethoxysilane is delivered anhydrously to a pressure of 0.4 Torr in the vapor deposition chamber. This silane vapor is left to react with the C₂ coated substrate for one hour. This process results in an epoxide terminated coating, with a contact angle of 50°. After deposition, the next step is to hydrolyze the epoxide groups. This is performed either in the liquid phase or the vapor phase, with 0.1M acetic acid. After epoxide hydrolysis, the contact angle is <20°. Contact angle measurements are taken on a silicon wafer using a Ramé-Hart goniometer model 190.

Example 3

Alternative Contact Angle Measurement

It is relatively easy to measure the contact angle on the flat silicon wafers using a goniometer. However, not all our substrates have such smooth and flat surfaces. Frits can be considered a chromatography column's most important substrate, since the fluidic surface area to mass ratio is higher in the frit than in any other column hardware component. In order to measure the solid-liquid wetting properties of frit porosity, and confirm the presence of a coating, we can use the bubble point test. The bubble point test is used to determine the largest pore diameter of a frit structure, and the bubble point pressure is related to this diameter with the following equation:

$$P=(2\gamma \cos \theta)/r$$

Where,

P=bubble point pressure, Pa (measured)

γ =surface tension of test liquid, N/m (known)

θ =contact angle between test liquid and pore material (calculated)

r=largest pore radius, m (calculated)

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This equation is from ASTM E128 and is derived from the equilibrium condition of capillary rise.

Using the bubble point test to calculate a contact angle requires two steps. This first is to test the frit in IPA, and assume a 0 contact angle, since IPA has excellent wetting characteristics. This will yield a maximum pore diameter. The next step is to repeat the experiment with water as the test liquid, and the known pore radius. This will yield the contact angle with water, relative to the assumed 0 degree contact angle of IPA. FIG. 18 displays the different bubble point pressures recorded versus coating composition. FIG. 19 displays the derived contact angles versus coating composition. These values correlate well with measurements taken with a goniometer on a flat silicon wafer.

Example 4

Corrosion Performance of Silane Coatings

ASTM G48 Method A is used to rank the relative pitting corrosion performance of various grades of stainless steel. It consists of placing a part in ~6% ferric chloride solution for 72 hours, and checking the mass loss of your component. The test can be run at room temperature, or at slightly elevated temperatures to increase the corrosion rate. The ferric chloride solution is similar to the environment inside a pit during "non-accelerated" pitting corrosion; an acidic, oxidizing, chloride containing environment. When an entire part of interest is submerged in the ferric chloride solution, pitting corrosion is greatly accelerated, with normal test times only being 72 hours. FIG. 20 displays the corrosion performance of a non-coated column tube, and various coatings on a column tube. The improvement ranges from ~10x to ~100x.

Example 5

HILIC-Fluorescence-MS of Phosphoglycans

A recombinant alpha-galactosidase was diluted to 2 mg/mL. A 7.5 μ L aliquot of the protein solution was then added to a 1 mL reaction tube containing 15.3 μ L of water and 6 μ L of buffered 5% RapiGest SF solution—commercially available from Waters Corporation (Milford, MA) (50 mM HEPES-NaOH, pH 7.9). The mixture was placed in a heat block at 90° C. for 3 minutes. Thereafter, the reaction tube was allowed to cool at room temperature for 3 minutes. To the reaction tube, 1.2 μ L of PNGase F was then added and incubated at 50° C. for 5 minutes. After incubation, the reaction was again allowed to cool at room temperature for 3 minutes. To a vial containing 9 mg of RapiFluor-MS reagent, 131 μ L of anhydrous DMF was added and vortexed to create a labeling solution. A 12 μ L volume of this labeling solution was next added to the reaction tube. This labeling reaction was allowed to proceed for 5 minutes to produce the final sample.

A fully porous amide HILIC stationary phase (1.7 μ m, 130 Å) was used in a 2.1x50 mm column dimension to chromatograph the samples at a flow rate of 0.4 mL/min and temperature of 60° C. The gradient flow conditions initiated with 75.0% organic solvent (Acetonitrile) and 25.0% aqueous eluent (50 mM ammonium formate, pH 4.4) followed by a 11.66 min linear gradient to 54.0% organic/46% aqueous eluent. The column was then cycled through an aqueous regeneration step at 100% aqueous mobile phase at a flow rate of 0.2 mL/min for one minute. After the aqueous regeneration, the column was equilibrated at initial conditions for 4 minutes. Species eluting during the above separations were detected serially via fluorescence (Ex 265/Em

425, 2 Hz) followed by online ESI-MS. Mass spectra were acquired with a Xevo G2-XS QToF mass spectrometer operating with a capillary voltage of 2.2 kV, source temperature of 120° C., desolvation temperature of 500° C., and sample cone voltage of 50 V. Mass spectra were acquired at a rate of 2 Hz with a resolution of approximately 40,000 over a range of 700-2000 m/z. FIGS. 4A-4C present comparisons of this HILIC separation of RapiFluor-MS labeled released N-glycans from a recombinant alpha-galactosidase as performed with columns constructed of varying coatings and materials. FIGS. 6A-6C present comparisons of this HILIC separation as performed with columns constructed of varying coatings and materials in conjunction with a sample needle and column inlet tubing constructed of varying coatings.

Example 6

RPLC-UV-MS of Phosphopeptides

A vial of phosphopeptide test standard (Waters Corporation, Milford, MA) was reconstituted with 50 μ L of 0.1% formic acid. A fully porous CSH C18 stationary phase material (1.7 μ m, 130 Å) was used in 2.1 $\times$ 50 mm column dimensions to chromatograph the samples at a flow rate of 0.2 mL/min at a temperature of 60° C. The gradient flow conditions initiated with 0.7% organic mobile phase (0.075% formic acid in acetonitrile) and 99.3% aqueous mobile phase (0.1% formic acid) followed by a 30 min linear gradient to 50% organic/50% aqueous. Species eluting during the above separations were detected serially via UV (220 nm) followed by online ESI-MS. Mass spectra were acquired with a Xevo G2-XS QToF mass spectrometer operating with a capillary voltage of 1.5 kV, source temperature of 100° C., desolvation temperature of 350° C., and sample cone voltage of 50 V. Mass spectra were acquired at a rate of 2 Hz with a resolution of approximately 40,000 over a range of 500-6500 m/z. FIGS. 7A-7C present comparisons of this reversed phase separation of the phosphopeptide standard performed with columns constructed of varying coatings and materials.

Example 7

RPLC/MS of Small Biomolecules (Nucleotides and Sugar Phosphates)

RPLC/MS analyses of two nucleotides (adenosine monophosphate and adenosine triphosphate) and two sugar phosphates (glucose-6-phosphate and fructose-6-phosphate) were performed by reversed phase separations with an organosilica C18 stationary phase according to the methods parameters noted below. FIGS. 8-11 present comparisons of these reversed phase separations as performed with columns constructed of varying coatings and materials.

LC Conditions

Columns: BEH C18 130 Å 1.7 μ m 2.1 $\times$ 100 mm

Mobile Phase A: 0.25% Octylamine in H₂O pH adjusted to 9 with acetic acid

Mobile Phase B: ACN

Column Temperature: 35° C.

Injection Volume: 10 μ L (100 ng/mL sample concentrations)

Sample Diluent: Water

Detection: Tandem quadrupole mass spectrometer operating in ESI negative ionization mode and with MRM acquisition.

Gradient Table:

Time(min)	Flow Rate(mL/min)	% A	% B	Curve
Initial	0.450	95	5	Initial
5	0.450	75	25	6
10	0.450	75	25	6
20	0.450	50	50	6
21	0.450	5	95	6
22	0.450	95	5	6
30	0.450	95	5	6

Example 8

LC-Fluorescence-MS of Highly Sialylated Glycans Using Charge Surface Reversed Phase Chromatography

RapiFluor-MS labeled N-glycans were prepared from bovine fetuin (Sigma F3004) according to a previously published protocol (Lauber, M. A.; Yu, Y. Q.; Brousmiche, D. W.; Hua, Z.; Koza, S. M.; Magnelli, P.; Guthrie, E.; Taron, C. H.; Fountain, K. J., Rapid Preparation of Released N-Glycans for HILIC Analysis Using a Labeling Reagent that Facilitates Sensitive Fluorescence and ESI-MS Detection. *Anal Chem* 2015, 87 (10), 5401-9). Analyses of these released glycans were performed using a Waters ACQUITY UPLC H-Class Bio LC system and a separation method based on a previously described charged surface reversed phase chromatographic material described in International Application No. PCT/US2017/028856, entitled "CHARGED SURFACE REVERSED PHASE CHROMATOGRAPHIC MATERIALS METHOD FOR ANALYSIS OF GLYCANS MODIFIED WITH AMPHIPATHIC, STRONGLY BASED MOIETIES" (and incorporated by reference). Specifically, RapiFluor-MS labeled glycans (e.g., glycans labeled with the labeling reagent discussed in PCT/US2017/028856) were separated according to a mixed mode separation using a fully porous (130 Å) 1.7 μ m diethylaminopropyl high purity chromatographic material (DEAP HPCM) in a 2.1 $\times$ 100 mm column configuration. Details of the method are described below. FIGS. 12 and 13 present comparisons of this mixed mode separation of sialylated glycans as performed with columns constructed of varying coatings and materials.

LC Conditions

Column: DEAP HPCM 130 Å 1.7 μ m 2.1 $\times$ 100 mm

Mobile Phase A: Water and 100 mM Formic Acid/100 mM Ammonium Formate in 60%

Mobile Phase B: 100 mM Formic Acid/100 mM Ammonium Formate in 60% ACN

Column Temperature: 60° C.

Injection Volume: 4 μ L

Sample Concentration: 10 pmol/ μ L

Sample Diluent: Water

Fluorescence Detection: Ex 265 nm/Em 425 nm (10 Hz)

Gradient Table:

Time(min)	Flow Rate(mL/min)	% A	% B	Curve
Initial	0.400	100.0	0.00	Initial
24.00	0.400	78.0	22.0	6
24.20	0.400	0.0	100.0	6
24.40	0.400	0.0	100.0	6
24.60	0.400	100.0	0.0	6
30.00	0.400	100.0	0.0	6

LC/MS of a Reduced, IdeS Digested Monoclonal Antibody (mAb)

Formulated NIST mAb Reference Material 8671 (an IgG1 κ) was added to 100 units of IdeS and incubated for 30 minutes at 37° C. The resulting IdeS-digested mAb was then denatured and reduced by the addition of 1M TCEP and solid GuHCl. The final buffer composition for the denaturation/reduction step was approximately 6 M GuHCl, 80 mM TCEP, and 10 mM phosphate (pH 7.1). IdeS-digested NIST RM 8671 (1.5 mg/mL) was incubated in this buffer at 37° C. for 1 hour, prior to being stored at 4° C. Reversed phase (RP) separations of the reduced, IdeS-fragmented mAb were performed to demonstrate the effects of employing different vapor deposition coated column hardware pieces, namely the column tube and the frits that enclose the stationary phase into its packing.

A C4 bonded superficially porous stationary phase (2 μ m, Rho 0.63, 290 Å) was used in a 2.1 $\times$ 50 mm column dimension to chromatograph the samples at a flow rate of 0.2 mL/min and temperature of 80° C. across a linear gradient consisting of a 20 min linear gradient from 15 to 55% organic mobile phase (aqueous mobile phase: 0.1% (v/v) formic acid in water; organic mobile phase: (v/v) formic acid in acetonitrile). Species eluting during the above separations were detected serially via fluorescence (Ex 280/Em 320, 10 Hz) followed by online ESI-MS. Mass spectra were acquired with a Synapt G2-S mass spectrometer operating with a capillary voltage of 3.0 kV, source temperature of 150° C., desolvation temperature of 350° C., and sample cone voltage of 45 V. Mass spectra were acquired at a rate of 2 Hz with a resolution of approximately 20,000 over a range of 500-4000 m/z. FIGS. 14-17 present comparisons of this reversed phase C4 separation of reduced, IdeS-digested NIST Reference Material 8671 as performed with columns constructed of varying coatings and materials.

Example 10

Ion Exchange Chromatography

NIST mAb Reference Material 8671 (an IgG1K) was separated using columns constructed from a 3 μ m non-porous cation exchange stationary phase packed into either uncoated or vapor deposition coated hardware. Separations were performed with an ACQUITY UPLC H-Class Bio instrument according to the experimental conditions outlined below. FIGS. 21 and 22 present comparisons of these separations and their resulting data as obtained with columns constructed of varying coatings and materials.

LC Conditions

Columns: 3 μ m non-porous cation exchange stationary phase in a 2.1 $\times$ 50 mm column dimension

Sample: NIST mAb Reference Material 8671 diluted to 2.5 mg/mL with 20 mM MES pH 6.0 buffer

Gradient: 20 mM MES pH 6.0, 10-200 mM NaCl in 7.5 min

Flow Rate: 0.2 mL/min

Column Temperature: 30° C.

Injection Volume: 1 μ L (2.1 mm ID columns)

Detection: 280 nm

Hardware Design: Hardware A—Identical to ACQUITY UPLC BEH C18 column hardware

Hardware B—A design with an alternative sealing mechanism and some alternative material compositions.

Oligonucleotide Ion Pair RPLC

Testing has shown that flow paths modified with the vapor deposition coatings of this invention are also helpful in improving oligonucleotide separations. Example 11 provides evidence of such as observed in the form of improved recoveries and more accurate profiling of a sample's composition, particularly with respect to the first chromatograms obtained with a column.

In this work, a mixture of 15, 20, 25, 30, 35 and 40-mer deoxythymidine was separated using columns constructed from a 1.7 μ m organosilica 130 Å C₁₈ bonded stationary phase packed into either uncoated or vapor deposition coated hardware. Separations were performed with an ACQUITY UPLC H-Class Bio instrument according to the experimental conditions outlined below. FIGS. 23A-F and 24 present comparisons of these separations and their resulting data as obtained with columns constructed of varying coatings and materials.

LC Conditions

Columns: 1.7 μ m organosilica 130 Å C₁₈ bonded stationary phase in a 2.1 $\times$ 50 mm column dimension

Sample: 20, 25, 30, 35 and 40-mer deoxythymidine (0.5 pmol/ μ L)

Column Temperature: 60° C.

Flow Rate: 0.2 mL/min

Mobile Phase A: 400 mM HFIP, 15 mM TEA in water

Mobile Phase B: 400 mM HFIP, 15 mM TEA in methanol

Gradient: 18 to 28% B in 5 min

Injection volume: 10 μ L

UV Detection: 260 nm

Example 12

RPLC of Citric and Malic Acid

It should also be pointed out that the benefits of this invention are not limited to only biomolecules or phosphorylated/phospho group containing analytes. In fact, numerous types of so-called "small molecules" can be seen to have their separations improved through the adoption of vapor deposition coated flow paths and column hardware. One notable class of small molecules corresponds to compounds having a carboxylic acid moiety. By their nature, these are ubiquitous compounds and some, like citric acid and malic acid, are important metabolites of living organisms, given that they are constituents of the Krebs cycle.

Herein, we have investigated the effects of separating citric acid and malic acid with untreated versus vapor deposition coated columns. Citric acid and malic acid were analyzed by LC-MS with columns constructed from a 1.8 μ m silica 100 Å C₁₈ bonded stationary phase packed into either uncoated or C₂C₃ vapor deposition coated hardware. Separations were performed with an ACQUITY UPLC I-Class PLUS instrument, and eluting analytes were detected with a Xevo TQ-S triple quadrupole mass spectrometer according to the experimental conditions outlined below. FIGS. 25A-D presents a comparison of these separations and their resulting data. It can be observed from these results that use of the vapor deposition coated column hardware led to improvements in recovery and peak shape and thus sizable increases in MS intensity. This is noteworthy as it highlights the fact that the vapor deposition coating can be used to facilitate the development of a more sensitive and more accurate quantitation assay of these and other chemically similar compounds, including but not limited to isoc-

itric acid, α -ketoglutaric acid, succinic acid, fumaric acid, lactic acid, aconitic acid, itaconic acid, oxaloacetic acid, pyruvic acid, pantothenic acid, biotin, and folic acid. It is reasonable to assume that even zwitterionic small molecules would benefit from this invention. This class of compounds includes but is not limited to amino acids and neurotransmitters. Likewise, it is envisioned that this invention will be advantageous to be used to separate and analyze compounds containing metal binding moieties, such as cobalamin and the various types of porphyrins. Lastly, these same compounds would exhibit improved separations whether analyzed by RPLC or other modes of chromatography, such as hydrophilic interaction chromatography (HILIC), ion exchange, or mixed mode LC separations (i.e. ion exchange/reversed phase or ion exchange/HILIC).

LC Conditions

Columns: 1.8 μ m silica 100 Å C₁₈ bonded stationary phase in a 2.1×50 mm column dimension

Sample: Citric acid

Malic acid

Column Temperature: 30° C.

Flow Rate: 0.5 mL/min

Mobile Phase A: 0.1% formic acid, water

Mobile Phase B: 0.1% formic acid, acetonitrile

Injection volume: 2 μ L

Gradient Table:

Time(min)	Flow Rate(mL/min)	% A	% B	Curve
Initial	0.500	100.0	0.00	Initial
0.25	0.500	100.0	0.00	6
2.00	0.500	75.0	25.0	6
2.50	0.500	5.0	95.0	6
3.00	0.500	5.0	5.0	6
3.10	0.500	100.0	0.0	6

MS Conditions

MS1 Resolution: 1 Da

MS2 Resolution: 0.75 Da

Capillary Voltage: 1 kV

Source Offset: 50

Desolvation Temp.: 600° C.

Desolvation Gas Flow: 1000 L/hr

Cone Gas: 150 L/hr

Nebulizer: 7 bar

Source Temp.: 150° C.

MRM (Citric Acid): 191.2>87.1

MRM (Malic Acid): 133.2>115.2

Example 13

Mixed Mode Chromatography of Pesticides

Glyphosate is non-selective broad spectrum herbicide which is widely used across the globe as a crop desiccant. Maximum residue limits (MRLs) are enforced globally on various commodities of interest because of the potential health consequences posed to consumers. Glyphosate and its metabolite aminomethylphosphonic acid (AMPA) require unique approaches for sample preparation and chromatography separation. Various methods can be employed for quantitation, whether they are based on reversed phase, porous graphitized carbon, ion chromatography, hydrophilic interaction chromatography (HILIC) or mixed mode retention mechanisms. No matter the separation mode, assays for glyphosate and other related herbicide compounds can prove to be problematic. First, polar pesticides are difficult to retain on reversed phase columns without derivatization. Second,

glyphosate interacts with active metal surfaces. As a result, it is notoriously observed in the form of a broad peak or one with pronounced tailing.

Herein, we have investigated the separation of glyphosate with untreated versus vapor deposition coated mixed mode HILIC columns. Glyphosate was analyzed by LC-MS with 1.7 μ m diethylamine bonded organosilica 130 Å columns constructed from either uncoated or C₂C₁₀ vapor deposition coated stainless steel hardware. Separations were performed with an ACQUITY UPLC H-Class Bio coupled with a Xevo TQ-XS triple quadrupole mass spectrometer according to the experimental conditions outlined below.

FIGS. 26A-B and 27A-B show a comparison of coated and uncoated column performance for glyphosate in a solvent standard. As seen in FIG. 26B, glyphosate appears as a severely tailing, broad peak. In contrast, as seen in FIG. 26B, glyphosate is separated with much improved peak shape on the vapor deposition coated column. It can be observed from these results that the use of the vapor deposition coated column hardware led to significant improvements in peak shape, reduced peak widths and thus sizable increases in MS intensity (FIGS. 27A-B). It is reasonable to assume that the vapor deposition coated column also yielded higher recovery. These results are noteworthy as they demonstrate a means to developing more sensitive and more accurate quantitation assays for glyphosate and other chemically similar compounds, including but not limited to pesticides such as Ethephon, 2-Hydroxyethyl phosphonic acid (HEPA), Glufosinate-Ammonium, N-Acetyl-glufosinate, 3-Methylphosphonicopropionic acid (MPPA), Aminomethylphosphonic acid (AMPA), N-Acetyl-glyphosate, N-Acetyl-AMPA, Fosetyl-aluminium, Phosphonic acid, Maleic hydrazide, Perchlorate, and Chlorate.

LC Conditions

Columns: 1.7 μ m diethylamine bonded organosilica 130 Å stationary phase in a 2.1×100 mm column dimension

Sample: Glyphosate

Column Temperature: 50° C.

Flow Rate: 0.5 mL/min

Mobile Phase A: 0.9% formic acid, water

Mobile Phase B: 0.9% formic acid, acetonitrile

Injection volume: 10 μ L

Gradient Table:

Time (min)	Flow Rate (mL/min)	% A	% B	Curve
Initial	0.500	10	90	Initial
4.00	0.500	85	15	2
10.0	0.500	85	15	6
16.0	0.500	10	90	1
20.0	0.500	10	90	1

MS Conditions

MS1 Resolution: 1 Da

MS2 Resolution: 0.75 Da

Capillary Voltage: 2.4 kV

Ionization: ESI-

Desolvation Temp.: 600° C.

Desolvation Gas Flow: 1000 L/hr

Cone Gas: 150 L/hr

Nebulizer: 7 bar

Source Temp.: 150° C.

MRM 1: 168.0>62.6

MRM 2: 168.0>149.8

ALTERNATIVES

There are a number of alternative methods and uses for the present technology. While the above methods have

generally been discussed with respect to chromatography or the use of a column. Other types of fluid components having an internal flow path may benefit from the present technology. For example, it is generally thought that capillary electrophoresis, such as capillary zone electrophoresis, exhibits relatively poor reproducibility. Much of the reproducibility issues can be reasoned to originate from irreproducible surface chemistry on the inner diameter of the tubular capillaries that are used to perform the separation. A vapor deposition coating on capillaries intended for a CE separation may therefore circumvent reproducibility issues as it can yield an inordinately thick, rugged coating. The inventions described herein may consequently be applicable to improving CE separations of both small and large biomolecules.

Moreover, while the examples of the described aspects employ comparatively hydrophobic coatings, with water contact angles ranging from 15° to 110°, it is reasonable to suggest that some separations could be enhanced through the application of hydrophilic coatings, including but not restricted to diol, amide/ureido type, and polyethylene oxide/glycol bondings.

Other analytes, not yet explicitly described, may also benefit from vapor deposition coated chromatographic flow paths, for instance phosphorothioated oligonucleotides. Nucleic acids inherently contain repeating phosphodiester bonds as part of their backbone. In some case, the phosphodiester backbone is replaced in part with a phosphorothioate backbone, which can impart in itself unique challenges for a separation. Similarly, intact and proteolytically digested antibody conjugates may benefit from methods entailing the use of vapor deposition chromatographic flow paths. Lastly, biomolecules containing histidine residues are likely to benefit from this invention as, like phosphorylated and carboxylate containing residues, they have a propensity for binding to metal.

What is claimed is:

1. A chromatographic device for separating analytes in a liquid sample comprising:

a sample injector having a sample injection needle for injecting the liquid sample into a mobile phase;

a sample reservoir container in fluid communication with the sample injector;

a liquid chromatography column downstream of the sample injector; the liquid chromatography column having fluid connectors; and

fluid conduits connecting the sample injector and the liquid chromatography column;

wherein interior surfaces of the fluid conduits, sample injector, sample reservoir container, and liquid chromatography column form a fluidic flow path having wetted surfaces; and

wherein a bioinert coating is deposited on at least a portion of the wetted surfaces by infiltrating vaporized bis(trichlorosilyl)ethane or bis(trismethoxysilyl)ethane through the fluidic flow path to form the bioinert coating.

2. The chromatographic device of claim 1, wherein at least portion of the fluidic flow path is formed of metal components.

3. The chromatographic device of claim 1, wherein the bioinert coating consists of a product of the silane hydrolysis reaction of vapor deposited bis(trichlorosilyl)ethane and water vapor or a product of the silane hydrolysis reaction of vapor deposited bis(trismethoxysilyl)ethane and water vapor.

4. The chromatographic device of claim 1, wherein the bioinert coating has a contact angle of at least about 15°.

5. The chromatographic device of claim 1, wherein the bioinert coating has a contact angle less than about 50°.

6. The chromatographic device of claim 1 further comprising a detector downstream of the liquid chromatography column and wherein the fluidic flow path further comprises the detector.

7. The chromatographic device of claim 6, wherein the detector is a mass spectrometer and the fluidic flow path includes wetted surfaces of an electrospray needle.

8. The chromatographic device of claim 1, wherein the fluidic flow path has a length to diameter ratio of at least 20.

9. The chromatographic device of claim 1, wherein the bioinert coating has a thickness of at least 100 Å.

10. The chromatographic device of claim 9, wherein the bioinert coating has a total thickness of about 500 Å.

11. The chromatographic device of claim 1, wherein the chromatographic device is characterized such that, when the analytes comprise phosphoglycans, phosphopeptides, glycans, or proteins, chromatographic separation of the analytes has improved recovery and peak shape relative to an uncoated wetted surface as measured in increased chromatographic peak area and reduced peak width.

12. A method of tailoring a metallic fluidic flow path for separation of a liquid sample comprising an analyte, the method comprising:

(a) infiltrating vaporized bis(trichlorosilyl)ethane or bis(trismethoxysilyl)ethane into a metallic fluidic flow path; and

(b) controlling temperature and pressure to deposit a bioinert coating on wetted surfaces of the metallic fluidic flow path, the bioinert coating having a thickness of at least 100 Å and a contact angle of at least 15°.

13. The method of claim 12, further comprising pretreating the wetted surfaces of the metallic fluidic flow path with a plasma prior to depositing the bioinert coating.

14. The method of claim 12, further comprising delivering water vapor to the metallic fluidic flow path for silane hydrolysis of the vaporized bis(trichlorosilyl)ethane or bis(trismethoxysilyl)ethane on the wetted surfaces.

15. The method of claim 12, further comprising modifying the bioinert coating with a silanizing reagent to functionalize the bioinert coating with a different silane than the vaporized bis(trichlorosilyl)ethane or bis(trismethoxysilyl)ethane.

16. The method of claim 15, wherein the silanizing reagent is a non-volatile zwitterion or an acidic or basic silane.

17. The method of claim 16, wherein the non-volatile zwitterion is sulfobetaine or carboxybetaine.